

Philosophical Propaedeutic

in

Outline

by

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Adamant homines scientias et contemplationes particulares, aut quia autores et inventores se earum credunt, aut quia plurimum in illis operæ posuerunt, iisque maxime assueverunt. Hujusmodi vero homines, si ad philosophiam et contemplationes universales se contulerint, illas ex prioribus phantasiis detorquent et corrumpunt.

FRANC. BACO.

[Men grow attached to particular sciences and speculations, either because they fancy themselves their authors and inventors, or because they have bestowed the greatest labour upon them and become most habituated to them. But if men of this sort betake themselves to philosophy and universal speculations, they distort and corrupt these out of their former fancies. — Francis Bacon.]

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Introduction.

The philosophical propaedeutic sets itself to answer two fundamental questions: *what is philosophy? what validity can be ascribed to philosophy as a science?*

To answer these questions the propaedeutic must characterize philosophy by pointing out the shapes which it has assumed at its various stages of development; the general fundamental concept will then be determined more closely as it runs through significant stages. The propaedeutic, in letting philosophy come to be for contemplation in its own way, must everywhere take up something historically given. But, although the historical is made use of, we can here give neither an epitome of the history of philosophy in its entirety, nor a separate detailed treatment of individual, arbitrarily chosen principal parts on their own account; the historical must be grasped and used in such a way that, within the connection determined by the nature of the matter, it illuminates the tasks that arise, and serves the totality.

To characterize philosophy as a science in its relation to the other sciences, it is doubtless necessary to enter more closely into an investigation of that which brings about the distinctive character of the various sciences, and thus to point out the principle of division on which their separation essentially rests; but since the determination of what is universally scientific is and remains the main thing, the treatment of the mutual relation of the diverse sciences cannot take up the particulars of that ramification to such a degree that there should here be any question of an encyclopædia of the sciences.

Insofar as philosophy itself is a science articulated into definite members, it will likewise, for the sake of the fundamental concept, be necessary to point out, within this science's own domain, the inner connection of the larger divisions with the articulating principle; but since every survey in which the individual parts are not at the same time treated with some detail necessarily becomes superficial, the propaedeutic must, instead of setting itself the task of an encyclopædic presentation of the disciplines of philosophy, rather lay out its principal parts in such a way that, by a more detailed treatment of each division on its own, it may find occasion to illuminate the systematic structure from one or another definite side, and thereby gradually open an insight into the architectonic relations of the whole.

In order to comprise all the separate considerations under one comprehensive point of view, the propaedeutic begins with the conceptual determination that philosophy is a scientific theory of consciousness.

The theory of consciousness shows what it means to become conscious of something, and must to that extent distinguish between consciousness itself and that which comes to consciousness. This distinction in turn demands the union of the distinguished sides. An empty consciousness, a consciousness that is conscious of nothing, is a contradiction: consciousness must have a content.

Although consciousness and its content are to that extent essentially the same — since the passivity and activity of consciousness are always determined in likeness with that which, under this passivity and activity, is taken up into consciousness — they are nevertheless in reality so different that the scientific orientation depends on whether attention is directed especially either to the consciousness determined by the content, or to the objectivity of the content set over against consciousness.

Since the side of consciousness is everywhere, and must be, the predominant thing in philosophy, it is necessary to investigate the fundamental relation a) between consciousness and its content in general, b) between consciousness and the particular content, c) between consciousness and the absolute content. The main parts of the propaedeutic corresponding to this are then I) Theory of Knowledge, II) Theory of Science, and III) Doctrine of Ideas.

I. **Theory of Knowledge.**

§ 1.

Knowledge presupposes identity of consciousness and content; as the object is, so too must consciousness be: “the like is understood by the like.” Insofar as consciousness is sensing, the object is sensible; insofar as it is thinking, thinkable.

The essential difference between sensation and thought can be recognized by the difference between the Positive and the Negative. The object of sensation is, without regard to whether the sense is outer or inner, always something immediate, existing

in the moment, and to that extent a Positive: the Negative cannot be sensed. By suspending immediacy and positing negation, thought arises; one can think *about* the immediate, but its immediacy cannot be immediately thought. Insofar as thought begins with the immediate, the understanding shows itself as the intellectual sense. The sensing understanding is not a thinking, but the immediate presupposition of thinking, a thinking in possibility: actual thinking begins with the Negative. In its pure, qualitative opposition to the immediacy of sensation, the negation of thought is itself immediate. The immediately Positive and the immediately Negative bound one another; by this bounding they mutually determine each other. When it is asked what the Positive is, the answer must be given by means of a Negative, *omnis determinatio est negatio* [all determination is negation]; likewise, on the other hand, the Positive must be presupposed when negation and the Negative are to be determined.

As sensing, consciousness is essentially passive; here it holds that “all our knowing is a finding, receiving, and beholding (*scimus, quia accepimus* [we know because we have received])”; as thinking, consciousness is essentially self-active. As regards self-activity, all our knowing (Cf. Sibbern: *Om Erkjendelse og Granskning*) is “a doing and producing (*scimus, quia facimus* [we know because we make])”. The sensible is objectified in representation and image, the thinkable in insight and conception.

Sensation is not merely heterogeneous with thought: what can be sensed cannot be thought, and conversely, what can be thought cannot be sensed; but the individual senses are also mutually heterogeneous: what can be seen cannot be heard, what can be heard cannot be felt. Nevertheless these opposites too determine one another mutually. The audible can be determined by the visible (sound-figures), the sensible by the thinkable. In the passivity of the sensing consciousness there is an activity, in the thinking activity there is a passivity: the representation is obtained only insofar as consciousness itself forms it; thought determines only insofar as it is led, compelled, and determined by that which is thought; the sensing consciousness is thinking, the thinking sensing; the acting I is passive, the passive acting (*scientia et potentia in idem concidunt* [knowledge and power coincide in one and the same]).

Since it is now one and the same consciousness that is both passive and active, both sensing and thinking, and since one and the same object can have both sensible and thinkable properties, it follows from this that there must not only be an infinite reciprocal relation between sensation and thought, but that this reciprocal relation

must even be able to set itself up so differently that now it is thought that is absorbed into sensation, now conversely sensation that vanishes into thought.

Insofar as thought is absorbed into sensation in such a way that only the sensible in the object is granted validity, we get a one-sided sensuous tendency: *Sensualism*; the corresponding one-sidedness then consists in this, that sensation is grasped as the fleeting presupposition of thought, so that only the thinkable in the object, the negative essence, the pure concept, is acknowledged as the True: *Idealism*. The infinite reciprocal relation of sensation and thought will therefore first become evident when Sensualism and Idealism have each contributed their share to a closer determination of the truth in a double-sided *Factual Knowledge*.

A.

Sensualism.

§ 2.

The consciousness of the individual begins with an immediate unity of sensation and thought; but the advance of knowledge is conditioned by thought's being brought out on its own in its opposition to sensation. Now so long as the equilibrium between the two expressions of consciousness is preserved, so long as the agreement between the sensible and the intellectual is still maintained, individuality is happy, and its cultivation beautiful. But with growing abstraction, as the imagination really begins to work over and transform the elements of sensation, thought becomes restless, a tension sets in, and the equilibrium is lost. The lost equilibrium — whose mark is that now the intellectual, now the sensible, now an overwrought spirituality, now a bold naturalness, has gained the upper hand — will then be re-established, and the equilibrium thus re-established be lost again.

Corresponding transitions are seen in the history of philosophy.

Greek philosophy begins among the Ionians (Thales, Anaximander, Anaximenes) with a naïve cognition of the non-sensible in the sensible; thought seeks the principle of existence, and also divines that the principle must be “something unbounded,” yet finds it most nearly represented by a given sensible material: “Water,” “Air.” This

going-together of thought with sensuousness is not yet Sensualism. Thought is here so far from feeling the pressure of the senses' preponderance that it is precisely on the point of raising itself above the parcelling-out of the sensible; it is Oneness that thought has here grasped with youthful courage, as it first attempts a soaring flight toward the universal.

Only when thought has left immediacy behind, has consumed its content, and is weary of mirroring itself in itself, does Sensualism emerge. Sensualism is a conscious opposition, an express counterweight to the empty conceptual essence: only after Greek thought (Pythagoreans, Eleatics, Plato) had volatilized the material of sensation, held fast to the pure essence, and tried its hand at all manner of abstractions, does Aristotle assert experience, and the repressed Sensualism then properly comes forward in the theory of knowledge of the Stoics and Epicureans.

The stages which in this respect characterize the directions of the movement in Greek philosophy repeat themselves on an even larger scale in the universal history of philosophy.

In contrast to the Middle Ages, antiquity on the whole exhibits the beautiful equilibrium and natural unity of sensation and thought; but in the Middle Ages abstract spirituality commits such an overreach that a reaction thereby called forth has been able to lead the modern age to a denial of spirit whose extent, sharpness, and inner transparency is unknown among the ancients.

In the Middle Ages, when the natural is only something corruptible, the sensible only something sinful, the present only a deception, it is naturally impossible to know things as they are. Frightened by fancy and feeling, the thought of science enters the service of faith, and is cowed by the supernatural; the cowed thought finds only a formal freedom in the logical. Hence the emptiness, the subtlety, the word-splitting, and the scientific barrenness in which Scholasticism ends. Instead of grasping a reality, and nourishing itself on the working-up of a digestible material, reflection fed upon itself, and was tormented by hunger.

The efforts that scientific hunger called forth drew minds to nature in two mutually opposed directions, a speculative and an empirical; yet in such a way that something of the one can be found again in the other.

The speculative direction. Insofar as there can be any talk of a speculative Sensualism, we find such among the Italian philosophers: Telesio, Cardano, Bruno, and others.

With a decided predilection for the forms of ancient philosophy, these thinkers were inspired by the assurance of nature's identity with spirit, by the contemplation of that divine life which animates the world-organism, the living unity that works through all things.

That Aristotelian categories are applied to the interpretation of natural phenomena is not in and of itself Sensualism; the Scholastics did the same, in good understanding with the Hierarchy. The Sensualistic, on the contrary, comes to light in the boldness, frankness, and unconditionedness with which an omnipresent natural life displaces the Beyond, and sets itself in the place of the supernatural. The realm of reason with which Giordano Bruno deals (*De la causa, principio ed uno*) is at the same time a realm of the senses, and the wonders to which Vanini (*De admirandis Naturæ, reginæ Deæque mortalium, arcanis*) points are owed not to "the eternal King," but to "the Queen of mortals."

The empirical path. By sober research, by careful observation of the phenomena (Tycho Brahe) and corresponding calculations, that path was at last marked out which alone could lead to a true knowledge of nature. The law of falling bodies on earth (Galileo) and the planetary orbits in the heavens (Kepler) showed clearly enough that the path had been found. The investigators who thus understood not merely how to open the eye, but also how to arm the eye (the telescope), then, in the name of science, asserted the right of the senses, and set in force facts whose incontrovertibility must at length subdue the imaginings of the Middle Ages, and shake Scholastic metaphysics in its foundations.

Were such efforts to be grasped, judged, and presented in a theory of knowledge framed to the needs of natural science, then it would doubtless above all be a matter of impressing two chief demands: thought's liberation from abstract theories (from *anticipationes naturæ* [anticipations of nature]), and the leading-back of knowledge to true experience (*interpretatio naturæ* [interpretation of nature]). By working against formalism, emphasizing "Induction," and bringing the concept of natural law into harmony with the demand of physics, Francis Bacon of Verulam (†1626; *Novum organum scientiarum*), though without a natural-scientific initiation, has nonetheless, in inspired traits, pointed out a new foundation, and marked the transition to a new age.

§ 3.

Advancing Empiricism soon discovered the misinterpretation of the a priori that lay hidden in the doctrine of “innate ideas.” Against this misinterpretation J. Locke (†1704) directed his “philosophical essay on the human understanding” (*An essay concerning human understanding in four books*), whose content we may here consider under the following points of view: the origin of “ideas,” the essence of “ideas,” and the knowledge of things.

a) *The origin of “ideas.”* There are no innate ideas. Even if it could be shown that there were propositions about whose universality all must agree, this would still prove nothing, so long as the universal acceptance of these propositions could be explained in another way. But so far are such propositions from being found, that nations at different levels of culture are even at odds about the very most important practical principles. If one sets forth the commandment “Whatever you would have others do to you, do also to them,” then everyone who has not heard the commandment before, but thinks it over, will of course ask: why? But to ask for the grounding of a commandment that is innate is a contradiction.

As regards the theoretical propositions, such as “Whatever is, is; it is impossible that one and the same thing can at the same time both be and not be,” children, idiots, and the uneducated are wholly unacquainted with their meaning. But if such principles really were innate to the understanding, then all would have to be aware of them from earliest childhood, for to be in the understanding is the same as to be in knowledge.

Likewise it is an evasion when, in order to defend innate ideas, one appeals to the fact that these ideas immediately become evident to consciousness as soon as a person begins to use his reason. The relation is precisely the reverse: it is not universal but particular propositions, such as “Sweet is not bitter,” of which one first of all becomes conscious when one begins to think. That theoretical and practical principles should be innate is on the whole just as absurd as that arts and sciences should be innate. It is therefore impossible to be born with ideas; the child’s soul is like a clean sheet of paper, as empty as a *tabula rasa* [a blank tablet].

Whence, then, does the understanding get its ideas? From experience. But experience is of a twofold kind: insofar as it is made by an observation of external objects conducted through the senses, we call it sensation, *Sensation*; insofar as it

rests on observation of our own understanding's activity, we call it the inner sense, *Reflection*. Sensation and Reflection are to be regarded as "the only channels that convey knowledge to the understanding, the only windows through which the light of ideas falls into the dark chamber of the soul."

b) The essence of "ideas." The ideas coming from experience are partly simple, partly compound.

The simple ideas furnish the material of knowledge, are, so to speak, the letters in the alphabet of existence. In that these simple ideas press themselves upon consciousness from without, the soul is passive, and takes up the impressions of the surrounding world as the mirror takes up the object's image; if, on the other hand, they are obtained through Reflection — as the idea of thought, the idea of will — the soul is active. The simple ideas are partly such as come to the understanding through a single sense: colors through the eye, tones through the ear, impenetrability through the sense of touch; partly such as are conveyed to it through several senses, as when space and motion are apprehended both by touch and by sight.

By the connections and compositions of the simple ideas the soul can now know the objects of the surrounding world, "just as in language, by manifold compositions of letters, one can form syllables and words"; but, in that the compound, complex ideas are referred to three chief classes — modes, substances, and relations — it becomes above all the task to give an account of the concept of substance.

The nominal essence indicates the relation between genus and species; the real essence, on the other hand, denotes the inner peculiarity of the existing thing. In the simple ideas and modes the real and the nominal essence are one and the same; with respect to "substances" they are different. That a surface bounded by three straight lines is a triangle expresses the unity of the real and the nominal essence. It is otherwise with the part of matter that forms the ring on my finger; on the real peculiarity of the non-sensible parts depend the properties of its color, weight, softness, hardness, and so on. After the observed properties the name "Gold" is formed, an expression that denotes the nominal but not the real essence.

By sensation and reflection we experience how a number of simple ideas come together in a peculiar connection, presuppose on that account a necessary ground of connection, and refer this ground to a substratum subsisting on its own, namely *substance*; but is it now possible to know substance?

c) *The knowledge of things.* As the bearer of the qualities by which the simple ideas are produced in us, substance can be known only in this, that a distinction is made between the properties that necessarily belong to the thing in itself, and prove the agreement of the simple ideas with the objects (*qualitates primariæ* [primary qualities]), and such properties — color, tone, taste, smell, warmth, cold — as do indeed proceed from the things' action upon the soul, but are nevertheless wholly different from the objective essence of the things (*qualitates secundariæ* [secondary qualities]).

In this way Locke would determine the opposition between real truth (*real truth*) and mental truth (*mental truth*); but although substance is distinguished from all other complex ideas precisely in this, that it has its archetype outside us — one does not connect a sheep's bleating with the representation of a horse —, and precisely therein makes known its objective reality, it is nevertheless impossible for us to penetrate into substantiality itself. What we call the essential does not bear upon the individual. "It is doubtless necessary that I be what I am: God and nature have formed me thus; but I know nothing that is essential to me, the individual, as individual: an accident, an illness, can bring about incredible changes." So the essential bears, then, upon the species; but it can by no means be the real essence that determines the species, since the real essence is unknown to us. We are referred to a collection of sensible qualities that we can observe; but even when our observations are conducted with the greatest possible accuracy, our knowledge is just as far from corresponding to that inner constitution of the thing from which these qualities follow "as a peasant's representation of the clock in Strasbourg is far from corresponding to the inner, artful structure of that remarkable machine."

What Locke had begun, Condillac (†1780; *Essai sur l'origine des connaissances humaines*) continued. By letting the two sources of knowledge — sensation and reflection — flow together in such a way that the latter vanishes into the former, and by laying a reinforced emphasis on sensation, Condillac arrived at the result that the so-called inner sense is only a reflex of the outer, and that consequently all expressions of spirit represent only modified sense-impressions. Once Sensualism, thus carried through, has first shown that the True is only something sensible, it lies near to reverse the proposition and say: only the sensible is the True. Here is a materialism for which everything spiritual is a figment (La Mettrie).

§ 4.

From presuppositions by which a one-sided standpoint is expressly designated, consequences can be developed partly in a coordinate, partly in a superordinate direction. To consequences in a direction coordinate with the standpoint belongs everything that unfolds its character, and raises its significance to an ever wider compass. In this sense Condillac and La Mettrie have driven the consequences of Sensualism to the extreme.

To consequences in a direction superordinate to the standpoint belong those that lead out beyond the standpoint itself, and thus either establish or at least introduce a new standpoint. In this sense Hume's (†1776) Scepticism (*A treatise of human nature*) is to be grasped as a higher consequence of Lockean Sensualism.

To test the validity of concepts, one must — herein Hume agrees with Locke — hold to experience. Those concepts that can prove their origin from sense-impressions (*impressions*), or at least from the reflections corresponding to these, ought to be acknowledged; those that cannot, are of spurious descent, and ought to be rejected.

According to Locke, "the idea of cause and effect" arises when, in considering the constant change of things, we observe how certain things — be they substances or qualities — begin to exist through the law-governed activity of another. But what validity now has this observation? whence do we know that two things stand to one another as cause to effect? *A priori* we cannot know it; for a priori knowledge is analytic, but the knowledge of the causal concept must be synthetic, since the effect is really something other than the cause. Neither do we know it from experience; for experience shows us only that the one fact partly follows the other in time (succession), partly touches the other in space (contiguity); but no experience can teach us that the one essentially follows from the other, as the effect from its cause.

To assert the validity of substance Locke had already referred to habit; this indication Hume follows: "Our inferences from experience are grounded solely on habit; we are accustomed (*the mind is carried by habit*) to see the one thing follow partly along with, partly after the other, and then habituate ourselves thereby to the representation that the one must be an effect of the other. Out of the relation of contiguity and succession our imagination forms the causal relation: *cum hoc aut post hoc, ergo propter hoc* [with this or after this, therefore because of this]."

That the threads of the investigation here run together in the question of the causal concept as in their nodal point, shows the sureness with which the task has been posed.

Once the causal concept has fallen, then the categories — force, substance, necessity, truth — fall as of themselves; the bonds of essence that were to hold things together must now all burst.

As proceeding from a cooperation of sensation and reflection, the concept “force” might, in Locke’s opinion, well have its validity; but how is it now possible to know force? It might at first glance seem as though the relation between soul and body furnished, in this respect, a reliable presupposition. In that certain bodily organs involuntarily obey our will, spirit does indeed act as force, and we need therefore only to become conscious of our voluntariness in order to become conscious of an expression of force. This is, however, a deceptive reflection; partly we do not know the means by which spirit acts upon the body; partly it is not, after all, all the body’s organs whose activity is subject to the will. Were force to be an object of experience, then — what is impossible — cause and effect would also have to be an object of experience.

By presenting substance as an Unknown, Locke had already introduced the doubt about its existence. How are we now to know substance? Were the validity of substantiality to admit of proof, then the inner necessary connection of the properties that in substance are united into unity would have to admit of proof; but with causality the necessary connection (*this idea of a necessary connexion*) has fallen away. To refer to the difference between original and derived properties does not help. The derived properties are all subjective, and thus have only existence in our apprehension; as for the original ones, they let themselves at last be resolved into one single one, namely “Solidity,” but Solidity is an empty word, since we, when all the other properties are suspended, cannot possibly make the Solid an object of any experience.

In reliance on the “inner sense,” Locke had still held fast to “the idea of will,” and acknowledged the self. But how should it now be possible to assure oneself of the reality of the I? Our feelings, thoughts, and representations cannot possibly be regarded as expressions of force of an I, when the category expression-of-force has no validity; they cannot possibly be referred to an acting self, as to their true cause, when the causal concept has no truth.

The remnant of reality that still attached, in Locke, to the concept of truth (*real truth*) has thus vanished: truth is gone, probability has stepped into its place. The I, the self, is then in reality nothing other than a figment, which arises in this, that an aggregate of many rapidly successive representations is laid under a substratum.

That a philosophy which subordinates thought to sensation, even to such a degree that concepts must lose their universal validity, that the laws of things must vanish before the rules of the association of ideas, the ground of reason before habit, truth before probability, the self-knowledge of the I before a deluding figment, is in itself neither tenable nor satisfying, goes without saying; nevertheless Hume's Scepticism, in contrast to dogmatic Materialism, is a higher power, a spiritual force whose attack upon thought can transform itself into a defense of thought.

B.

Idealism.

§ 5.

If Sensualism can be recognized by this, that it examines the senses in order to judge thought, then Idealism can in turn be recognized by this, that it makes thought the judge of the senses; if Sensualism sets things above consciousness, then Idealism reverses the relation, and sets consciousness above things.

In that Descartes (†1650; *Principia philosophiæ*) begins with doubt, he wishes, like Bacon, to cast off prejudices, and free the mind from blind trust in authority; but, in that by doubting everything (*de omnibus dubitandum est* [everything is to be doubted]) he can also draw into doubt the validity of natural things and of experience as well, he goes further than Bacon.

Since doubt is a work of thought, thought is, in the midst of doubt, raised above doubt; everything can be doubted, but the validity of the doubting thought cannot be drawn into doubt; to doubt doubt is to confirm doubt.

Empiricism, which begins with an unconditional reliance on the immediate sense-impression, ends consistently in Scepticism; for Scepticism is an activity of spirit in

which thought, after having stared itself blind on the sensible, incessantly comes to itself as a stranger, tears itself down, and cannot know its own essence.

In the idealistic doubting of everything, thought begins precisely by finding itself, in that it begins by removing the foreign, and shaking off the appendage of tradition. This doubting is then so far from losing itself in Scepticism that the thinker rather wishes to assure himself of the validity of everything that withstands the testing of thought. The thinker then not only sees the unity of his thought with being (*cogito ergo sum* [I think, therefore I am]); but also gains certainty that consciousness is borne, penetrated, and filled by an absolute being, by the Unconditioned, whose image is in the soul, whose forms (*ideæ innatæ* [innate ideas]) shine with a clarity that can proceed only from the Perfect.

In Descartes, however, Idealism is still restricted in this, that over against the being that is in thought there stands a material being opposite thought; to suspend this substantial separation, and transform thought and being into attributes in one and the same substance, is an advance owed to Spinoza (†1677). Insofar as the question here is of the validity of thought and its identity with being, Spinozism is decidedly Idealism.

As regards the soul's a priori self-activity, Leibniz (†1716) went a step further. In his "Nouveaux essais" he works against Lockean Sensualism, and frees the assumption of "innate ideas" from the misunderstanding that could justify Locke's objections. To the Sensualists' *nihil est in intellectu, quod non antea fuerit in sensu* [nothing is in the intellect that was not previously in sense], there is then added a *nisi intellectus ipse* [except the intellect itself]. That "ideas" are innate must not be misinterpreted as though at birth they lay ready-made in the soul, since on the contrary they are in it only according to disposition. This view is in essential connection with the doctrine of monads; as monad the soul is originally self-active, can receive no impression from without, but on the contrary develops a world in its interior.

Whereas Locke compared the soul to an unwritten tablet, Leibniz compares it to "a block of marble in which the veins preform the shape of the statue." The principle of contradiction (*principium contradictionis*), the proposition of sufficient reason (*principium rationis sufficientis*), and the so-called *principium indiscernibilium* belong among "the innate ideas." How Idealism, by thinking over the relation between the senses and the sensed, and pointing out the contradictions that everywhere accompany the cognition of an external, can transform the whole outer world into

a world of spirit, has been shown in particular by G. Berkeley (†1754; *A treatise concerning the principles of human knowledge*; and *Three dialogues between Hylas and Philonous*). — Convinced that it is the assumption of matter's unconditional existence that especially favors Scepticism and Atheism, Berkeley reckons this assumption among the most meaningless and confused that can ever arise in a human head.

Sensible things are such as are immediately (*immediately*) apprehended by the senses, whereas the cause of things, which cannot be sensed, must expressly be thought, and, in being thought, be in consciousness. If I see that the one part of the sky is red, the other blue, then I infer that there must be a cause of this color-difference; but the cause cannot be said to be a sensible thing (*or perceived by the Sense of Seeing*). Nor can I, when I hear various tones, be said to hear the cause of the tones; just as little as I, when by feeling I experience that a thing is hot or heavy, can be said to feel the cause of the heat or the weight.

Sensible things are aggregates of sensible qualities (*Combination of sensible Qualities*); these sensible qualities are not outside the soul, but in the soul. As little as one can ascribe to the needle a sensation of the pain that its prick causes us, so little can one ascribe warmth to the fire. To have a high degree of warmth, to be hot, is to be in pain; were the fire really hot, then it would have to burn itself as it burns us, it would then have to be in pain; but that is absurd.

Not only of the derived properties, but of the original ones too, it holds that they exist only in our consciousness. Were properties such as extension, figure, weight, density, motion something outside consciousness, then one and the same thing would have to be both large (namely when seen near) and small (insofar as seen at a great distance), both smooth and rough, both straight and crooked, and the same motion both slow and fast; but that is absurd.

Just as absurd is the opinion that the named properties should have a substratum; the substratum on which extension and the other properties should depend would, in order to bring about extension, have to have an extension already beforehand, and so on to infinity.

The infinite divisibility of matter is generally assumed. There is then in every finite matter an infinite number of parts that the senses do not yet apprehend; but the more the eye is sharpened, the more parts it sees, and the larger each part becomes: could the eye be sharpened to infinity, each smallest part would become infinitely

large. This materialism contradicts itself; for it follows from this that every body is in itself infinitely extended, that is to say: without any shape or figure. It is therefore not an essence of matter, but consciousness itself, that forms all changes in the bodies of which the visible world (*the visible World*) consists: matter is an idea.

As for mathematical natural philosophy, mathematics forgets to give a sufficient account of its principles. Arithmetic reckons with units (*Unities*), but is not able to show what “the unit” really is; in geometry the infinite divisibility of finite extension plays a chief role; but finite extension is, as already shown, only an idea in the soul. Since it is now evident that no one is able to separate infinitely many parts either in a finite line or in a finite surface or in a finite body, everyone can, by making clear to himself the idea of extension, convince himself that the alleged infinity is not present. What in everyday speech we call things, we must then, according to the given explanations, call ideas. It is doubtless a hard saying, if for daily use we should eat ideas, drink ideas, and clothe ourselves in ideas; but usage is one thing, truth another. That things outside us are transformed into ideas in our own soul is not to be understood as though the sensible representations rested on our own arbitrariness. We can arbitrarily open and close our eyes, but the shapes and colors that show themselves to our opened eyes affect us with necessity.

Our sensible sensations and representations must doubtless be referred to a cause independent of us. But since cause and effect must be of like kind, and since a thing cannot possibly impart to us what it does not itself possess, it is just as absurd to assume that a thing that does not itself have sensation should be able to arouse sensation, as that a brain-impression should be able to become a representation. There exist therefore no bodies, but only spirits; it is not a material cause, but an almighty spirit, that through all sense-images, representations, sensations, and impressions lets the finite spirits impart themselves to one another.

By thus explaining the sensible and abolishing the material, this Idealism has transformed itself into a Spiritualism that annihilates nature, appeals to the supernatural, and culminates in Theology. *It is therefore plain, that nothing can be more evident to any one, that is capable of the least Reflexion, than the Existence of GOD, or a Spirit who is intimately present to our Minds, producing in them all that variety of Ideas or Sensations, which continually affect us, on whom we have an absolute and intire Dependence, in short, in whom we live and move, and have our Being.* (*Principles of hum. knowl.* London

1734. S. 166).

§ 6.

Among the many mutually crossing approaches to a theory of knowledge, Scepticism and Dogmatism meet as conflicting results. In Scepticism (Hume) thought is at variance with itself, and experience stands as an unexplained fact; in Dogmatism (Chr. Wolff) the unity of being with thought is grasped so immediately that the rigid schema of the understanding does indeed dazzle by its proofs, but lacks the grounding. Against these mighty one-sidednesses I. Kant (†1804; *Kritik der reinen Vernunft*) comes forward.

Descartes's Idealism, which makes the existence of objects in space unprovable and doubtful, Kant calls the problematic; Berkeley's, which presents the external existence of things as false and impossible, he calls the dogmatic. Dogmatic Idealism is unavoidable when space is to be regarded as a property belonging to things in and for themselves; for then every thing for which such a space should furnish the condition is an absurdity. The ground of this Idealism is suspended in "the transcendental Aesthetic." Problematic Idealism, on the other hand, is first refuted when proof is furnished that even our inner experience, not doubted by Descartes, is possible only under the presupposition of outer experience.

Although the Kantian critique of reason, as a systematic whole in itself, has by no means come into being out of finite regard for this or that view, it can nonetheless, if one will, be grasped in such a way that the two parts of "the transcendental Doctrine of Elements," namely the Aesthetic and the Logic, may chiefly be said to be directed against Scepticism, its third part, namely the Dialectic, against Dogmatism, and "the transcendental Doctrine of Method" against both. We therefore treat, in this connection, the Kantian theory of knowledge insofar as it characterizes itself, by its opposition to the named extremes, as a "*transcendental*" Idealism. Thus:

1) Against Scepticism.

That all our knowledge begins with experience is beyond all doubt; but from this it does not yet follow that knowledge in each and all has its source from experience. There are a priori sensible intuitions, a priori categories of the understanding, and a priori ideas of reason.

a) “The transcendental Aesthetic” deals with the pure sensible intuitions: space and time. Space is the form of the outer sense, time of the inner. If we abstract from the objects outside us, we get space back as the pure, universal form; if we abstract from the changes observed by the inner sense, empty time remains.

That intuitions of space and time are a priori can be shown partly from a metaphysical, partly from a transcendental point of view. The metaphysical consideration proves that the intuitions space and time cannot possibly be borrowed from experience, since one must precisely presuppose them in order to be able to make an experience; it proves further that space and time are not concepts but intuitions: the individual spaces are contained as parts in universal space, likewise the individual times in universal time, whereas the universal concepts, on the contrary, always comprise the particular ones as members under themselves, never as parts within themselves. The transcendental consideration proves the a priori origin of space and time in an indirect way, in that it shows that certain sciences, such as, for example, pure mathematics, would be impossible if there were no universal and necessary propositions about space and time; but universal and necessary propositions are never had from experience, they are had only *a priori*.

b) “The transcendental Analytic” treats the pure concepts of the understanding. The understanding is, negatively determined, a non-sensible faculty of cognition; outside intuition there is only a cognition by concepts. To intuition as receptivity corresponds the spontaneity of thought: intuitions rest on “affections,” concepts on “functions.” Intuition bears immediately on the object, the concept, on the contrary, mediately by means of another representation. The understanding judges; the judgment is a mediate cognition, a representation of a representation.

When we then abstract from the content of a judgment, and look only at the bare form of the understanding, we find that the function of thought lets itself be brought under four titles, each of which again contains three moments. In judgments there lie, namely — insofar as they are divided, according to Quantity, into universal, particular, and singular — the categories: totality, plurality, unity; insofar as they are divided, according to Quality, into affirmative, negative, (limiting), the categories: reality, negation, limitation; insofar as they are divided, according to Relation, into categorical, hypothetical, and disjunctive, the categories: substance and inherence, cause and effect, reciprocity; and finally, when they are distinguished, according to Modality,

into problematic, assertoric, and apodictic, the categories: possibility, actuality, and necessity.

That these twelve categories, from which the rest let themselves be derived, are not owed to experience, but are *a priori* concepts, is seen from this, that these too are just as necessary conditions for the possibility of an experience as the pure intuitions. In and for themselves the categories are empty forms that get their material conveyed only through intuition; only when the sense-content of intuition is absorbed into the form of the understanding does experience come about. Yet a difficulty meets us here. The objects, which are of sensible nature, are heterogeneous with the concepts: how then does it come about that *a posteriori* sense-objects can be subsumed under pure concepts of the understanding, and judgments *a priori* be formed therefrom? This happens in that a third, which on the one side is *a priori*, on the other side sensible, steps between, namely, the pure intuitions: space and time. In order in this respect to bring the concept into intimate union with intuition, the imagination forms “the transcendental schema.” Thus Quantity has for its universal schema the time-series or number, so that the pure concept of magnitude meets intuition, in that we in imagination produce units, and thereby go from unity through plurality to totality. Quality’s schema is the time-content: the Real fills a given time, and, when the content of reality is gone, negation is intuited in the emptiness of time. The schema of the categories of relation is the time-order, for substantiality is the persistence of the Real in time, the members of causality follow one another with necessity in time, and reciprocity is the regular coexistence of two or more expressions of substance in time. The schema of Modality is the time-comprehension: the schema of possibility signifies a representation’s agreement with the conditions of time in general, the schema of actuality an object’s existence in a given time, and that of necessity an object’s being at all times.

Since the objects we can observe all together fall in time, they must necessarily also fall under those schemata of time. This is proved in “the Analytic of Principles.” At the head of the analytic judgments stands the principle of contradiction; but the highest principle of the synthetic judgments lays down that every possible experience “must stand under the conditions that are necessary for bringing the manifold impressions of intuition to synthetic unity.” α) All phenomena are then, since they can be apprehended only under the forms of space and time, extensive magnitudes; “the Axioms of Intuition.”

β) All phenomena are, since every sensation can be strengthened and weakened in a definite degree, intensive magnitudes; “the Anticipations of Perception.” γ) All phenomena are, according to the schema of the categories of relation, subject to necessary determinations of time; the substantial remains in time; the accidental changes in time; the temporal change of accidents expresses a relation between cause and effect; “the Analogies of Experience.” δ) Finally all phenomena are, in agreement with the schema of the categories of modality, determined in such a way that what agrees with the formal conditions of experience is possible; what agrees with its material conditions, actual; what exhibits universally valid conditions of experience, necessary; — “the Postulates of Empirical Thought.”

c) “The transcendental Dialectic” investigates the ideas of reason.

Our cognition begins with the senses, passes thence over to the understanding, and ends in reason as in its highest. If the understanding is the faculty of bringing phenomena under a unity by means of rules, then reason is in turn the faculty of bringing the rules of the understanding to unity under principles. The understanding develops the conditioned, reason points out the unconditioned; the understanding furnishes the premises, reason draws out the conclusion. From the three logical inferences of reason, namely: the categorical, the hypothetical, and the disjunctive, the three speculative ideas of reason are derived: the idea of the soul as a thinking substance (the psychological idea), the idea of the world as the aggregate of all phenomena (the cosmological idea), the idea of God as the highest condition for the possibility of all (the theological idea).

That the ideas of reason are not abstracted from experiences is seen immediately from this, that in the whole world of experience no object can be pointed out that corresponds to any of those ideas. That these ideas nonetheless have validity, namely a priori validity, is understood from this, that they are the indispensable lodestars of the understanding, the regulative principles of cognition. We order our subjective faculties by means of the idea of the soul, the infinite series of conditions by means of the idea of the world, the ascending steps of causes by means of the idea of God.

2) Against Dogmatism.

The a priori is, according to Kant, solely in our consciousness, and has nothing to do with the essence of things. Already “the transcendental Aesthetic” warns us against

regarding space and time as being outside us. Were one to set up the proposition that things are in space and time, then it would be a delusion; not in themselves, but only for us are things in space and time, and that in such a way that the phenomena for the outer sense are in space as well as in time, the phenomena for the inner sense only in time. Just as emphatically “the transcendental Analytic” impresses that the categories of the understanding have universal validity only insofar as they hold for us, in that we apply them to things as objects of experience, but not to things in and for themselves. In that we see objects through subjective forms, and apprehend them in subjective concepts, we are like those who see the colors of things through painted spectacles.

The discord between thinking and being that Kant thus brought about by impressing from first to last the separation of the thing as object of experience and the thing in itself (*das Ding an sich* [the thing in itself]) — a separation that is further illuminated by the way in which, with respect to *Phænomena* and *Noumena*, he demonstrates “the Amphiboly of the Concepts of Reflection” — is well suited to give uncritical Dogmatism a fundamental blow; this blow becomes especially palpable in the critique that the ideas of reason must undergo in “the transcendental Dialectic.”

The Dialectic, “*eine Logik des Scheins* [a logic of illusion],” is to clear up the illusions in which reason, by applying the categories of the understanding to the unconditioned, must necessarily entangle itself. This “transcendental illusion,” this optical self-deception of reason, shows itself in the different ideas of reason from a different side.

a) The Paralogisms of pure reason. Rational psychology makes the soul into “a soul-thing with the attribute of incorporeality (immateriality); into a single substance with the attribute of incorruptibility (incorruptibility); into a numerically identical intellectual substance with the predicate of personality; into a spaceless, thinking being with the predicate of immortality.” These propositions of rational psychology are sheer fabrications. At the ground of the four paralogisms belonging here can lie nothing other than the simple, in and for itself empty representation: “I think”; but this “I think” is neither intuition nor concept, it is only an act of consciousness that accompanies, bears, and connects all my representations and concepts. That I who think can always be regarded as subject, a something that does not merely, as a predicate, hang upon thought, is an apodictic, indeed even identical, proposition; but

it does not signify that I as object am a being or substance subsisting for myself. The proofs of immortality rest on deceptive inferences. "I can indeed separate my pure thinking ideally from my body; but from this it by no means follows that my thinking can also persist really, after being separated from the body."

b) The Antinomies of pure reason. To determine the chief sides from which the idea of the world shows itself, one is guided by the categories given in the table. In respect of the Quantitative, there is then here something to lay down about the totality of the world's times and spaces, and as regards Quality, something about the divisibility of matter; in respect of Relation the causality of the effects given in the world must be pointed out, and, as regards Modality, the conditions of the contingent are pursued until at last they are absorbed into the necessity of the whole.

At each of these points, assertion stands against assertion (antithesis against thesis) with proof of equal force. Against the thesis: "The world has a beginning in time and is enclosed within the limits of space," holds the antithesis: "The world is infinite in respect both of time and of space"; against the thesis: "Every composite substance in the world consists of simple parts," the antithesis: "No composite thing in the world consists of simple parts"; against the thesis: "Besides causality according to natural laws there holds in the world a causality of freedom," the antithesis: "there is no freedom, but everything in the world happens solely according to natural laws"; against the thesis: "To the world belongs something that, either as a part of it or as its cause, is a simply necessary being," the antithesis: "There exists no simply necessary being as world-cause, neither in the world nor outside it."

These cosmological antinomies then show us the dialectical struggle, the self-contradiction, in which reason entangles itself when by the idea of the world it wishes to think the objective essence of the world.

c) The Ideal of pure reason. "The category" is empty, but can nonetheless find its object in experience; "the idea," which has no object of experience at all, is thus more remote from objective reality than the category. "But still more than the idea, that seems to be removed from objective reality which I call the Ideal, and by which I understand the idea, not merely *in concreto* [in the concrete], but *in individuo* [in the individual], i.e. as a single thing determinable and determined by the idea alone." After having shown how reason determines its Ideal in different ways, namely as primal being (*ens originarium* [original being]), insofar as it has it in itself, as the

highest being (*ens summum* [highest being]), insofar as it has it above itself, and as the being of beings (*ens entium* [being of beings]), insofar as everything is conditioned by it, Kant passes over to an attack on the “speculative reason” of Theology, and in his critique of the dogmatic proofs of the existence of God drives the separation of thinking and being to the extreme.

Among the three proofs, the ontological, the cosmological, and the physico-theological, the emphasis must doubtless fall especially on the first. “On the impossibility of an ontological proof of the existence of God” the following should here be brought out. The proof is in brief this: “The most real being is possible; among the many realities belongs also existence; it therefore lies in the concept of the most real being that this being itself must necessarily exist.” One clarifies this by examples and dazzles by examples. When I suspend the predicate in an identical judgment, and keep the subject, then quite rightly a contradiction arises, and therefore one says that the predicate belongs to the subject with necessity. But if I suspend the subject together with the predicate, then no contradiction arises; “*denn es ist nichts mehr, welchem widersprochen werden könnte* [for there is then nothing more that could be contradicted].” To posit a triangle and yet suspend its three angles is a contradiction; but to suspend a triangle together with its three angles is no contradiction. So it is with the absolutely necessary being. God is almighty; almightiness cannot be suspended when you posit God; but when you say: God is not, there is neither almightiness nor any other predicate. Being is, as copula in the judgment, no real predicate. The actual contains in the concept no more than the merely possible. A hundred actual dollars contain not the least more than a hundred possible ones; for in the opposite case the concept would not be adequate. A concept loses not a single one of its properties because its object does not exist, and gains not a single property from its object’s being found existing.

“On the so famous ontological (Cartesian) proof, then, all trouble and labor is wasted; a person would, by mere ideas, become no richer in insight than a merchant in fortune by adding a few zeros to his cash balance.” From thinking to existence there is no speculative transition.

3) Transcendental Idealism.

Since Kant expressly understands by “a transcendental cognition” one that “does

not properly have to do with objects, but only with our cognition of objects, insofar as this is to be possible *a priori*,” it is already seen from this why he himself calls his Criticism a transcendental philosophy. This philosophy will now doubtless in its way assert the validity of the world of experience; but so would Berkeley’s Spiritualism too, in its way. The manner in which the moment of subjectivity becomes the overreaching thing in all our cognition characterizes the Kantian transcendental philosophy as Idealism.

In what relation this transcendental Idealism has expressly set itself both to Scepticism and to Dogmatism is expressed especially in “the Doctrine of Method,” which treats the pure reason’s *Discipline, Canon, Architectonic, and History*.

“Were one,” it says, “to ask the cold-blooded David Hume, properly created for the equilibrium of judgment, what then moved him, by scruples that only a laborious brooding could devise, to undermine the imagination so comforting and useful to men, that their rational insight suffices when the concept of the highest being is to be determined, then he would probably answer: Nothing, but the intention of bringing reason further in self-knowledge, and therewith a certain displeasure at the compulsion one would impose on reason, in that one prides oneself on it, and yet hinders it from making a candid confession about the weaknesses that it can discover in itself only by serious examination.”

Unsatisfying as Scepticism is in and for itself, so justified is it over against an uncritical Dogmatism that does not know what the understanding is capable of, and has not tested the limits of the possibility of cognition according to principles. “*Und so führt der Skeptiker, der Zuchtmeister des dogmatischen Vernünftlers, auf eine gesunde Kritik des Verstandes, ja der Vernunft selbst* [And thus the sceptic, the disciplinarian of the dogmatic rationalizer, leads to a sound critique of the understanding, indeed of reason itself].”

In pure cognition as science, reason must on the contrary always proceed dogmatically, i.e. conduct strict proofs from secure principles *a priori*; but to conduct strict proofs from secure principles *a priori* is not the same as (*Wolffs methodus mathematica* [Wolff’s mathematical method]) to imitate mathematics. Mathematics’s thoroughness rests on definitions, axioms, and demonstrations; philosophy’s on explications, deductions, and acroamatic proofs. It does not befit philosophy “*mit einem dogmatischen Gange zu strotzen und sich mit den Titeln und Bändern der Mathematik auszuschnücken*,

in deren Orden sie doch nicht gehört [to strut with a dogmatic gait and adorn itself with the titles and ribbons of mathematics, in whose order it does not after all belong].” The apodictic propositions (be they now provable, or else immediately certain) divide into Dogmata and Mathemata. A direct synthetic proposition from concepts is a *Dogma*, whereas a similar proposition by construction of concepts is a *Mathema*. Since there is now in the speculative use of pure reason not a single Dogma, every dogmatic method too — be it now borrowed from the mathematician, or in another way become a peculiar mannerism — is in itself unfitting. “*Denn sie verbirgt nur die Fehler und Irrthümer, und täuscht die Philosophie, deren eigentliche Absicht ist, alle Schritte der Vernunft in ihrem klärsten Lichte sehen zu lassen* [For it only conceals the faults and errors, and deceives philosophy, whose proper aim is to let all the steps of reason be seen in their clearest light].”

§ 7.

“In every experience two factors are set against each other: the I and the thing, the intelligence and its object. If one abstracts from the I, then one gets a thing *in and for itself*, and must then grasp the representation as a product of the object; if one abstracts from the object, one gets an I *in and for itself*, and must then hold to the self-active consciousness that produces its own content. The former is *Dogmatism*, the latter *Idealism*. These two are mutually incompatible; a third is not given: here one must choose. To judge between Dogmatism and Idealism, let one note the following: the I occurs in consciousness, the thing in itself is a fiction of consciousness. In that Dogmatism is to explain the origin of the representation from the thing in itself, it performs the dubious feat of going behind consciousness, and invoking being; but being effects only being, not consciousness; therefore only that standpoint can be true which is chosen, not in being, but in consciousness, not in the object, but in the I itself.” In this spirit J. G. Fichte (†1814) has drawn a consequence of Kantianism and (*Erste Einleitung* 1797) introduced his “*Wissenschaftslehre* [Doctrine of Science].”

While Fichte dissolves “the thing in itself,” holds fast to the I, and carries through *subjective Idealism*, the Kantian doctrine is illuminated from an opposite side, in that J. F. Herbart (Metaphysik) fixes attention on “the thing,” grasps even the I as a thing for itself, a “real with many properties” (Psychologie), and in his way develops a *critical Realism*. “The material of experience is the foundation of philosophy; but

as a mere given it still stands outside philosophy. What then is philosophy's first step, the act with which it begins? Experience has forced upon us concepts that do not let themselves be thought. They are doubtless thought in a way by everyday understanding; but it is an obscure and confused thinking that does not see through the conflicting distinguishing marks. The sharpened thinking, metaphysics's logical analysis, discovers in the concepts of experience (space, time, motion, and so on) contradictions, conflicting, mutually excluding marks. What is to be done here? These concepts cannot be thrown away; they are given to us, and we have only the given to hold to. Neither can they be kept, since they are unthinkable, in themselves impossible. The only way out is: we must rework them. *The reworking of the concepts of experience is the task of Speculation.*"

Returning, after this indication of the Herbartian direction, to Fichte, we must, in order to understand subjective Idealism, well imprint on ourselves the highest principles of "the Doctrine of Science"; for in these principles "the Doctrine of Science" is theory of knowledge.

α) *Thesis*. If one grants that the proposition $A = A$ is self-evident, then one grants at the same time consciousness's power to posit something. One posits not that A is, but only that, if A is, then it is A . If we now, instead of looking at what is posited, look at the positing one, or rather: at the identity of the positing one and what is posited, then the proposition $A = A$ transforms itself into $I = I$. Whereas instead of $A = A$ we could not say that A is, we can now, instead of $I = I$, say: I am. This self-grounded is the ground of all action in the human spirit, and thus the original distinguishing mark of the activity of consciousness in and for itself.

β) *Antithesis*. The proposition $\text{Non-}A \text{ not} = A$ is, as a free primal act, according to its form likewise unconditioned, but on the contrary, according to its content, its matter, conditioned, because the antithesis presupposes the thesis, because when a $\text{Non-}A$ is to be posited, an A must already beforehand have been posited. By the first principle, $A = A$, the form of the primal act was a positing; by the second it is a counter-positing. What $\text{Non-}A$ is, is not yet known; only this is known, that $\text{Non-}A$ is the opposite of A . But now A is posited by the I ; here only an I is presupposed: what is opposed to the I is thus $\text{Not-}I$.

γ) *Synthesis*. The third principle is, since it is determined by the two foregoing, almost entirely capable of receiving a proof. The task for the act expressed by the

third principle is given with the two foregoing, but not the task's solution. On the one side the I is completely suspended by the Not-I: the I cannot be posited insofar as the Not-I is posited. On the other side the Not-I, posited by the I, is itself an I, posited in consciousness. The I is thus not suspended by the Not-I; it is suspended, and yet nevertheless not suspended. This is the contradiction. For the solution of this contradiction we must seek to find an *X* in which the mutually excluding principles can be united. But how can being and not-being, reality and negation, be united with one another without annihilating one another? Answer: by mutual limitation! The *X* asked for is thus a barrier, and self-limitation the primal act of the I. Insofar as the essence of I-hood, the pure I, is grasped as a principle that develops itself in the empirical I's reciprocal action with a world of barriers, subjectivity has to such a degree gained the preponderance that herein is contained a natural summons to lay an emphasis of ideality on the objective too.

In his work "*Von der Weltseele*" F. W. J. Schelling (†1854) has, still with Fichte's presuppositions, sought to show how the concept of matter necessarily proceeds from the nature and intuition of the human spirit. "The mind (*das Gemüth* [the mind]) is the unity of an unlimited and a limiting force. Absence of barriers would, as well as absolute limitedness, make consciousness impossible. Only in that the force that strives out into the unlimited is limited by the opposite, while the limited one is conversely freed from its barriers, is feeling, observation, cognition thinkable. A corresponding thing is seen in nature. Not matter as matter, but the forces whose unity matter constitutes, is here the first. Matter is the ever-becoming product of attraction and repulsion. Matter is what it is by force; but force is the immaterial in matter, force in nature is that which can be compared with spirit. Nature and spirit must originally be united in a higher identity. The spirit's organ for the apprehension of nature is intuition; it is intuition that takes into possession the space bounded and filled by attractive and repulsive forces."

With these thoughts Schelling has already trodden the objective ground of natural philosophy. "All knowledge," it says (*System des transcendentalen Idealismus*), "rests on the agreement of a subject with an object. The aggregate of the merely objective is nature; the aggregate of the merely subjective is the I or the intelligence. For the union of both sides two ways are possible; either one makes nature the first, and asks how the intelligent comes to it, i.e. one strives to interpret it in pure forms of

thought — natural philosophy; or one makes the subject the first, and asks how the objects proceed from the subject — transcendental philosophy. All philosophy must aim either at making an intelligence out of nature, or a nature out of the intelligence. Just as transcendental philosophy has to subordinate the real to the ideal, so natural philosophy must seek to explain the ideal from the real. But both are only the two poles of one and the same knowledge, which mutually seek each other; therefore one must also, when one starts from the one pole, necessarily come to the other.”

“All the parts of philosophy must be presented in one continuity, and the whole of philosophy must be presented as what it is, namely as the continuous history of consciousness, for which what is deposited in experience serves only, so to speak, as monument and as document.”

Schelling’s transcendental Idealism can now be regarded as forerunner of Hegel’s “*Phänomenologie des Geistes* [Phenomenology of Spirit]”; what Schelling promised, Hegel (†1831) fulfilled.

“Consciousness has in general, according as the object shows itself, three stages. The object is namely either an object standing over against the I, or it is the I itself, or something objective that nevertheless belongs just as much to the I, to thought.” The three stages are thus: Consciousness, Self-consciousness, Reason.

a) *Consciousness*. At the first stage consciousness is sensing, observing, understanding.

α) *The sensing consciousness is immediate certainty of an external object*. The expression for such an object’s universality is, *that it is, and is this, now according to time, here according to space*. The object is thereby wholly different from all other objects, and for itself sufficiently determined. But this Now is no more, another has stepped into its place, which however is also a Now: the Now is, in its vanishing, something abiding, namely the universal. This Here, which I mean and point out, has a right, a left, an above, a below, a behind, a before, to infinity. This Here is thus not the single, immediately determined, but: now here, now here, now here! an aggregate of Many, and in these Many nevertheless the One, thus the universal.

β) The universal, insofar as it still hovers between sensuousness and reflection, is object for the *observing* consciousness; *here is the thing with its properties*. The sensible properties are at once immediately each for itself, and yet in relation to one another, belong to the thing as this single thing, and are yet so universal that they, separated

from one another, can be found in manifold other things. Thing and properties change, and yet have in this change their subsistence: the changeable is an outer, the subsisting an inner.

γ) Insofar as the object is an inner with its outer, a force in its expression, it is there for the *understanding* consciousness. The understanding sees in the changing outer of things a series of phenomena whose inner consists in a system of abiding relations: the laws. Things are under the laws, and the laws are the laws of things; things and laws are two sides of the same: the True, the essence of things, is thought, the concept.

b) *Self-consciousness*. In that consciousness turns toward thought as thought, toward essence as essence, toward the inner as inner, it gets its own reflection and thereby itself as object. Self-consciousness has in its formation or movement three stages: desire, the relation of ruling and serving, universal self-consciousness.

α) *In desire self-consciousness relates to itself as single, solely existing self-consciousness*; it is a selfless object that is to satisfy the craving; but, in that the object is consumed, and satisfaction sets in, consciousness has in its self-feeling understood that it, as subjective, must at the same time essentially be objective.

β) The concept of self-consciousness as a subject that is at the same time an object finds its complete expression in this, that self-consciousness sees in its object another self-consciousness. In that the two self-consciousnesses set over against each other are each according to their concept independent, the tension arises between being master and being servant. The servant is selfless, insofar as his self is a moment in the master's consciousness; but the master too is selfless, insofar as his being master rests on his having a servant; he is master only by means of the servant. With this reciprocity self-consciousness makes the transition to articulated universal will, to positive freedom.

γ) *In universal self-consciousness the individual knows himself* as serving the will of the many, in that he essentially serves himself. Here there is not a sum of wills, but the many wills are moments in the universal will. This self-consciousness is the foundation for all virtues: those of love, honor, friendship, bravery, sacrifice, distinction, and so on.

c) *Reason*. Reason is the highest union of consciousness and self-consciousness in knowledge of the object and knowledge of itself. Reason is the certainty that its

determinations are just as much objective, determinations of the essence of things, as that they are the subject's own thoughts. Fichte's "I" and Herbart's "real" have hereby become moments; reason's knowledge is not merely subjective certainty, but truth; but truth is unity of certainty and objectivity, of knowledge and being.

"The goal, absolute knowledge, or the spirit knowing itself as spirit, has for its path the recollection of the spirits as they are in themselves, and accomplish the organization of their realm. Their preservation, in respect of their free existence appearing in the form of contingency, is history, but, in respect of their conceived organization, the phenomenal knowledge's science; both together, the conceived history, form the recollection and the Golgotha of absolute spirit, the actuality, truth, and certainty of its throne, without which it would be the lifeless, solitary; only out of the chalice of this realm of spirits foams up for it its infinity."

C.

Factual Knowledge.

§ 8.

If we now look back on the theories of knowledge dealt with here, the sensualistic as well as the idealistic, then it is quite striking to us that not a single one among them all satisfies "sound understanding" in its consequences, while they nevertheless in their cooperation — insofar as each on its own most forcefully asserts its side of the matter — as it were unite in giving "sound human understanding" its right.

From this it cannot now be concluded that "sound understanding" is wiser than philosophy, but only that it, with the sureness of instinct, holds fast to the members whose mutual relation philosophy wishes to explain. When philosophy, then, instead of clearing up the relation of the members, explains away or bungles an essential member, "sound understanding" senses mischief, and protests.

"Sound understanding" readily grants that one cannot become conscious of a surrounding world without taking up its impressions and corresponding images into one's

circle of representation; but it maintains at the same time that the world we become conscious of must, as truly as it is a real world, have a real existence independent of our consciousness. The idealist may now exert himself to prove the indissoluble unity of world and consciousness as much as he will: “sound understanding” pro- tests. That our sensations and representations proceed from immediate sense-impressions, the understanding will readily grant; but when the sceptic draws from this the consequence that cause and effect exist only in our imagination, that the laws are not laws, “sound understanding” stands still, and begins to protest.

A philosophy that is in open contradiction with “sound understanding” is false; a philosophy that immediately conforms to “sound understanding” is superficial. Philosophy will not arbitrarily contradict the natural understanding, but only show that the so-called sound sense, without having any inkling of it, incessantly contradicts itself.

“Sound understanding” demands that I should immediately become conscious that there is something outside my consciousness; but in becoming conscious of something, it is of course in my consciousness; insofar as it is outside, I cannot become conscious of it: thus I am to become conscious that what is in my consciousness is not there; what a contradiction! When the sun shines, and the trees are mirrored in the stream, “sound understanding” means that the light is brightness, whether or not there is anyone who sees it, and that the trees really are mirrored in the stream, whether or not this mirroring is reflected in a seeing eye. But the question is how one is to convince oneself of this. By sight? That is a contradiction, since here it is a matter of the phenomenon, not insofar as it is seen, but insofar as it is not seen. By thought? That is a contradiction; the phenomenon does not let itself, as phenomenon, be thought, for in that case the one born blind could also think it. By the testimony of others? An empty evasion; since here it must go for the one as for the other. Only for the seeing one is light brightness; only in that the brook and the tree are mirrored in the eye can the tree itself be mirrored in the brook.

It falls naturally that one thinks back to the time when he did not yet exist; and just as naturally that he thinks of the moment when he will again have ceased to be. Nonetheless there is here in both a strange contradiction. That consciousness can be conscious of its own existence is evident; but how can the existing consciousness become conscious of itself as non-existing? In reality? No, that is a contradiction, for

when consciousness really is, its non-being is really excluded; and, when consciousness is not, it cannot become conscious of anything. In fantasy? But when consciousness sees its non-being only as a fantasy, how can it then convince itself that the real either has corresponded or will correspond to this fantasy? “Sound understanding” may readily advance with its explanation, but the matter is that the explanation contains just as many contradictions as that which it is to explain.

What is truth? Agreement between what consciousness takes to be real, and this real itself. And untruth? Naturally, conflict between what is posited in consciousness and what is in reality. If now all reality were at once gone, and the empty consciousness with its fantasies and thoughts stood alone behind, it would be able to carry on the most arbitrary play with figments. That truth is gone when the real world-content is gone, sound understanding will surely readily grant; but the real as merely real is, when consciousness itself is gone, not the True either. A stone, a star, any real object whatever, is in and for itself neither something true nor something untrue. In order that the difference between true and untrue can arise, a judgment must be passed on the real object; but in order that a judgment can be passed, there must absolutely be a judging consciousness. But if consciousness is an indispensable requisite for truth, the real for itself cannot possibly be the True; how then can “sound understanding” suppose it can think a time when there were in *truth* real atoms, real elements, a real earth, but yet no, no consciousness at all? Real objects that could not possibly be objectified; what a contradiction!

The outer, real world is, in its pure unconscious reality, a nothing; if anyone will maintain the opposite, he must maintain it by force of consciousness, and thus contradict himself.

What is an Eternal? When a universally valid proposition is known, it does not look at all as though the True first came into being in the moment of cognition; the True shows itself on the contrary to consciousness as that which has always been. That every thing is what it is; that two magnitudes, each of which for itself is equal to one and the same third, are mutually equal; that a ground must necessarily have its consequence, an effect its cause — all these so-called “eternal truths” present themselves to us as those that have been prior to all finite knowledge; and yet they are in themselves nothing when one takes knowledge away: how can the consciousness that comes to be in time become conscious of the non-becoming, the eternal?

What does contradiction mean? An existing nothing is a contradiction; a round square is a contradiction; a flying horse is also a contradiction. Nonetheless these contradictions are of different kinds. A flying horse one can represent to oneself, but not think. The unthinkable lies not straightforwardly in the logical, that the predicate “flying” contradicts the concept “horse,” for this concept could of course be extended; but in this, that the whole concept thus extended stands in conflict with the conditions and laws of experience. It is otherwise with the round square; here the contradiction is seen immediately in the representation itself. To represent the square as round is impossible: to think a round square is, on the other hand, possible, insofar as it is possible to think the predicate’s *conflict* with the subject. The contradiction is thus not merely that the predicate’s conflict with the subject makes the judgment meaningless; but the contradiction is in a higher sense this, that the contradiction itself, this unthinkable, nevertheless lets itself be thought. This becomes especially clear when the task is to think the existing nothing. It is indeed nothing, but in that consciousness posits this negation, nothing has precisely existence in thought, thus a thought existence.

By pondering a little on these many contradictions “sound understanding” will best learn to understand that the eager strife the philosophers carry on about “the principle of contradiction” is more than a strife about words.

Herbart and Hegel are both agreed that the concepts of experience contain a contradiction; they are furthermore agreed that the contradiction is to be suspended; but about the relation of the contradiction to the essence of things, indeed about the very essence of the contradiction, they are in the highest degree at odds. In that Herbart holds fast to the principle of contradiction, he sees in the contradiction an error that one is to avoid, a delusion that one is to remove. If the concepts of experience contain contradictions, then the fault lies not in the objective nature of things, but in the knowing subjectivity. The thinking consciousness must therefore, by reworking the concepts of experience and doing away with the contradiction, see to remove the false apprehension. Zealous against the contradiction, Herbart even accuses the Hegelian philosophy of Empiricism, since it “takes up the concepts from experience unchanged, and, despite the insight into their contradicting nature, precisely because they are empirically given, regards them as justified, indeed even for their sake remodels logic.” Hegel, on the contrary, with his “*Alles ist ein Widerspruch*

[Everything is a contradiction],” lays the contradiction into the essence of things, so that a being in finitude is necessarily a being in contradiction (solution of the Kantian antinomies). On contradiction rests not only the pain of life, but also the drive that drives existence forward: contradiction calls forth motion, and motion is in turn the solution of the contradiction. As little as existence can exclude contradiction, so little can thought exclude it: the pulse of thought is dialectic, the contradiction the nerve of dialectic.

§ 9.

How intimately contradiction is united with thought is seen in the logical judgments. The word “leaf” can, as a word in language, point to a sensibly given in its immediate singularity, but also, as a conceptual expression, designate something universal. An exclamation such as “this leaf!” is in linguistic form imperfect, and gives a definite meaning only when someone at the same time points to a real leaf. If the eye is now fixed on “this leaf,” thought can be united with the sensible, in that a judgment is uttered: “this leaf” is green. But “this leaf” is not green, for it is yellow; “this leaf” is thus green and yet not green; “this leaf” is both green and yellow; here is a swarm of contradictions. In the first place the predicate “green” is, in relation to “this leaf,” something universal, whereas “this leaf” itself is something single. The judgment thus states that the single is a universal; but the single is nevertheless not a universal, but a single: here is a contradiction. What then is “this”? “This” is a leaf, “this” is a table, “this” is a horse; “this” is thus not “this,” but something other than “this”: sheer contradictions.

Through the history of philosophy runs a series of controversies that all point to the contradiction present in the matter. In Stilpo’s utterance: “the cabbage that is offered for sale here, is not; for cabbage existed already 1000 years ago,” lies the meaning that the individual in its sensible immediacy cannot be thought. The Scholastics’ controversy about the relation between *res* [things] and *universalia* [universals] points to the same contradiction, which also, as regards the more recent philosophy, repeats itself in the opposition between Sensualism and Idealism.

Locke and Hume are both attentive to the delusion that can arise in this, that the universal is held fast for itself in the word; and, since the objective Idealism had advanced so far that Hegel’s Dialectic was assumed to have transformed the sensible

into a moment vanishing in the concept, Feuerbach (*Philosophie der Zukunft*) came forward with a critique that demanded unconditional Sensualism.

For the understanding of Idealism's one-sidedness we bring out in this connection the relation between the sensible and the thinkable, between the individual and the universal, between the real and the essential.

a) The sensible and the thinkable.

That the sensible is an other of thought, a barrier of reflection, Idealism must of course grant; but, in that reflection shows that thought precisely needs this barrier, the otherness, the barrier itself, is nevertheless thought after all, and the sensible, the Negative, which thought thinks in thinking itself, is thereby melted down into a moment in the concept. This ambiguity can for a part be veiled in the pure, objective Logic, where "matter," this illogical, nonetheless by the help of Dialectic at last becomes just as logical as "form"; but in "natural philosophy," where reflection strikes against the fact, the error is laid bare.

"By the old, general thought that every body consists of four elements, or the more recent Paracelsian one that it consists of mercury or fluidity, sulphur or oil, and salt (Jacob Böhme called it the great Trinity), and by many other thoughts of this kind, the refutation has in the first place been made easy, in that under those names one would understand the individual, empirical materials that are most nearly designated by these names. But it is not to be mistaken that they should much more essentially contain and express the conceptual determinations. One must therefore much rather admire *die Gewaltsamkeit, mit welcher der Gedanke, der noch nicht frei war, in solchen sinnlichen besondern Existenzen nur seine eigene Bestimmung und die allgemeine Bedeutung erkannte und festhielt; er darf darum auch nicht auf experimentirende Weise widerlegt werden*: [the violence with which thought, which was not yet free, recognized and held fast, in such sensible particular existences, only its own determination and the universal meaning; it may therefore also not be refuted in an experimental way:]" (Hegel, *Naturphil.* S. 271). This apprehension is quite in the spirit of Idealism, and that Hegel too will not have it restricted to an assessment of that thought "*der noch nicht frei war* [that was not yet free]," but even expressly applied in favor of his own interpretation of the phenomena, is evident from the way in which he himself, e.g., introduces the "lunar" and the "cometary" principle; for precisely in this "lunar" and this "cometary" is seen "*die Gewaltsamkeit, mit welcher der Gedanke in solchen*

sinnlichen besondern Existenzen [the violence with which thought, in such sensible particular existences],” recognizing and holding fast only “*seine eigene Bestimmung und die allgemeine Bedeutung* [its own determination and the universal meaning].”

The fundamental error (πρῶτον ψεῦδος) lies in the misprision of the sensible. The sensible is by no means suspended because it is grasped as thought’s other; the element of sensuousness is more than a logical negation, it is for thought the real barrier, the in-itself-unthinkable, which is therefore to be grasped and determined in relation to the immediate sense.

“According to the moment of its ‘being-in-itself,’ matter would be the single point; according to the moment of its ‘being-outside-itself,’ it would most nearly be a multitude of mutually excluding atoms. But, in that these, by excluding one another, relate to one another, the atom has no reality, and the atomistic as well as absolute continuity or infinite divisibility is only according to possibility present in it.” In this Hegelian apprehension it does indeed at first glance look as though thought penetrated matter. By the logical opposition of “*Insichseyn* [being-in-itself]” and “*Außersichseyn* [being-outside-itself]” and its dialectical treatment one does indeed escape the difficulties that Atomism, this stumbling-stone of Speculation, otherwise brings with it when it is to be clarified in a way satisfying to those versed in the matter; but the difficulty is only got around, not suspended: in that the element of sensuousness asserts its right, the relation between matter’s discretion and continuity takes up an unthinkable, a for reflection impenetrable; for the sensible is thought’s barrier.

To convince ourselves from yet another side how unthinkable the sensible is, one need only cast a glance at the Herbartian view. With objective Idealism, Herbart’s Intellectual-Realism namely has this in common, that it, though in its way, will make the sensible thinkable. Immediately apprehended, the sensible is only a semblance; but this semblance presupposes an in-itself-being. This being, an independent reality resting on itself, can be known only by thought. The otherwise material atom is here transformed into an atom of thought. Of such peculiar, positive, unchangeable thought-atoms, qualia, there is a great multiplicity. “Just as well as a mathematical point can” — in spite of *principium indiscernibilium*! — “coincide with another at the same place, just as well can indeed these absolutely positive thought-atoms penetrate one another, and coincide in one and the same point;” but such thought-atoms are

then round squares: the sensible does not let itself be dissolved.

b) The individual and the universal.

In “*Phänomenologie des Geistes*,” “this Here” and “this Now,” which involuntarily designated the punctual, were transformed by reflection into “the universal,” and the punctual suspended. But, in that by this “Now!” this “Here!” this “There!” one points to a sensible, the universal is again suspended, and the punctual comes into force. “This here!” is of course not the same as “this there!”; the universal is in the word, but the word points, by this pointing, out beyond itself to the individual. It is with respect to this of significance that Locke and Hume have investigated the origin of concepts.

In the universal concept the representation vanishes; with the determination of the individual it returns again: to mistake the representation is to disdain the fact, and to mistake the individual.

In the working-up of the element of representation two factors are active: the productive imagination and the reflecting thought. The involuntary representing forms the image in agreement with the object; hereby is conditioned the factual agreement between the circle of representation and the given surrounding world. The free imagination, on the contrary, brings about disagreement between reality and consciousness, in that it in various ways transforms the representations. If we now ask whence the thought, restlessly working on the dialectic of its concepts, gets its most pliable, most spiritual, and thus most convenient material: from the factual individual-determinations, or from the representations transformed by imagination and the freely hovering universal-images? the answer cannot be doubtful.

To what degree the objective Idealism has mistaken the element of representation by exchanging facticity’s images for hovering intuitions can be seen from many examples. “The first qualified matter is matter as pure identity with itself, as unity of reflection into itself; to that extent it is only the first, itself still abstract manifestation. Existing in nature, it is the relation to itself as independent against the other determinations of the totality. This existing, universal *Self* of matter is *Light*, as individuality the *Star*, and this as moment in a totality, the *Sun*.” (Hegel *Naturphil.* S. 130).

So long as this intuition of the nature of light keeps itself in pure generality far from all finite facticity-determinations, one may readily grant it an ingenious application of logical categories; but if, on the contrary, a connection to definite phenomena is demanded, the whole apprehension will show itself so hollow and hovering that here at

no point is there even a possibility of a penetration into the factual. Where Speculation enters more closely into a characterization of given objects, and thus engages with the individual, the misunderstanding becomes still more palpable. “Clay-earth is, like silica, the immediate, simple, unopened concept, thus the first different earthy — the possibility of combustibility... , talc or bitter-earth is the subject of salt; hence comes the bitterness of the sea. It is an intermediate taste that has become fire-principle...” (Naturphilosophie S. 416).

Instead of developing the concepts by sharp analysis of the individual, and reducing the particular in such a way that the content-rich universal can find its determinations confirmed in the given, Idealism commits the blunder of letting the universal-forms move at such a distance from the facts that these, like phenomena in the distant horizon, take on hovering outlines, and, according as “the concept” finds it serviceable, become either rocks, or animals, or clouds.

c) The real and the essential.

That one can be right in something essential and yet in reality wrong is just as comprehensible as that something can be true in general, and yet in its application to the single become untrue. That thinking is identical with being is essentially true; but that our conscious thinking and the finite existence of things should be identical is really untrue; that what exists must express the category of existence is essentially true; but that it is only the category of existence that exists is really untrue. Those who speak of essence, and those who speak of the real, speak a different language.

“The universe is to be thought under the image of a line, at whose midpoint falls $A = A$, at whose ends falls on the one side $\overset{+}{A} = B$, i.e. an overreaching of the subjective, on the other side $A = \overset{+}{B}$, i.e. an overreaching of the objective, yet in such a way that relative identity takes place even in these extremes. The one side is the Real or Nature, the other the Ideal. The real side develops itself according to three potences, in that potency designates a definite qualitative difference of subjectivity and objectivity: the first potency, A , is matter and the force of gravity — the greatest preponderance of objectivity; the second potency, A^2 , is light, an inner — as gravity an outer — intuiting of nature. Light is the higher expression of the subjective. It is the absolute identity itself. The third potency is the common product of light and the force of gravity, the organism, A^3 . The organism is just as original as matter.”

If now a strict empiricist, a man who knows of no speculation, suddenly came

to hear this Schellingian explanation of nature, he would undoubtedly regard the whole as a figment of the brain. Hereby, however, it is not proved that what is stated in those speculative propositions is in itself meaningless; hereby it is only given that the empiricist does not understand the language. If someone who would explain the world's maintenance suddenly seized Lagrange's formula: $\int \frac{da}{dt} dt = \Sigma \frac{p}{in - i'n'} \sin(in - i'n' + q)$, with the assurance that the solar system is maintained solely on the ground that i and i' can never simultaneously be equal to 0, so that the denominator $in - i'n'$ can never become 0, because n and n' are incommensurable; then even those who otherwise think rationally about real things, but have no concept of mathematics, would judge this mathematical explanation just as severely as the empiricist that idealistic one.

That ignorance and lack of philosophical sense unite to tear down Idealism means nothing; in order to understand the philosophical language, one must just as surely study philosophy as one must, in order to understand the mathematician, study mathematics. But even when we seek to understand philosophy philosophically, it cannot be denied that the objective Idealism has, with its profound cognition of essence, sinned against reality in a twofold way, partly by abstracting from real points of departure, partly by wishing to force a real content into unreal concepts. "The Newtonian theory, according to which light is to spread in lines, or the wave-theory, according to which it is to spread wave-like, as the Eulerian ether or as the trembling of sound, are," says Hegel (Naturphil. S. 141), "material representations, *die für die Erkenntniß des Lichts nichts nutzen* [that are of no use for the knowledge of light]." Hereby, then, there is abstraction from all real points of departure, and every controlling investigation made impossible.

Instead of entering into the course of the real investigations, and by the concept's dialectical superiority leading the empiricist, by force of his own acknowledged pre-suppositions, to a cognition of the essence of the matter, this Idealism has not only, with an arbitrary apprehension of such phenomena as refraction and colors, but from first to last, wished to correct the expressions and procedure of Empiricism, without, however, rightly grasping, let alone seeing through, the problems of Empiricism.

§ 10.

When it is a matter of a cognition of the real, the theory of experience must

have the decisive voice. Humean Scepticism makes, in its zeal for experience, true experience impossible. By proving the — though only subjective — universal validity of the *a priori* forms, Kant now doubtless brings philosophy into closer understanding with Empiricism. But, although the theory of experience to a certain degree readily grants Kant the impossibility of knowing the interior of things, it is far from grasping time and space as merely being in our consciousness. When the astronomer speaks of the moon's distance from the earth, he surely does intuit this distance from the earth, not as a merely subjective mode of apprehension, but as a real, objective spatial relation; when it is said that Neptune perturbs Uranus, the reciprocity between the planets is surely not sought in our understanding alone, but in the nature of things. The separation Kant makes between the phenomena of experience and the things themselves is a schoolroom theory that has not the least significance in the theory of experience. By asserting the objective validity of the *a priori* forms, Hegel, on the contrary, has — despite all his offenses against reality — in this respect pointed out the essential agreement of philosophy and Empiricism.

If this agreement is now also to be carried through in the form of the concept from the standpoint of reality, the rational and yet empirical connecting links between the *a priori* forms and the immediate facts must let themselves be pointed out. It is not enough that a universally valid relation between ground and consequence, conditions and the conditioned, cause and effect, and so forth, is acknowledged *a priori*; the task is to assure oneself of the real relations.

The relation between cause and effect is a relation of necessity that has its abiding, universal expression in law. The cause for itself is not a law, though it is an *a priori* law that everything must have a cause; the effect for itself is naturally not the law either; but the law is that the effect must set in when the cause takes place. On these *a priori* presuppositions rest observation and experiment.

When a real cause, a real ground-relation, a real law, is to be found out, one must begin with an assumption. That the earth's attraction is the cause of the stone's fall, that a liquid's evaporation is the cause of cold, that every body has its definite melting point, that ebb and flow are owed to the influence of sun and moon, all such assumptions point originally back to a subjective estimate, to impressions of experience, which, however often they may be repeated, cannot possibly prove the necessary, or establish the universal. In order then to see how far the theory of

experience is, by its “observations” and “experiments,” able to secure the universally valid, we consider especially: abstraction, induction, and the real law.

a) Abstraction.

The abstracted reminds of that from which it is abstracted; thus pure space reminds of matter, and demands, so to speak, its lost content back. To space’s discretion correspond matter’s atoms, to space’s continuity matter’s mass. The universal-determinations that lie in geometry’s lines, surfaces, and bodies have validity too insofar as they are stated of the material. That the straight line is the shortest path between two points, that two straight lines can cut each other only in one point, and so on, are propositions that can be abstracted from the material, and applied to the material. Insofar as matter is grasped as the abstract, the moment of experience here coincides with the *a priori*.

But even where the moment of experience steps forth more strongly — as when the space-filling matter is ascribed a force of resistance, a force of expansion, gravity, and so on — we are by no means, as Hume supposes, merely referred to the association of ideas and habit. “All bodies have gravity, all bodies have force of cohesion, all bodies can be divided”: a simple denial of these propositions’ universal validity has nothing to signify; they are supported by millions of impressions of experience. These propositions are indeed in their generality abstract, but this abstract is a reflex of the concrete; he who therefore will deny the universal validity must expressly adduce facts, and take upon himself the *onus probandi* [burden of proof].

What the incessant reciprocal determination of the element of experience and the *a priori* signifies is shown by physics. “It is important that the application of the most simple mathematical truths in natural science be carefully justified, so that no doubt is left about nature’s agreement with thought” (H. C. Ørsted); and it is likewise important that the propositions of natural science be grasped mathematically; for that which especially furthers the strict investigations, in which the consequence becomes recognizable on the presupposition, and this in turn on that, is precisely the exact form in which mathematics (Cf. *Philosophie og Matematik* pp. 12–14) grasps the functions in question.

The difference between the abstract, philosophical and the mathematical-physical atomic theory then also consists essentially in this, that the latter at once grasps reality in its punctuality, whereas the former never gets beyond the one-sidedly universal.

Mathematics too doubtless begins with abstractions, but it shows at the same time how the abstractions can be integrated. The mathematician speaks not merely of atoms and the atoms' union in molecules, but formulates besides the expression in such a way that it precisely suits an investigation of the real. Let the number of molecules be, e.g., m , i.e. the mass, v the volume of the mass, which encloses a number of molecules; one then has in the formula $\frac{m}{v} = \rho$ an exact expression for density, and herein a presupposition for the determination of the specific gravities of the various bodies.

If now v encloses a number of n molecules, one gets $\frac{v}{n} = e^3$, and e^3 thus as the calculating expression for the molecule. If further, in a body, m , the finite part of the mass, becomes infinitely small, thus: dm , then one obtains $\frac{dm}{dv} = \rho$, whereby then also the finite sum of the parts, Σ , passes over to $\int$, the infinite sum; one thus has for the *homogeneous* body, in that ρ is a function of the various places in the mass, the formula: $\rho \int dv$, and the formula: $\int \rho dv$ for the *heterogeneous*, and herein a fundamental concept, from which countless specifications agreeing with experience let themselves be derived.

The mathematical punctuality makes possible an analysis of reality that in sharpness can measure itself with Dialectic's analysis of essence.

b) Induction.

In abstraction the individual case is disregarded; in induction the individual gets its significance again. How the individual members lead over to the universal, insofar as this has existence in them; how the parts by their peculiarity determine the whole, so that the essence of the whole finds its complete expression in the reality of the parts – this is shown by induction. “It is the remarkable thing about this kind of inference that the conclusion is never a singular or particular, but always a *universal* judgment. The categorical inference infers from a rule to a case; the disjunctive, on the contrary, from the cases to the rule. There the rule is what is presupposed, here what is attained by inference. Categorically one can only partially, hypothetically only negatively (*modo tollente* [by the mode of denying]), infer from the particular to the universal. The disjunctive mode of inference is the only one that permits an affirmative and at the same time universal inference from the particular to the universal. Therein lies the magic power of induction: it lets us, from the juxtaposition of observations and facts, know the law.” (*Die Theorie der Induction* von E. F. Apelt. Leipzig 1854. S. 18).

With respect to the development and application of induction, Socrates, Aristotle,

and Kepler can be said to designate the turning-points.

“Two things,” says Aristotle, “can no one with fairness dispute to Socrates, induction and the determination of the universal concepts.” How Socrates proceeded, in that by consideration of a multitude of individual cases and examples he sought to clarify the ethical concepts, is well known. But, although this Socratic ἐπαγωγή is doubtless a progression from the individual and particular to the universal, it is nonetheless — as Apelt too remarks — not what we now understand by induction, but rather an analyzing art of abstraction.

As logician and natural philosopher, Aristotle is called upon to give induction a more definite form, and then sets ἐπαγωγή partly against συλλογισμός, partly against ἐπιστήμη. In the first case ἐπαγωγή is the inference from the particular to the universal, from the parts to the whole, from the cases to the rule; in the second case ἐπαγωγή also designates the beginning of the universal (ἀρχή ἐστὶ καὶ τοῦ καθόλου); whereas ἐπιστήμη only draws out rational inferences from the universal. Ἐπαγωγή leads to the principle, ἐπιστήμη develops its content from the principle. To this Apelt now remarks: “Induction (ἐπαγωγή in opposition to συλλογισμός) never leads to the principle, one presupposes it already; it leads rather to a law that ties a definite circle of phenomena to a principle. The regressive, epagogic train of thought that leads to the principle (ἐπαγωγή in opposition to ἐπιστήμη) is no inferential connection, but an abstraction. This confusion of induction and abstraction has in the history of philosophy remained standing until recent times.”

As the one who in recent times has rightly grasped and scientifically applied induction, Kepler is named; for Bacon of Verulam only emphasizes its significance, but does not know its application. Kepler’s discovery of the Mars orbit happened by application of induction; the solution of this task Apelt treats with great acumen and expertise. “The art consists in finding out, from a series of numerical values for two connected variable magnitudes, the law of their mutual dependence on one another, i.e. the form of the function. Herein lies the great difficulty of the task.” The logical form of the Keplerian induction is this: “The Mars orbit contains in itself the Mars positions:

$$O_1, O_2, O_3, \dots O_n.$$

At the position O_1 , i.e. at the time when Mars’s true anomaly is v_1 , the magnitude of

the radius vector is r_1

$$\begin{array}{ccccccc} \text{„} & \text{„} & \text{„} & v_2 & \text{„} & \text{„} & \text{„} & r_2 \\ \text{„} & \text{„} & \text{„} & v_3 & \text{„} & \text{„} & \text{„} & r_3 \\ & & & \vdots & & & & \\ \text{„} & \text{„} & \text{„} & v_n & \text{„} & \text{„} & \text{„} & r_n \end{array}$$

The Mars orbit is a curve whose polar equation is: $r = \frac{a(1 - e^2)}{1 - e \cos v}$. But $r = \frac{a(1 - e^2)}{1 - e \cos v}$ is the ellipse's polar equation. The Mars orbit is thus an ellipse, in one focus of which the sun stands."

c) The real law.

How induction, beginning with the observation of the individual, leads from the individual to the total, in that it leads through the facts to the universal, and thereby to the cognition of the real law, can be seen from the adduced example.

That the Mars orbit really is an ellipse is not proved by observing all the individual points in the orbit; such an observation is impossible, since the orbit contains infinitely many points. Nonetheless induction gives sufficient certainty. This certainty is not superficially supported on a probability-calculation that only wishes to approach the elimination of chance: no, here is an essential inference that grounds a doctrine.

To grasp the probative force of induction, one must well note that it is not to decide whether the phenomenon in question, here the planet's motion, is subject to a law or not (for that is decided a priori); but rather to establish *which* law has here in reality found its expression.

Between the a priori presupposition that here is a law, and the empirical question: which law? the distance is so great that Apelt with reason recommends "heuristic maxims" to lead the observer onto the right track. The "maxims" themselves are in this connection without special interest; whereas the following instructive direction (Anf. Skr. S. 50) ought not to be passed over.

"In the categorical inference the three termini are each time given; in induction, on the contrary, the major concept must often first be sought. When one is to seek something, one must first know *what*, and *how*, and *where* one is to seek it. Induction seeks the law on which a definite circle of phenomena depends. One must then evidently know beforehand that the law's necessity, and not chance (lawlessness), rules in this circle. Thus, e.g., the course of the stars is bound to a law, the number of

the earthly elements, on the contrary, not. But the mere presupposition of lawfulness is not yet sufficient; one must also have a certain representation of the character of the law one will seek. When Kepler sought the figure of the Mars orbit, he knew already prior to his induction, i.e. a priori, that the planets do not go their way through the heavens according to whim and chance, but that they describe around the sun an orbit that lets itself be designated according to the laws of geometry. One must, if one will not miss the way to truth, know whether the postulated lawfulness is merely geometrical, or mechanical, or teleological. Thus some seek even today teleological grounds of explanation for the phenomena of life in physiology, others physical (mechanical and chemical) ones, because one is not yet agreed about the laws prevailing in this domain.” That Galileo, who emphasizes the principle of matter’s inertia, represents abstraction, as Kepler induction, and that both directions of mind are united in Newton (*Philosophiæ naturalis principia mathematica*), is an ingenious remark of Apelt; whereas the speculative interpretation Hegel (Naturphil. S. 94–124) gives of the fall-motion and the Keplerian laws is just as unstable as his judgment of Newton’s theory of gravitation is unmotivated.

Neither the sensualist nor the idealist is able to know the real law; the former not, because he denies the a priori; the latter not, because he overlooks the fact.

§ 11.

That a cognition of the real laws is an inviolable condition for knowing the existing things, scarcely anyone will draw into doubt. But the question how a cognition of the real relates to a cognition of the essence indicates in this connection a new point of view, from which the opposition between Kant and Hegel can be more closely assessed. According to Kant it would then be possible to carry it even very far in cognition of the real, without thereby in the least coming closer to the essence of things; according to Hegel it would conversely be possible to know the eternal essence, without troubling much about the difference of the real things. How this is to be understood will first become clear to us as we more closely investigate the relation between things and laws, things and their properties, essential knowledge and real objects.

a) Things and laws.

Even when one grants what is universally valid in the laws, it nevertheless makes an essential difference whether one grasps these laws as essenceless rules without

exceptions, or sees in their fulfillment an expression of the nature of things. In the theory of experience a double language is spoken. On the one side it is said: "We do not know what matter is in itself; force is to us an unknown, we know only its expressions; the essence of light is unknown to us, we know only the phenomenon; we do not know the cause of heat, we know only its effects." On the other side it is said again: "it is not by speculations and theories, but by observations and experiments, that we penetrate into the essence of things, and get to know the hidden forces. We know nature's laws; these laws constitute the universal in the manifold, the eternal in the changes; these natural laws are true thoughts of nature, so that what we know in nature is nothing foreign, but the same as that which lives in ourselves."

These two views stand in open conflict with each other. If it is true that we know only the phenomenon, not the essence, only the expression, not the force, only the effect, not the cause; then it is untrue that the laws of things express the essence of things. If it is true that the natural laws, as revelations of an eternally active reason, express the essence of things, then the impossibility of knowing the essence, the force, the cause, is only a fiction.

Such questions as: what is space in and for itself? what is matter in and for itself? what is force in and for itself? can betray much unclarity in the a priori. Matter in and for itself, pure matter, is just as surely an abstraction as pure space. One cannot ask *whence* or *how* matter has come into space, for the being of matter and of space is in principle the same. Here is an essential identity in a real distinction; as the sensible content, matter forms a real opposition to the outer form. It is so far from the case that the intuition of an empty space should be merely subjective, that it on the contrary shows space's objective validity as a real distance between two bodies removed from each other. As little as matter can be in itself without being in space, just as little can force be in itself without being in matter. Name whatever force one will, it will always be determined by a peculiar relation in which the one matter can find itself to the other. What is force of cohesion? An expression for the resistance a body makes against the separation of its parts. Here, then, there is in the body not first a multitude of parts in and for themselves, and besides a force in and for itself; no, the force of cohesion is an expression of essence of the material parts; the strong cohesion expresses that the parts are something in themselves only in this, that they are something for one another mutually. As it goes with the force of cohesion, so it goes also with the force

of gravity, with the magnetic, electrical, chemical forces, and so on.

If all force is thought away, matter is only an empty abstraction; if matter is thought away, force is an abstraction. If one cannot say that it is matter that originally gives force existence, since matter, in order itself to be, must presuppose force; so one cannot, on the other side, assume either that force gives matter existence, since every expression of force must presuppose an existing matter from which it can proceed. One need not brood over the question how force originally came into matter, since the being of force and of matter is in principle the same.

As for chemistry's so-called elements, it is just as contradictory to wish to determine their nature in and for itself as to wish to speculate out what the chemical attraction is solely in and for itself. Oxygen, hydrogen, sulphur, such elements are only in and for themselves what they express in their mutual relation and combinations. It is not a defect in our cognition that we cannot see what the chemical force is unconditionally in and for itself; for it is, as an unconditioned in and for itself, a nothing, an abstraction, a figment. That Kant in his critique of reason had to strike upon an inexplicable "thing in itself" is comprehensible; but to wish, after all that the Hegelian Dialectic has clarified, still to hold fast to such a "thing in itself," and thereby to suppose one avoids the dialectical contradictions, is only to deceive oneself: a thing in itself, a thing behind all conditions, is a lawless thing; for things are under the law only insofar as they are with inner necessity for one another mutually, and condition one another reciprocally; a thing in itself, an unconditioned thing, is not merely a contradiction, but, when it is held fast, a hardening of abstraction, an absurdity.

If it is now evident that things lay bare their essence only in their mutual reciprocity, and if the law is the formal expression of necessity that for this reciprocity indicates the definite consequences of definite presuppositions; then it is doubtless also incontrovertible that the law is more than a universal rule without essence, that a real cognition of law is at the same time a cognition of essence, that the different laws indicate different sides of the essence of things.

b) Things and their properties.

As the essence is known by its phenomenon, so the thing must be known by its properties. The thing is the existing unity of the properties, and the properties are in turn the inner determinations of things in existing plurality. But, although the property is what is proper to the thing, the property of the one thing can in reality

become recognizable only by its finite relation to the property of the other thing; for the illumination of the matter it is required that the reciprocity of the properties be seen from the real side.

The difference that in reality is between the immediate and that mediated through many intermediate links, between the partial, which is apprehended singly, and the total, whose inner connection demands all-sided, accurate, persevering research, brings it about that the distance between a preparatory cognition of the phenomena and a concluding cognition of essence steps clearly forth.

By merely holding to the optical phenomena, one still gets no representation of the chemical effects of light; from the mere sensation of heat one cannot explain to oneself that heat expands bodies. How does it now come about that light, the same cause, can have so many and highly different effects? Can the same thing that is luminous also be warming, or is there perhaps a peculiar heat-substance? Can light be transformed into heat, and heat into light, and wherein then consists the transformation? When such questions are set in motion, it is seen that he who really wills a true cognition of the essence of light and heat can scarcely, like Hegel, overlook “the Newtonian theory” and “the Eulerian ether.”

What significance the hypotheses disdained by Hegel must have even for the cognition of essence is seen already from this, that it is a matter of an investigation of matter’s proper discretion (*Fechner: Atomenlehre. Leipzig 1855. S. 18–32*) when it is a matter of an investigation of heat’s identity with light.

That the emanation-doctrine presupposes atoms of finite size goes without saying; but neither can the undulation-theory be carried through except under a similar presupposition. In refraction the broken white ray spreads itself into a narrow fan of colors; this phenomenon can be brought into mathematical agreement with the vibration-hypothesis only when the ether-parts are assumed to be separated by finite discretion. The narrow fan of colors rests, namely, on the fact that the different rays have different refrangibility, and that the distance of the ether-parts is not vanishing in comparison with the breadth of a light-wave. The same view — that, namely, the ether-parts are in finite separation — is further confirmed when the usual light-phenomenon is compared with the polarization-phenomenon. In a usual light-ray the vibrations have all possible directions, in a polarized light-ray they are parallel, in both they are transversal. But the continuation of transversal vibrations is possible only under the

presupposition of the ether's discretion.

As now the light-phenomenon can in all respects be interpreted according to the vibration-hypothesis, so can the heat-activity too. The propagation of heat by contact is indeed so different from radiation that at first glance it should seem impossible to explain such diverse phenomena from the same general presupposition; but Fourier has shown that the phenomena let themselves be reconciled when one assumes, namely, that the bodies are not continua, but consist of separated small parts. The conductivity of the bodies then becomes dependent on the vibration-speed of the conducted heat, thus on the temperature: heat and the heated characterize each other mutually.

That the heat-rays act most strongly when they fall perpendicularly on the surface of the body, and on the contrary more weakly when they fall in an oblique direction, is inexplicable so long as one continues to start from matter's continuity; if, on the contrary, one presupposes a finite discretion, the phenomenon is not merely explicable, but even calculable, in that the ray-effect is weakened according to the law of the sine.

In chemistry and crystallography not only does the atomic theory here pointed out find peculiar confirmation, but the doctrines of light and heat at the same time undergo a new testing. If we then look at the matter in its wholeness, the essence of light can be known only from the totality of the phenomena; but the totality has an infinite compass: the one totality reaches into the other. To investigate light as the pure light in and for itself is impossible; the properties of light are known only by the properties of the things that are affected by light; the same holds of heat, of magnetism, of electricity, and so forth. These "potences" are not merely what they are, each for itself in the reciprocity of their properties with the properties of the things in question; but they are at the same time for themselves what they are in their reciprocity with one another mutually.

c) Essential knowledge and real objects.

In relation to the universal essence Hegel is right against Kant; but reality is right against Hegel. Instead of knowing the real relation between thing and properties, Hegel has already in "*Phänomenologie des Geistes*" laid out the development of concepts in such a way that in reflection one proceeds from the thing's lower to objectivity's higher category. But by this logical progression it is so far from the case that the problem of reality becomes recognizable, that it is rather veiled. The universal is the objective, but the universal is identical with thought; the objective is, in its peculiarly

determined objectivity, an object, but objectivity is identical with its concept: the object must be objectified by the subject. Along this path one never gets beyond the universal identity of thinking and being.

If, on the contrary, we let the existing, the thing with its properties, set itself in sensibly determined opposition to the subjective reflection, the problem of reality comes into force. In this point Locke sees more clearly than Hegel.

Strictly speaking, the “substratum’s” identity with itself in its many properties and accidents cannot be asserted for any real object with a priori necessity. What is this? “Silver! one sees it at once by the peculiar luster.” — But the luster can of course dazzle; another body could possibly have a similar whiteness. By one or two properties an object cannot be known; we must then, in the case of silver, test a multitude of properties: malleability, ductility, specific gravity, crystal form, melting point, attraction to chlorine, to sulphur, to oxygen, and so on. Each property must be tested sufficiently, and a sufficient number of properties must be tested: here is a task of reality in its infinity. It can be granted to the Hegelian analysis of concepts that it, against Hume as well as against Kant and Herbart, clarifies the dialectical identity of the thing and its properties, of substance and its accidents, of the concept and the object in general, and thereby a priori secures for the theory of experience an objective foundation: that the silver-substance is in essential identity with the properties of silver cannot be doubted without the law of reality being violated, and Empiricism contradicting itself; but *which* properties are here united, and *how* they are united in this real substance, must be tested by observation and experiment; the real concept fetches every moment from Empiricism.

By force of the empirical research’s inner original connection with the a priori, there is then an essential knowledge; but according to the heterogeneity of thinking and sensation, real cognition is subject to finite conditions, and all our knowledge bounded. The bound of knowledge is twofold: a *relative* one, insofar as here there is always something we do not yet know, but which we can nonetheless in time get to know; an *absolute* one, insofar as a finite separation of the content of consciousness and of reality is grounded in our organization: here, then, is something we in this existence never get to know.

II. Theory of Science.

§ 12.

Where the theory of knowledge ends in thought-confusing or merely negative results (Sophistry, Scepticism), there can naturally be no talk of true science. As the theory of knowledge is, so must the theory of science become. If the theory of knowledge draws a critical bound between phenomenon and essence (*Kritik der reinen Vernunft*), the theory of science will on all its domains remind of the bound, and as in a *reservatio mentalis* [mental reservation] always tacitly understand the impossibility of knowing “the thing in itself.” If the theory of knowledge culminates in subjective Idealism (*Wissenschaftslehre*), the scientific doctrinal structures, insofar as they are the work of the principle, will transform themselves into mirrors of reflection for “the pure I.” If the theory of knowledge raises itself to the absolute knowledge of objective Idealism (*Phänomenologie des Geistes*), the sense-things will fall, but the concepts of essence stand as dialectical realities.

On true insight into the conditions of cognition rests the true insight into the essence of the sciences. Since now the conditions of cognition are not merely inwardly subjective, but at the same time outwardly objective, the essence of science, in its different breakings-through and peculiar appearances, is besides at the same time so closely grown together with the conditions of given ages, with the aim of life, with the events of history, that in order to understand the theory one must look at the age.

What philosophy was for the aristocratic antiquity, what theology was for the hierarchical Middle Ages, that is physics for the bourgeois present: cultural history and the history of the sciences illuminate each other.

How movements in the outer relate to movements in the inner is shown by the discoveries in the 15th, and the Reformation in the 16th century. To assess the intellectual advances one must, however, distinguish between the impulses and awakenings by which great leading thoughts are called forth, and the quiet, persevering activity of spirit that tests the points of view and gives the content scientific elaboration.

Though science may at times slumber, because minds are dulled in the handed-down, and the age lacks inspiring elements; so on the other side thought can also be overstrained and the forces go astray, when the ferment is too strong and the impressions too excessive (Cardano, Telesio, Campanella, Bruno).

As for the scientific interest itself, the enthusiasm can either exclusively hold fast to theoretical ideals (Copernicus, Le Verrier), or keep closer to practical results (J. F. Böttger, Daguerre); both directions proceed from the spirit of science, for the theoretical will prove its reality in the practical.

But although science lives a vigorous life only when it finds itself in sound and true understanding with the powers of life, it cannot be drawn down into the series of means, and subordinated to the self-interested regards of finitude. Only that science is cultivated which is cultivated for its own sake. To will the scientific merely for the sake of utility is precisely not to will the scientific. Self-interest deceives itself; it wills only the means; but the means fails to appear where spirit is lacking: the mere craftsman's eye does not see so deeply that it sees the new possibilities.

Where, on the contrary, scientific insight is in every direction recognized as a spiritual power, and, without regard to self-interest's petty Wherefore?, is embraced with the interest of infinity, of ideality, science will not remain standing as a pale theoretical shadow, but, like a luxuriant tree, bear good fruits beneficial to life. The discoveries of natural science in the present century (Galvani and Volta; Ørsted and Humboldt) furnish a series of grand examples of true reciprocity of theoretical and practical interests.

A.

The Science of the Sciences.

§ 13.

Since science has its root in human life, where the ideals of fantasy go before those of thought, it is natural that the different sciences come into the world at first in

the swaddling-clothes of imagination, and are burdened with unscientific admixtures. Thus astronomy was long grown together with astrology, chemistry with alchemy, philosophy with mythology.

It doubtless lies in the drive of science to lift the pupation, break its husk, and purge out the foreign constituents. But cultivation costs time, the purging exertion; the hindrances that the human spirit has here to overcome are of different kinds.

a) An *outer hindrance* proceeds from the natural strife between egoism and truth. The intellectual love of truth has at all times waged combat, not only against the prejudices of the ages and the thoughtlessness of the multitude, but above all against their egoism, whose power and influence saw itself threatened by the rising light. So long as a given condition is of such a character that the spiritual guardians of the population have an interest in holding to the untrue, they will not easily convince themselves of the true. Science is doubtless objective; but the subjective does come into the conviction: insight is conditioned by prospect. Were there still in the countries considerable positions for astrologers, would there then not also still be those who felt themselves called to study astrology — of course not for the sake of prestige, but because they had really convinced themselves of the deeper speculative in astrology? Those who will the truth must be zealous for the truth; science too has its martyrs (Vanini, Galileo, J. G. Fichte).

b) Of a nobler, though not less dangerous, nature are the *inner hindrances* that the richly gifted, overreaching subjectivity can prepare for itself. In enthusiastic striving after deeper insight it easily happens that the thinker confounds the subjective representation with the objective object, and then supposes he finds that in reality which nevertheless is found only in his own fantasy. In the greatest minds, the noblest natures — a Plato (*Timæus*), a Kepler (*Mysterium cosmographicum*), a Schelling, a Hegel — mixtures can be pointed out, whereby partly in themselves unscientific elements are worked into scientific form, partly in themselves scientific constituents are found placed in an unscientific, fantastic connection.

c) To this comes still the tension between science and view of life, which is increased by an ever growing *conflict between the real and the ideal*.

The more science in the finite sense holds fast to the real, the more easily will it undeniably avoid the fantastic, but the more difficult will it on the other side be to satisfy the ideal demands of personality. The stronger the scientific upswing is toward

the ideal, the more easily will science doubtless satisfy the demand of spirit, but the more easily will it then also expose itself to overrunning the conditions of reality, and be caught in illusions.

What then is science? By what does one everywhere and under all shapes know the scientific from the unscientific? How ought the personal view of life to relate to the objective, impersonal results of science? For the answering of these questions is required a comprehensive theory of science, a science of the sciences.

§ 14.

From the relation between the theory of knowledge and the theory of science it follows that the true science of the sciences can be none other than philosophy. With this it agrees exactly that the true comprehensive theory of science can come about only in that the fundamental interests of the diverse sciences are represented in the deliberations from which it proceeds. Under “the comprehensive discussion and debate of all against all” (Sibbern) each science then has its vote, and each its contribution. That philosophy works up, clarifies, and organizes these contributions into an intellectual whole proves that it in this respect does not decide the matter on its own, but works as the sciences’ common organ, as their own universal consciousness, the essence that is their kinship, the light in which they know and cognize one another mutually.

It is therefore not presumptuous arbitrariness when philosophy sets the different doctrinal structures before the tribunal of the theory of science, and demands that with respect to each one of them in particular an account be given both of the foundation of the structure and of the manner of building; it is not self-taken authority when, after carefully held examination, it permits itself to assign to those theories which in foundation and manner of building have more likeness to aesthetic castles in the air than to scientific doctrinal structures “a peculiar place” outside science.

Philosophy, with all this, does only its duty, in that it, bound by the laws of cognition, does not act according to whim, but as the lawful organ of the sciences, the submissive handmaid of scientificity.

If we set together the great representatives of the development of spirit — philosophy, theology, and physics — with one another mutually; then not only can the first, insofar as it is set in opposition to the two others, by no means wish to exempt itself

from scientific justification, but reasonably neither can the latter.

Were physics, dazzled by its present popularity, to cast contempt on the investigating theory of science, and, instead of heeding the light of thought that will shine through its ground and essence, only to rely on the immediately given, and invoke the incontrovertible authority of the senses, it would only thereby set itself lower than the medieval theology. Theology too can speak of the given, use the popular, and invoke an incontrovertible authority; and so long as there is in human nature a spark of a Higher, the authoritative pronouncement of faith will always be able to outweigh the authoritative pronouncement of sensuousness.

Where, then, thus the authoritative pronouncements of Supranaturalism and Naturalism — now alternately, now partially, now in open enmity, now in mystical agreement — step into the place of the theory of science and thereby give themselves scientific importance, it is doubtless natural that philosophy is either straightforwardly despised, or at least regarded as an intellectual plaything; but then scientificity too is in danger.

But does theology then perhaps want no theory of science? If one could at once hold to Anselm and yet profit from Descartes, could get *credo ut intelligam* [I believe in order that I may understand] to say *de omnibus dubitandum est* [everything is to be doubted], and conversely, *de omnibus dubitandum est* to proclaim itself as the right *credo, ut intelligam*, then theology would surely readily unite with philosophy in commending a thorough and sound theory of science; but on these conditions it is an impossibility for philosophy to unite with theology, since the matter itself is a contradiction.

Although philology, criticism, and history — which in and for themselves belong under science — constitute constituents of theology, they by no means constitute the theological in theology. Insofar as the named disciplines are treated scientifically, they are treated untheologically; insofar as, on the contrary, they are treated theologically, they are treated unscientifically. The theological in theology is known, namely, by this, that the cause from which all effects in the realm of nature as in that of spirit are partly immediately, partly mediately derived and explained, is not sought in the nature of things or the essence of reason, but in the absolute will, in the Almighty, who creates out of nothing. But even the purest, most acute, and most consistent application of such a principle (Occasionalism; Malebranche) makes science unscientific.

When in the theory of science it is a matter of a principle, the matter cannot be decided by a More or Less, as though one, by following “the middle way,” would suppose one could still get through scientifically by assuming a little of the supernatural, if only one did not make too much of it.

The question of a principle is a question of an unconditioned, all-conditioning presupposition; such a question must therefore always be answered with Yes or No. Either theology must once and for all acknowledge the supernatural causality, and has then at the same time repelled the objectively rational foundation, or else once and for all acknowledge this foundation, and has then at the same time renounced every support in the supernatural. In the first case theology will thus cease to be science, in the last case to be theology.

§ 15.

That life and science cannot relate outwardly and indifferently to each other follows immediately from both’s intimate connection with culture and humanity; that life and science are nonetheless independent rests on a principled separation of the subjective and the objective.

In science subjective moods — inclination, disinclination, fear, hope, and the like — have nothing to signify. Insofar as the question here is of the real, we will not know how this real, according to our opinion, ought preferably to be, but how it is in itself, and necessarily must be; insofar as the question here is of the possible, the question here is not how far this possible best satisfies our hopes, but how far it agrees with the conditions of existence.

Even when science works in the service of the interests of life, it works by force of the objective. It is not subjective feelings, not pious wishes, but insight into the essence of things, that leads off the lightning-flash, and sets the steam-carriage in motion.

Science is doubtless impossible without subjectivity; but the subjectivity is serving. Here one senses, here one feels, here one thinks; but the merely subjective in sensation and thought must everywhere be purged out as the contingent, the disturbing, the untrue; every presupposition is objective. Even there where the content touches subjectivity as immediately as in the aesthetic, science sees only the true, insofar as it sees in the matter the incontrovertible, that which is independent of arbitrary

predilection, the universally valid.

From the standpoint of life, on the contrary, subjectivity rules; here all the objective is serving. In infinite reciprocity with the surrounding world personality works, even in devotion and sacrifice, for subjective interests, subjective aims, subjective ideals. Everywhere where subjectivity here is in relation to another, it is through this other in a peculiar relation to itself. On selfhood, individuality, character, personality life lays infinite emphasis. That subjectivity now, among several possible tasks, also chooses science by no means suspends the opposition of life and science. One thing it is to live for a science, another to lay science at the foundation of life.

Insofar as subjectivity, in rising conflict with the objective world, develops the personal forces, we can here distinguish the following chief stages.

a) *The relative subjectivity.* At this stage the objective is set equally as means and as end. By setting it as end, subjectivity knows its bound and relates in serving fashion; by setting it as means, it satisfies the self, and asserts freedom. Out of interest of life for the self and the work, for the worker and the labor, subjectivity will then here make manifold agreements with the objective, venture into many a strife, suffer many defeats and win many victories, in that it runs through the scale of development from the useful to the exalted, from bourgeois industry to heroic exploit.

b) *The infinite subjectivity.* By breaking through the objective barrier and overstepping the bound of the finite, the consciousness of man strives to swing itself up to infinite freedom. Since reality cannot be denied, and a material surrounding world not annihilated, the self-liberation must be ideal. In vigorous one-sidedness a choice is made between two mutually opposed expedients. Partly the objective is under all shapes so reduced to a mere means that the enjoying individuality can in each moment make itself the lord of things (Hedonism; Aristippus); partly the objective, as the universally rational, is so breathed into consciousness that all finitude is transformed into something indifferent; purified and hardened in infinite resignation, the acting subjectivity will determine itself, in that it is determined by fate (Stoicism; Zeno — Epictetus).

An inner negative infinity, a dialectical unity of ethical earnestness and carried-through irony, is the deeper presupposition and unattained ideal of these one-sidednesses. (Socrates).

c) *The absolute subjectivity.* The formal self-liberation ends in the negative; sub-

jectivity seeks life, but finds death. Only that life which overcomes death is the true, the eternal life. That the individual's innate egoism is its own original guilt, a sin that deserves death; that the pure, guiltless subjectivity has voluntarily borne the sin, and saved the guilty from death; that the natural being is sheer impotence, only the will of spirit unconditioned power; that the present is only the means, the future the end: such presuppositions demand faith, but exclude knowledge.

In open conflict with all that scientific insight can in any way acknowledge as objective, faith is so far from obtaining any support from science that it precisely by collision with the objective knowledge must be thrust back into itself, and confess that its only ground is a need of subjectivity, an inner incontestable demand. At this stage all the objective is transformed into a point of transition for the subjective. Here it is not a matter of life's relation to the lifeless, but of the living's relation to the absolutely living; here the question is not of the will's relation to the will-less, but of the willing's relation to the absolutely willing.

Life is subjective, the eternal, true life absolutely subjective; the will is subjective, the eternal, unconditioned will absolutely subjective; the objective is thus, as objective, not the True: "Subjectivity is the truth" (S. Kierkegaard).

§ 16.

In that the strife between faith and knowledge is thus driven to its extreme, the intellectual reconciliation is at the same time introduced: the absolutely heterogeneous principles cannot, in the essential sense, do each other harm. The faith that hates knowledge, because it must strengthen itself by ignorance, is a false faith. Reality will be reckoned with; it does not help to close one's eyes: if the earth really goes round, then it goes round just as well with the believing as with the non-believing. In that faith declares that subjectivity is the truth, it doubtless loses all significance for science; but, in that science declares that objectivity is the truth, it at the same time renounces all influence on faith.

Faith and knowledge can then without contradiction be united in one consciousness; their mutual strife does not proceed from the principled. The strife is psychological; it is human frailty, which has neither resignation enough to reconcile itself with science's objective, nor inwardness enough to hold fast to faith.

That faith hates science is untrue; that it makes subjectivity independent of science

is, on the contrary, true. To set faith highest and yet keep science is possible; to set science highest, and yet preserve faith, is impossible. But although science itself must see that it can neither acknowledge the absolute subjectivity nor prescribe laws to it, it is nevertheless in its interest to have this subjectivity illuminated as all-sidedly as possible.

The separation of scientific and unscientific is made difficult by this, that what according to its nature must be rejected by science nonetheless becomes an object of scientific treatment, and so in a way does go along in science. Madness, e.g., is incompatible with science; nonetheless the symptoms of madness and the treatment of the mad become an important object for psychology and medical science. As a fetus of superstitious fantasy the myth is in conflict with the scientific; nonetheless there is a science of myth, a scientific mythology.

If the most diverse can be set under scientific illumination; if even the most subjective can become objective, insofar as it is objectified in reflection; then from this point of view nothing can be objected either against a science of subjectivity forming itself, a science of the subjective.

That such a science of subjectivity (psychology, aesthetics, ethics, philosophy of religion) must not, however, be confounded with what one otherwise is wont to call a science of faith lets itself be seen when we clarify to ourselves the difference between the scientific in the wider and in the narrower sense.

Scientific in the wider sense is everything that becomes an object of science's treatment: the scientific lies in the treatment; scientific in the narrower sense is, on the contrary, only that whose validity science sees, whose truth it can assess, test, and acknowledge.

By using the ambiguity that lies in the category "scientific," the doctrine of faith sometimes succeeds in giving itself the semblance of science; it is then the task of the theory of science to point out the most important sophisms belonging here.

First sophism: "If faith is not to rest on subjective notions, and be given over to the whims of individuals, it must be possible expressly to point out what one is to believe, and in whom one is to believe: faith must have its object. To investigate the objects of faith thoroughly is to investigate them scientifically. The doctrine of faith rests on such an investigation; thus the doctrine of faith is a science."

This sophism gives the matter the appearance that the objects of faith are secured

by science, because they are tested in science. The matter is the reverse. The objects of faith dissolve into vapor as soon as science is first permitted to apply its criticism. It is just as impossible for scientific criticism to acknowledge the Gospel of the Risen One as it is impossible for science to acknowledge the myth of Hercules. To acknowledge a miracle by force of scientific criticism is a contradiction, since the possibility of a miracle makes scientific criticism impossible: where the wonder speaks, the understanding must be silent; where the miracle begins, science must end.

Second sophism: "Every peculiar science must necessarily have its presuppositions. Just as logic presupposes and justifies the validity of thought, physics the validity of experience, and aesthetics the validity of the beautiful, so the doctrine of faith must for its part presuppose and justify the validity of faith and the content of faith. He who proves that the doctrine of faith is not a science because it has its peculiar presuppositions proves at the same time that there is no science at all; but here it holds: *qui nimium probat, nihil probat* [he who proves too much proves nothing]. Thus the doctrine of faith is a science."

This sophism dazzles by the ambiguity that lies in bringing along a presupposition. What makes the doctrine of faith unscientific is by no means the circumstance that it brings along presuppositions, but precisely the circumstance that the presuppositions it brings along conflict with science. It presupposes the miracle, it presupposes the absolute validity of subjectivity and of subjective conviction; with these presuppositions it has, so to speak, forced its way in upon science, and broken the scientific measure. If the consequences are to be scientific, the presuppositions must also be so; on unscientific ground no scientific structure can be raised: natural science is possible, but miracle-science is impossible.

Third sophism: "The doctrine of faith's supernatural presuppositions have, each for itself, doubtless only validity for the believer; but between these propositions mutually there is a logical and to that extent objective connection. Since the connection is objective, and the propositions of faith precisely find their true interpretation and deeper grounding in this connection, the grounding itself is objective, and the doctrine of faith thus, by force of this grounding, a science."

The dazzling in this sophism rests on this, that a logical connection without more becomes an objective connection, and this in turn an objective grounding. "All men are triangles; Caius is a man, thus Caius is a triangle. In every triangle the sum of the

angles is equal to two right ones; Caius is a triangle, thus the sum of the angles in Caius is equal to two right ones.” Between these propositions there is here undeniably a logical connection, but is there therefore here an objective scientific grounding? From the standpoint of life there can be an eternal difference between the Holy and the Profane, the Evangelical and the Mythical; but, when science once cannot assess this Holy, cannot master this Evangelical, the difference is in the eyes of science as though it were not at all.

Since the propositions of faith are each for itself unconditionally subjective, they are in science each for itself just as arbitrary as the most fanciful fantasies. If there is then a scientific grounding of doctrines of faith, then there is also a scientific grounding of doctrines of myth, a scientific grounding of nonsense.

B.

The Particular Sciences.

§ 17.

That philosophy calls itself the science of the sciences must not be misunderstood as though it hereby would give itself out as an oracle of the sciences, the only, proper, and true science. How such a misunderstanding arises, what confusions it brings with it, ought to be heeded.

Every knowledge is determined by its object, the peculiar material that cognition works up. A comprehensive apparatus of knowledge, a methodically acquired wealth of knowledge, gives learning, but learning is not yet science; only with the development of principles and the rational working-through of the material comes the scientific. But, although to a certain bound there can just as well be science without learning as learning without science, every penetrating cognition of the connection of real facts will always require a peculiar union of learning and science. All so-called positive sciences are learned sciences.

The special is always subordinate to the universal; but in order to enrich our insight into the universal, we must begin with the special. The more a chief science expands,

the more subsidiary subjects it will secrete off. It goes with science as with Alexander's conquests; the greater the empire becomes, the nearer is the turning-point when the ring is changed and the division takes place. If now such a division, in the case of science, is not to signify the empire's downfall, but a progress of the constitution, the smaller divisions must be strictly subordinated to the greater, and these to the organized whole: the totality is the truth.

If now the totality of truth is grasped as an a priori-metaphysical essence, then it is understood from this how the philosophy of absolute knowledge has been able to set itself as the one knower upon the throne and look down on the learned sciences as on its subordinates. Learning is means, science is philosophy. The particular sciences must now either work as day-laborers, or see to learn philosophy, and explain Speculation to themselves.

"Speculation is All, is to behold in God, to contemplate that which is. Science itself has worth only insofar as it is speculative, that is, contemplation of God as he is. But the time will come when the sciences more and more will cease, and the immediate cognition set in. Only in the highest science does the mortal eye close, in that no longer man sees, but the eternal seeing itself is seeing in man." (Schelling).

But as much as these and similar proclamations testify to a fusion of religious and scientific enthusiasm, still such an apocalyptic reference to a turning-point, when "the eternal seeing itself" will become "seeing in man," cannot replace for science the lack of composure and methodical sobriety that is an inviolable condition for advance in knowledge.

One-sidedness has, as the theory of knowledge shows, its deeper ground in the misprision of the element of representation that characterizes the objective Idealism.

Just as light in given cases must presuppose a flame, the flame a combustion, and the combustion combustible materials; so absolute knowledge must doubtless presuppose diverse materials of representation, but only in order to let them burn up in the Dialectic from which the concept shines. If now the material of representation is in this way treated, not as an intellectual building-material, still less as the flesh and blood of thoughts, but only as fuel; then it follows of itself that one does not take it so exactly with the individual pieces of the fact. The consequence of this is naturally mistakes in the individual and arbitrariness in the apprehension of the total; but, even when these blunders are corrected, the standpoint will be no less false.

To value the element of representation, one should remember that the intellectual light, just as little as the physical, can furnish colors and visual images unless it is refracted in the different media, thrown back by the different objects, caught up and clarified in definite impressions. Even when the presentation of the categories corresponding to a positive content keeps nearest to the universal, one notices an essential difference between the emptiness of the barren abstractions bereft of factual knowledge and the play of colors, the natural freshness, with which concepts spring forth that are worked through by intuition, and testify to autopsy.

§ 18.

In his treatise “*Ueber den Begriff der Wissenschaftslehre*” (*Sämmtliche Werke*, Berlin 1845) Fichte has raised the question how far the theory of science can be assured that it on the whole has exhausted human knowledge. The answering of this question reduces to the following circle: “When the proposition *X* is the first, highest, and absolute principle of human knowledge, then human knowledge is a single system; for the last follows from the proposition *X*. Since now there is to be a single system in human knowledge, the proposition *X*, which really (according to the science set up) grounds a system, is the principle of human knowledge as such; the system grounded on it is that single system of human knowledge. Beyond this circle one cannot get.” Further it is said (S. 65) of the relation between the theory of science and the particular sciences: “One sees why only the theory of science has absolute totality, whereas all particular sciences will be infinite. The theory of science contains merely the necessary; all the other sciences bear on freedom, both with respect to our own spirit and to the nature independent of us.”

From the standpoint of absolute knowledge Hegel has given a presentation of the different sciences’ mutual connection.

“In a usual encyclopædia the sciences are taken empirically, as they are found. In it they are to be completely enumerated, and ordered in that the like, that which coincides under common determinations, is set together. The philosophical encyclopædia is, on the contrary, the science of the necessary connection determined by the concept, and of the philosophical coming-to-be of the sciences’ fundamental concepts and principles.”

“The sciences are, according to their mode of cognition, either empirical or purely

rational. Absolutely regarded, both are to have the same content. It is the aim of scientific striving to raise that which is known empirically to the always True, to the concept, to make it rational, and thereby to incorporate it into the rational science.”

“The rational expansion of the sciences is at the same time an expansion of philosophy itself. The whole of science then divides into 3 chief parts: 1) *Logic*. 2) *Natural science*. 3) *Science of spirit*. The sciences of nature and of spirit can then be regarded as the system of the *real* or particular sciences, in distinction from the pure science or logic, since they are the pure science’s system in the shape of nature and spirit.”

As transition from logic to natural science, mathematics is set up with respect to space and time. Natural science itself falls into: a) *Mechanics* with respect to matter and motion. b) *Physics of the inorganic* with respect to light, magnetism, electricity, the physical bodies, and the chemical process. c) *Physics of the organic*, under which then again belong geology, which treats the formation of the earth; botany and plant physiology, whose object is vegetable nature; physiology, *comparative* anatomy, zoology, which treat healthy, animal nature, as medicine the sick. The science of spirit too falls into three sections: a) Spirit in its concept, *Anthropology*, *Psychology* (language – *Philology*). b) The practical spirit, which expresses itself in deed and action; under this *the doctrine of right*, *political science*, *history*. c) Spirit’s completion in art, religion, and science, to which correspond *Aesthetics*, *doctrine of religion*, *Philosophy*.

But although the particular sciences are thus set up against one another mutually and against philosophy, their ideal independence is nevertheless only a semblance. The encyclopædia “is properly a presentation of philosophy’s universal content; for what in the sciences is grounded on reason depends on philosophy, whereas what is in them of that which rests on arbitrary and external determinations, or, as it is said, is positive and statutory, lies outside it.”

From his Kantian standpoint Apelt (*die Theorie der Induction*, S. 65–70) has indicated the principled in the separation of the sciences’ diverse fundamental character.

“There are,” it is said, “two different sources from which our cognitions spring, sense and pure reason. From the former comes sense-cognition, from the latter the pure rational cognitions. The latter fall into mathematics and philosophy. There are as a consequence three different kinds of cognitions: *empirical*, *mathematical*, *philosophical*. Each of these three modes of cognition has its peculiar character: contingency and intuitability belong to the *empirical*, intuitability and necessity to the *mathematical*,

necessity without intuitability to the *philosophical*.”

“In mathematical cognitions we become conscious of the forms of composition of things, in philosophical ones of their forms of connection *in abstracto* [in the abstract]. Kant called the former: forms of the figurative synthesis, i.e. intuitable forms of combination, the latter: forms of the intellectual synthesis, i.e. only thinkable forms of combination, laws.”

“The consideration of the different system-forms’ relation to one another and to the different kinds of cognition leads to a scientific architectonic, i.e. to a doctrine of a system of all human sciences.”

“The three different modes of cognition correspond to the three different system-forms. The pure philosophical form of cognition belongs under the categorical system; the pure mathematical form under the hypothetical system, and the pure empirical content under the conjunctive system. Philosophy is a cognition of mere concepts. But with such a mode of cognition no other systematic derivation is possible than by a categorical subordination of peculiar cases under the generality of the rule. Mathematics is a cognition by concepts of pure intuition. In this intuition the least composite is the most original and the ground from which new consequences are derived by ever new compositions. The derivation of the composite from the uncomposite happens here by hypothetical connection of thought. Finally the pure empirical has only to do with the apprehension of the individual given by sense-intuition. Each fact rests on itself, and the one stands beside the other in a conjunctive system that co-orders parts into a whole.”

The critically apt in this deduction is to be acknowledged; but it shows itself here, as everywhere, that the merely critical, without true peculiarity with the dialectical, is in philosophy unfruitful, and leads to schematism. The separation of sense and pure reason, on which this whole architectonic is grounded, has in reality no significance. The separation rests on an abstraction, but the abstracted will again be integrated by that from which it is abstracted. That sense-intuition is heterogeneous with thought is incontrovertible; but these heterogeneous are for reflection dialectical moments in the same real cognition. The sensible and the thinkable, the individual and the universal are in the thing not the same and yet the same. In reality all is concrete. The cognition of reality is conditioned by concretions; but the intellectual concretions are in turn conditioned by the integration of the abstract forms.

No one who has even merely a general concept of differential and integral calculus will, when he reflects philosophically, be able to deny the dialectical relation between the thought line, the surface's negative bound, the determinate negation that cannot be integrated, and the intuited line, the rectangle's differential, the positive bound that can be integrated into a finite rectangle (Cf. *Philosophie og Mathem.* S. 65–67; 75–76).

The immanent Dialectic will dissolve the particular sciences' independence, and transform them into moments in philosophy's system; the delimiting criticism will restrict philosophy to a peculiar rubric, and supposes on the whole to help by schemata and rubrics. Both efforts are one-sided. Though a science for itself, philosophy cannot by any critical line of demarcation be separated off from the other sciences; and although the other sciences have each in itself the universally scientific, they nevertheless do not have it in such a way that "what in the sciences is grounded on reason" should therefore be able to be said to be "dependent on philosophy."

Mathematics's difference from philosophy is not determined straightforwardly by the intuition and non-intuition of quantity, but by the peculiar analysis to which mathematics subjects the quantitative. (*Phil. og Mathem.* S. 9–14).

As for the positive sciences, their independence is immediately given with the material peculiar to each in particular, without its being thereby given that philosophy should not also in its way be able to appropriate the same material. The harder the opposition is between the object and reflection, the more remote will the positive science lie for "the absolute knowledge's" philosophy. It then makes here an essential difference whether the given object of thought is aesthetic, historical, or physical.

The aesthetic object, the given work of art, is doubtless always from one side or another related to something sensible, and must to that extent be apprehended empirically. But the empirical is here so shone-through by consciousness that the spiritual rules. The objective Idealism has then also without more vindicated for itself aesthetics as part of philosophy. Meanwhile aesthetics, with its special analyses, its autopsy, its history of art, and so on, has another side, from which it, without giving up its scientific character, may well be said to occupy an independent position over against philosophy.

That history, with its facts and its apparatus of learning, stands still more independent, is evident; nonetheless the historical object, as belonging to the past, the life of spirit, and memory, still has so ideal a tinge that Idealism itself can move within these

domains as though it were only on its own. Only physics is hard, stands unconditionally against, and permits no idealistic encroachments. Against the physical object, against reality's polygon, "the absolute knowledge" strikes, and betrays its weakness.

§ 19.

History narrates, philosophy reflects: here is a bound that cannot be leveled by a More or Less. The narration can doubtless take up reflection, and conversely; but the former is nevertheless a mark of history, the latter of philosophy.

To narrate is to determine the universal by the individual in such a way that thought is dramatized in events, and the inner goes up in the outer: here is the element of representation predominant. To reflect is to determine the universal in such a way that events furnish concepts, that intuition is deepened into the inner, and representation dissolves into thought.

From this difference in the manner of presentation a difference in the treatment of the whole and its parts is a natural consequence. The narrating history moves through the particulars of the parts step by step forward to a mediate making-visible of the whole; the reflecting philosophy begins at once with insight into the total, apprehends the particular parts as essential moments in the content of the whole, and cannot give itself over to the detail. Even when philosophy interests itself most of all in the individual, this individual is always shone-through by the total. Philosophy can from different sides illuminate the total's self-mediation through the special; but it does not tolerate the special's immediacy, and cannot follow the narrating history along mediacy's long detour. Philosophy has, as the philosophy of history, another aim than learned history; but for the same reason learned history then also has another aim than philosophy.

If here now, with respect to the aim and the means, the task and its treatment, there is a principled difference between the historical and the philosophical, then history and philosophy can neither advance in converging lines, so that they approach a fusion, according as they approach a completion. Presentations in which the historical and the philosophical equally counterbalance, and then also in a way halve each other, are only intermediate forms, neutral domains, which with their temporary fruitfulness do not concern the principles.

Learned history and the philosophy of history can mediately further each other,

correct and enrich each other, and thereby come into an ever more intimate understanding; but precisely this understanding will confirm the difference. To separate true from false, history must be critical; to point out the causal connection of events, pragmatic; to find and present the spirit whose interests and plan the events serve, philosophical. In all three respects the learned historian may well need philosophical cultivation; but his activity of spirit in no respect goes under philosophy. Everywhere referred to facts and grown together with the given, the historian's thought, far from losing itself in the philosophically universal, will on the contrary be articulated by the intuitable, and completed in artistic individuation: historiography is an art.

If a historian has made some chief part the object of an all-sidedly penetrating study of sources, he has thereby come so near to a past reality that he as it were sees it face to face; will then not his fantasy be so filled with images, and his mind so crisscrossed by individual impressions, that he can scarcely, without preceding ferment, order his material and begin his presentation of what has been intuited?

Essentially seen, the tension rests on a peculiar reciprocal relation between the real and its depicted. In the depiction the real is to be rendered; with all separation of essential and inessential there can here not, as otherwise in an art, be thought of aesthetic idealization. It is a real personality, a real state, a real age, whose physiognomy is to be seen in given features, unadorned as it is, with wrinkles and freckles. And yet spirit is to be present, and imprint itself in the depiction, that it not be overloaded with the trivial, and run out into the real's endless prolixity.

With such a task in view the historian will see that there is no peculiar form that completely exhausts reality's content, but that in the presentation there is incessant strife between loss and gain. What here in one respect is gain, is in another respect loss; this holds not only of the plan on the whole, but repeats itself even in the execution of the individual. That the party spirit does not always need to do violence to the facts in order to get history on its side, it owes doubtless nearest to its own sophistical suppleness, but yet also essentially to the circumstance that the objective in history, however strictly it is held fast, necessarily obtains a coloring from the objectifying consciousness. Different presentations of the same subject can, according to the individuation, be equally justified.

But if the historical form is so artistically individual that one and the same universal task (e.g., the history of the French Revolution) first finds its complete solution when it

is solved in several peculiar ways; then it is seen from this to what degree the element of representation must protest against a fusion of learned history and the philosophy of history.

§ 20.

It is not the whim of chance, and not human arbitrariness, that may draw the bound between the particular sciences. Each science is in itself an essential totality; its bounds, divisions, and elaboration are conditioned by the material, governed by thought, and determined by the comprehensive task.

In the method of every science the universally scientific is seen: everywhere a reciprocal determination of the immediate and the mediate in cognition, of detour and shortcut, of necessary movement of the matter and free movement of thought (Philosophie og Mathematik. S. 44–46); likewise every science's systematizing striving will require definitions, divisions, subordination of lower and higher unities under one another mutually and under a highest unity, and so on. But the peculiar material and the peculiar aim in which it is worked up will nonetheless in the method as in the formation of system bring with it such decisive peculiarities that each science thereby not only becomes a doctrinal structure in a peculiar style, but is even in an inner, infinite way organized into a relative monad, an intellectual individuality.

To what degree the peculiar material acts upon the formation of system will especially become manifest from those sciences whose content is expressed in typical natural forms. Here is, in the mineral kingdom, the dead form: the crystal; in the plant kingdom the living form: the organism; in the animal kingdom the ensouled form: the individual.

a) *The dead form.* In its typical purity the substantial form of bounding is mathematical. By laying a mathematical body (e.g., the cube) at the foundation for the determination of the axes and the corresponding section, pure crystallography can with the sureness of an exact science round itself off into a whole articulated into particular systems — the spheroidal, the rhombohedral, the pyramidal, the prismatic. Hereby it becomes not only possible to lead an infinite manifold of derived special forms back to their fundamental form valid in each system; but even the combinations, alterations, and distortions that proceed from formations in free nature are so far from bringing the whole into confusion that they on the contrary become intelligible by

the systematics of the pure types (Applied crystallography).

It is otherwise when the question is not merely of the substantial forms of bounding, but at the same time of the corresponding substances' chemical and physical properties, of the mineral's characterization. Here the preponderant regard to the chemical substantiality (Berzelius) comes into open conflict with an equally preponderant regard to the individuality expressed in the physical properties (Mohs), the strict demand for systematic grounding in conflict with the schema and its many rubrics.

A more equal regard to chemical and physical distinguishing marks can now doubtless lead to ingenious combinations of system and schema; but only gradually, as science penetrates into the essential connection of the substances' chemical and physical properties, will the schematic go up into the systematic.

b) *The living form.* Although the real bound that separates the organic from the inorganic does not immediately let itself be pointed out, the essential difference between the mineral and the plant is nonetheless clear enough. Whereas the crystal, the dead form, goes up in the mathematical, the mathematical (three-numbered plant, five-numbered plant, the difference-angle of the stem-leaves, and so on) is, on the contrary, as mathematical of subordinate significance in the living forms. In the different species as well as in the individual specimens of each species the organic runs through such a series of metamorphoses and corresponding articulations that here one can rightly speak not merely of incessant metabolism, but also of lower and higher stages.

The sexual system, the so-called artificial plant-system, with its classes and orders (Linnaeus), is a grand, comprehensive, on the whole practically applicable schema; but only when a plurality of essential properties and organ-relations is taken into the grounding of the division does the natural, i.e. the proper, system begin to form itself. Strict holding-fast to a metaphysical principle and arbitrariness in the apprehension of the given (Reichenbach) has, however, instead of bringing the system to completion, on the contrary shown what difficulties are here still to be overcome.

c) *The ensouled form.* Whereas the individual unity in the case of the plant either is not at all, or is at least so protean that it vanishes into phytons and cells, the animal is, on the contrary, by force of the concept's necessary demands (sensation, will), originally set as a definite individual. The stricter articulation, whereby the organism's individual members are developed for themselves, advances on the whole

with a rising centralization, which gathers the organs' different activities into the sensing, self-determining unity of life.

It is then a matter of course that the distinguishing marks of the species and the typical separation into lower and higher orders — *molluscs*, *articulates*, *vertebrates* — must in the animal kingdom appear with a quite different clarity and orienting validity than analogous divisions in the plant kingdom. Strict Empiricism (Cuvier) has here too stepped into opposition to natural-philosophical Speculation (Geoffroy St.-Hilaire). But, in order that the system can be brought to completion, the reality of the genus-concept must first be decided, and the connection of the higher classes with the lower determined in such a way that chief forms and intermediate forms show themselves as naturally necessary members in the ascending chain of ensouled unities.

If now the universally scientific is everywhere by material and task grown together with the peculiar, then each science must also exclusively look to its ideal, and cannot possibly take its scientificity in fief from philosophy.

How little philosophy is entitled to ascribe to itself the thought and the developments of concepts that belong to the learned sciences is expressly seen from this, that categories such as substance, individual, species, genus, and the like are developed in the real sciences in a quite different way than in an objective logic or speculative natural philosophy.

The laws of thought are doubtless everywhere the same; but to hold fast to the laws of thought for themselves in their pure generality is precisely to keep them in a questionable one-sidedness. The strict thinking has in the one science a quite different character than in the other: the particular sciences presuppose particular talents.

§ 21.

Insofar as the particular sciences have in material and aim a finite delimitation, they are subject-sciences [Fagvidenskaber]. How peculiarly the different subject-sciences' view is formed becomes manifest when they, by treating one and the same subject, approach mutual contact. In the determination of soul, life, and life-principle a thoroughgoing difference shows itself between the strictly empirical physiology, for which the organism is "a living machinery" that does not even need a life-force for "machine-master" (G. Valentin), and the psychology grounded on self-observation,

which grasps the soul as constituting essence, and lets life well up from “an inner source.”

The strife that can thus arise between real subject-sciences must, however, above all not be confounded with the strife between the theory of science and a miracle-science. One thing is science’s strife with the unscientific; another is the strife between scientific one-sidednesses. One thing is a strife against confusing appendage of an unwarranted one; another a strife for true equilibrium between warranted ones. One thing it is that the archaeologist and the geognost, each equipped with his apparatus of factual proofs, meet in strife at a burial-mound; quite another that a medieval theologian, without acquaintance with the natural marks, invokes a supernatural, and mixes into the investigation about the human races and “our first parents.”

In order that the individual science can advance, it must be supported by those most nearly related to it. In that mathematics is a chief science for itself, it is at the same time an auxiliary science for mechanics and astronomy. Chemistry, mineralogy, and geognosy unite as auxiliary sciences in geology. The helping sciences are in turn helped by those they have themselves helped. To what degree botany and zoology further geology, and are in turn furthered by it, is conspicuous. In the great circle of coordinate tasks no subject has an exclusive title to be the chief subject: in the organism of the sciences the means is aim and the aim means. As easy as it is to grasp the particular sciences’ mutual reciprocal relation in general, just as difficult can it in turn be to determine this relation with respect to the individual science in particular. Physiology furnishes a telling example. The doctrine of digestion presupposes insight into the chemical processes; how many constituents ought now to be taken up from chemistry? The doctrine of sight presupposes optics, the doctrine of motion mechanics; how much ought now to be borrowed from these sciences? Thus one could go on.

Traditions and conventions have in such cases no decisive voice. Every expansion and restriction must proceed from the nature of the matter. But precisely in the development of the matter lies the difficulty. The one science can doubtless refer to the other; but in the mere reference the relation is external. If the point to which reference is made lies outside the matter, then the reference is, strictly speaking, superfluous. If, on the contrary, the point in question is an essential part of the matter itself, then it ought also to that extent to be taken up into the development of the matter.

But in what form is it to be taken up? As a borrowed proposition? But the borrowed

proposition is like a dead, mechanical admixture, a torn-off branch that is attached to a foreign trunk. If the attachment is to be organic, if a grafting is to take place, then the borrowed proposition must bring along so much of the grounding connection that thereby a part of the one science is carried over into the other. But in this way the one science will come to take up a multitude of extracts from other sciences; and yet every *juxtapositio* [juxtaposition] is in conflict with the essence of science.

The diverse contributions a science receives from others it must assimilate; references, borrowed propositions, extracts, all constituents conveyed from without must be worked up in such a way that the externality vanishes, and the patching-together is suspended. To suspend the patchwork and work together the heterogeneous contributions requires now doubtless a peculiar organizing talent, but then also above all a clear and distinct insight into the relation between subsidiary subject and chief subject. This relation becomes especially involved when sciences that in and for themselves are chief subjects partly furnish a comprehensive foundation, partly, for a definite purpose, are treated as subsidiary subjects.

Mathematics, which in itself is a chief subject, and for astronomy and mechanics a comprehensive foundation, must, insofar as it serves chemistry, be treated as a subsidiary subject. Chemistry, which in itself is a chief subject, and for mineralogy and physiology a comprehensive foundation, must in many peculiar cases expressly be treated as a subsidiary subject. The category foundation indicates in this connection an intermediate determination between chief subject and subsidiary subject.

To treat a science as a chief subject presupposes, however, only one decisive regard; but to treat a chief science as a subsidiary subject demands a plurality of regards, a doubling of reflection that makes the task difficult.

The bound between the different sciences is in its essence dialectical; but in the subject-sciences this dialectical cannot become the object of any peculiar exhaustive reflection; instead of such a reflection there steps a living leading thought, a peculiar decisive purpose, a definite plan. The aim expressed in the plan is a fundamental concept determining all peculiar demands, penetrating all parts of the presentation. Where this fundamental concept is not clear, where the definite aim is not at all points present to consciousness, the presentation will become now too prolix and overloaded with unnecessary detail, now again — of course “for lack of time!” — too superficial and survey-like.

The principles according to which the branches of science, with their chief and subsidiary disciplines, are grouped under a university's different "faculties" cannot possibly act solely according to abstract-theoretical regards; for that science is in an all-too-many-sided relation to real life.

The reciprocal relation between science and life is mediated by the state and its institutions. In the different faculties several often very diverse sciences are worked together into peculiar wholes calculated for definite spheres of activity. If these circles of knowledge are rounded off according to all-too-meager practical regards, science will be violated, and the scientific spirit lost. If these wholes are organized according to all-too-high-flown theoretical ideals, the interests of life will step into opposition to the scientific, and science lose.

The reciprocal relation between theory and practice must stand clear to consciousness as the plan is laid. Once the plan is laid, and the means assigned, the carrying-through ought to happen with strictly persevering consistency. The interest now falls decisively on the theoretical, precisely because this theoretical has taken up the regard to the practical as a moment integral to its plan. If the study of theory can really ground a station in life, then it also deserves that fresh vital forces be applied to bring it to its best possible completion within fixed bounds. The student who in this respect, without doing his duty, would filch the final advantage, is false toward science and false toward the state. For the state will not merely have a number of businessmen as practically usable organs; it will in these organs see the scientific spirit, the higher cultivation, represented.

C.

Philosophy and the Particular Sciences.

§ 22.

When the ancient problem how a science is attained is set forth thus: "The knower can, insofar as he already is knowing, get nothing to know; but neither can the ignorant

one get anything to know, since he, in order to come to a knowledge, would first have to see his ignorance, and, in order to see his ignorance, already beforehand be a knower,” the problem must doubtless at first glance look like a sophism. The matter nonetheless has, in the unconditioned sense, its correctness. One can, as regards finite expertise, get much to know that one did not know beforehand; but from an absolute not-knowing to knowing there is no finite transition.

As for the *a priori*, Plato (Phaedrus) has already pointed out the essential (ἀνάμνησις). Every advance in knowledge is point for point conditioned by an immediately evident; since now nothing can be evident except in that it goes together with our own consciousness, and reminds us of what we already know, the advancing development of concepts rests on an intellectual recollection, a reflecting recall, in which we incessantly find the True as in ourselves, and, so to speak, take our own knowledge into possession.

He who learns mathematics gets essentially nothing else to know than what he originally knows, and yet by learning mathematics he precisely makes advances in knowledge; the same holds of the *a priori* in aesthetics, ethics, and dialectic.

With respect to the *a posteriori*, the advance of knowledge will doubtless be conditioned by this, that here new knowledge is constantly gathered, that new facts are tested, new discoveries made, and so on; but is the matter herewith decided? If the advance of knowledge in relation to the *a posteriori* is grasped solely from this side, it is easy to dismiss philosophy. If it cannot even exhaust the *a priori* — since mathematics owes it no thanks — how much less can it then engage with positive facts! If the historian sets it a task: it cannot answer, for a historical task is answered only historically. If a chemist turns to it: it has no knack for experimenting, makes no discoveries in chemistry. If the astronomer wants help with an observation: it scarcely understands him. If the physician asks it for advice: it is ignorant. Even what it has on its own speculated out concerning the relation between soul and body, the physiologist perhaps cannot with difficulty even use.

But in these affairs too it is a matter not merely of getting something new to know, but also of knowing what one already knows, of getting one’s knowledge thoroughly, inwardly, all-sidedly brought to consciousness.

Although every science that begins with the first principles, and goes systematically forward, doubtless develops its content in such a way that the otherwise prepared

beginner can follow; so it is nonetheless unmistakable that one first fully understands the beginning and the earlier when one, after having come to the conclusion, looks back. The recall whereby the foregoing is illuminated by the following is an inwardizing of the matter, a doubling of reflection, which, with respect to finite points, does not indeed enrich us with anything new, but nonetheless in infinite degree heightens our knowledge.

If there is now, also in relation to facts, an advance of knowledge that consists not in a multiplication of the things that are illuminated, but in the intensity of the light that illuminates the things; then it depends in this respect likewise on reproduction, on inwardizing, on recollection (ἀνάμνησις), and thus philosophy has to that extent the same validity in the *a posteriori* as in the *a priori*.

In positive matters, in the affairs of experience, philosophy does not go before, but follows after. Everywhere where Speculation with its “intellectual intuition” has wished to anticipate Empiricism, lift the mystery, and unveil the hidden, it has only made itself ridiculous. But if anyone now concludes from this that philosophy is just as superfluous when it follows behind as it is unwarranted when it will go before; he can, by dwelling on what he already knows, without difficulty convince himself of his mistake.

To dismiss philosophy one could adduce that the higher knowledge owed to afterthought’s reflecting retrospect is attained not by philosophizing, but by repeated study of the subject-sciences in question. A deeper insight into the essence of mathematics is won not by philosophizing, but by incessant repetition of mathematical analyses; a higher consciousness of the chemical is won not by speculating, but by again and again revising what is won, and making the recognized truths quite living, penetrating, and clear to oneself. By thus going through science after science, everywhere with sharpened eye for presuppositions, principles, and the inner infinity of a systematic connection, afterthought will then bring the heightened light, and enrich us with an insight that can disdain philosophical abstractions.

If now this yield of consciousness is brought out for itself, in such a way that it can take on the shape of the concept and be clarified in categorical form; then we have full compensation for the vanished philosophy.

But in this compensation for the vanished philosophy, philosophy is precisely present. Every relation to an object is in science an objective relation of consciousness.

One can by consciousness set thought above sensation, the ideal above the real, but not deny consciousness's barrier. One can by consciousness set sensation above thought, the real above the ideal; but one cannot by consciousness set anything in consciousness's place. To become thoroughly, all-sidedly, inwardly conscious of all real and all possible relations of consciousness is philosophy's task. In this its unity of essence with the knowing consciousness, philosophy is precisely present when one proves its non-being, precisely in exaltation when one sees through its degradation, precisely vigorous when one considers its weaknesses — a spiritual power, a superior potency, an infinite Proteus invincible to the subject-sciences taken together.

That positive knowledge does not without more replace philosophy becomes not so seldom evident when learned researchers, especially natural researchers, give up the strict method, write popularly, and impart to the cultivated an improvised philosophy.

§ 23.

Over against the other sciences philosophy is doubtless a particular science, but nonetheless no proper subject-science. Although such ideal objects as thought, truth, life, the good must be said to be philosophical; thought is nevertheless not an object for philosophy in the same sense as the starry heaven is an object for astronomy, the plants for botany, the animals for zoology.

The subject-sciences are solid, always have fixed work, ample livelihood, and secure recognition in the learned republic; philosophy is in comparison with this like an itinerant without finite provision, a hovering shape that sinks and rises, dies away and lives up again, according as the spirit wills it. The subject-sciences are the rich landowners who have the objects, and live in the manifold of knowledge; philosophy is like that Eros, born of Poros and Penia (Symposium, Plato), always in want, owns nothing, brings nothing but the gift of penetrating everywhere, and being sated with a glance at the abundance of the rich.

He who cultivates a subject-science has, with all his exertion of spirit, nonetheless always an unmistakable advantage in that his material has a finite bound. It is otherwise with him who will cultivate philosophy. Instead of enjoying the impression of independence, the intellectual ἀντάρκεια, which persevering occupation with even merely a single branch can grant the capable specialist, a continued study will on the contrary convince the philosophizer that he cannot take a step on the path of

real cognition without feeling his knowledge's dependence on the learned sciences; whereas the specialist will soon learn to help himself as an independent researcher, it can be the philosopher's humble position incessantly to knock on a foreign door, incessantly to begin anew, and so be an apprentice his whole life.

With all its idealistic mobility philosophy is nonetheless a science determined in the concept. If each of the subject-sciences can teach us how it will be studied, then philosophy too can for its part offer a similar guidance.

It is natural that the study of a subject-science is united with the study of the science's history; but insofar as here there is an immediate material, the treatment of the material will step into the foreground. One then readily begins by studying first a part of natural science itself, and next the history of natural science; first historical events to a certain extent, and then the history of historiography. Science itself will in such cases stand above its own history.

In philosophy the relation is reversed; here the history of the science is an essential content for the science itself. The history of philosophy in union with the philosophical chief works then forms the foundation with which here one must begin.

The next step is the study of the real sciences. To behold the works of art, live with the poets, and make oneself familiar with the historians requires general cultivation, but no peculiar guidance. This is, on the contrary, the case when the talk is of the proper real sciences. That insight into the nature of things is attained only by real investigation of the natural things is shown by the theory of knowledge; but the mass of knowledge that must here be presupposed is so great, the all-sidedness and the compass in which the studies in this respect would have to be driven so considerable, that one, without looking to how much a single person can accomplish, may only look at the science itself.

That a philosopher should be perfectly at home in all possible sciences, and, like Aristotle in his day, possess a whole age's apparatus of knowledge in all its particulars, is not only impossible, but also in the matter itself superfluous. The philosophical analyses of reality are doubtless in so intimate connection with the facts of experience and corresponding observations that handbooks, surveys, popular lectures, and the like do not suffice; but it is nonetheless also a law that philosophy must not take up more material than the Dialectic can dissolve, and the concept assimilate. It then depends not so much on the greatness of the mass of knowledge as on choosing among

the mass's parts, sifting the material, and, so to speak, finding the ore from which a knowledge can be smelted out.

As an inviolable presupposition the theory of knowledge must demand that the philosophizer, even if he is not a mathematician, nevertheless learn to understand the language of mathematics, and, although he will not become a chemist, make himself familiar with the chemical laws, designations, and formulas. Without such a preparation the philosophizer will never penetrate into the kernel of the matter, never see through the exact adducing of proof, never understand what physics really means when it speaks of force and matter. Mathematics and chemistry are, so to speak, the keys to the secret recesses of real science; only he who has these keys is an initiate, and only for the initiate does the treasure show itself. In order that a study of so many and different branches of science shall now not either lose itself in bottomless thoroughness, or pass over into loose diletantism, it is naturally necessary that the philosophizer be conscious of a task, that he in all this see a definite goal, and have the endurance to pursue it.

As for the task in its generality, it will be determined by the reciprocal relation between the real and the ideal, between analysis of representation and analysis of concepts, between the principles and the finite presuppositions.

As for the task in its peculiarity, partly an individual question arising under peculiar circumstances, partly a comprehensive regard to philosophy's organic chief parts can determine the compass and direction of the real studies.

But even when the task is held fast, even most strictly, the real studies still bring considerable difficulties peculiar to the relation with philosophy. For an analysis of concepts it is namely not enough that the philosophizer knows a number of the points essentially attractive to thought; he must have such points in such a manifold, possess them with such freedom, and move with such sureness, that he, without striking against the real, can move his moments of concept as though they were not piecemeal-gathered contributions, but inventions of his own fantasy.

How difficult it is to acquire such sureness can easily be seen. From many sides Empiricism illuminates the inner, mysterious connection that is between things' otherwise so different properties. The universal is here known in the special, but this special in turn interrupts the cognition of the universal. Let one, e.g., order the elements in the so-called trilogies; let one compare barium (*A*), strontium (*B*), calcium

(C), and note the relation between the chemical numbers: $\frac{A+C}{2} = B$. If one now looks at the solubility of baryta, strontia, and lime, then one finds that the baryta has the easiest, the lime the heaviest solubility; whereas the strontia stands midway between. Let one test another trilogy: sulphur (A), selenium (B), and tellurium (C), whose chemical numbers will likewise give: $\frac{A+C}{2} = B$. If one now looks at the specific gravities, these relate thus: the sulphur has the greatest, the tellurium the least, and the selenium stands midway between. Let one think of the relation between the specific heats and the chemical numbers; let one think of the crystals' relation to light and their expansions under heating; let one think of the law according to which a rise and fall of the boiling point of the polymeric hydrocarbons (CH_2 , C_2H_4 , C_4H_8 , $\text{C}_{16}\text{H}_{32}$, ...) takes place, in that the point rises for each carbon atom that is added, and sinks for each added hydrogen atom, and so on.

But while here, under this penetration into the interior of things, a multitude of parallels, a series of presuppositions and consequences, come to light, there can nevertheless in this connection scarcely be made a single a priori inference independent of the test of experience. That the specific gravity of iron oxide is less than that of iron is comprehensible; but, if one drew from this the general inference that the higher metal oxides on the whole must have a smaller specific gravity than the lower, one would err: the specific gravity of stannic oxide is, e.g., greater than that of stannous oxide; that of plumbic oxide greater than that of plumbous oxide. If we look at the relation between the two properties, hardness and specific gravity, then it seems natural that on the whole they should correspond to each other: the diamond, which is far harder than graphite, really also has greater specific gravity; yet let one guard against a hasty inference.

When isomorphic substances vicariate, it is seen that hardness and specific gravity by no means straightforwardly correspond to each other. If iron steps into the garnet instead of lime, or lime instead of iron; then it holds: the more iron, the greater specific gravity, but conversely: the more lime, the greater hardness. Every essential connection doubtless contributes to the answering of a Wherefore; but here comes every moment a Wherefore that cannot yet be answered.

For the philosophizer the subject-knowledges cannot in and for themselves have significance, and yet it is not so seldom that precisely those points which are quite properly fruitful for the analysis of concepts lie hidden under a detail of facts and

special investigations, which thus must nonetheless be gone through.

Finally a deeper appropriation of the intellectual that offers itself in a definite circle of facts can first set in when this intellectual is illuminated from other and corresponding circles: *Opposita juxta se posita magis illucescunt* [opposites placed side by side illuminate each other more].

If the total can never be fully developed except by means of the individual, then the reverse also holds, and first and last in philosophy, that a penetrating study of the individual cannot possibly be carried through without a corresponding insight into the total.

§ 24.

But even if the philosophizer has now also conquered the difficulties of real study, drawn sublimates from the positive sciences, and sees himself able to work together the constituents: in what form is he then to impart what has been won? to whom ought he to turn, by whom will he be understood? For the illumination of the difficulty here pointed out we grasp the matter, first from an objective side, in that we look at the manner of presentation, next from a subjective side, in that we take account of understanding.

The manner of presentation. As for facts and corresponding laws, philosophy must treat the given as a given, and not by a kind of sophistical turn let the matter get the appearance that things which are due solely to experience nonetheless proceed with inner a priori necessity from the pure concept. The untruth that in this respect crept into the Hegelian method has doubtless its ground not in any conscious dishonesty, but in overstrained zeal of reflection and a consequent disproportion to the immediate.

Immediacy ought doubtless to be lifted; but immediacy must not be suspended in such a way that the given changes its nature, and is transformed from a positive fact into a metaphysical essence; immediacy must be suspended in such a way that the given, as given, furnishes an ideal contribution to the comprehensive development of concepts that the principle itself sets going. Insofar as the comprehensive development of concepts is and remains the main thing, the given then lays off its contingency, does not remain in thoughtlessness, but shows its ideal, obeys the concept, and comes when thought calls.

The point at which the conceptual side of the given each time coincides with

the leading concept of the development is a point of insight. At this point there will always gather so much light that the proposition which is taken up as a given from another science neither presents itself as a dead borrowed proposition, nor brings with it, for its final explanation, a circumstantiality that halts the pace of development and burdens thought with the all-too-mediate.

To conduct separate proof for each peculiar proposition is impossible and unnecessary; to illuminate each proposition in such a way that the side that is essential for the connection is also clarified in the connection, that the point on whose understanding it each time depends can also each time be understood, is possible and necessary.

Every scientific presentation of deeper content always has a multitude of luminous points in the foreground, a multitude of obscure points in the background, and between both manifold transitions. Where several sciences treat the same thing, each will make its own distinct; but they will never make the same distinct in the same way. Likewise with respect to clarity. What one development according to its purpose must lift to clarity, another may according to its purpose let remain in obscurity. The logical categories obscurity, clarity, and distinctness are of dialectical application.

Another difficulty with respect to the given is peculiar to philosophy. This science may not, as a positive subject-science does, give itself over to the manifold of the finite, enumerating, narrating, describing, and so on. Nonetheless it must in its way take up the outwardly manifold, insofar as it will give itself a real content.

To remedy this difficulty one has sometimes seized on the desperate expedient of laying out a schema of concepts, holding to the formal order, letting the reasoning within each of the rubrics run free, and then filling the emptiness with a multitude of heterogeneous examples fetched from all positive sciences; this is perhaps the worst method anyone can apply.

The task is to let the form take up the content in such a way that the development of concepts becomes a development of the matter. In this development of the matter the positive elements must then be chosen, distributed, ordered, and treated not merely in such a way that the individual facts are placed where they are singly needed, but also in such a way that the whole aggregate of real moments is worked through by reflection, that each concluding totality can express the content of the matter in the form of the concept. Through concluding totalities in different — partly coordinate, partly mutually enclosing — circles the development will then go to the last conclusion.

Understanding. Where tradition and habit once give a certain manner of teaching prestige and support, it is often striking to what degree one can suppose one explains the inexplicable and understands the incomprehensible (half-scientific religious books); where, on the contrary, the principle of cognition demands that in something handed down several changes necessary for understanding must be made, it is again just as striking to what degree obstinacy can strive against, and find the in-itself-understandable incomprehensible. That opinions about the understandable and the incomprehensible can be so divided, that one can make such highly different demands on what is to be understandable, follows from this, that the category understanding is so infinitely dialectical.

It is an easy matter to reproach a philosophy with incomprehensibility. General cultivation does not as a rule go deep enough to follow a stricter analysis of concepts: the understanding of the individual presupposes expertise. But if special expertise is an inviolable demand, then the consequence of this must be that each specialist holds to his piece, and lets the rest stand as an incomprehensible. That such a parceling-out of understanding must make philosophy meaningless is then understandable.

As for the presentation, it can become incomprehensible just as well by holding to the general as by going into the special. If special analyses are predominant, then one doubtless notices the involved, the thorny, the prickly; but if the universal-forms are predominant, then one gets perhaps still more troubles with “the abstract,” “the nebulous,” “the hovering,” “the ambiguous” (turns of phrase, word-philosophy, phraseology). It depends in such things not on the finite What, but on the infinite How.

To clarify to oneself the relation between understandable and incomprehensible, one must, on occasion of the demands that can with right be made on the presentation of what is to be understood, then above all also not forget the demands that can with right be made on him who will understand, such as: necessary preparation, persevering attention, scientific sense.

Insofar as each science has its peculiar structure of thought, it consequently also has its peculiar kind of understandability. In the mechanical, understanding comes otherwise than in the chemical, in the exact otherwise than in the dialectical, in mathematics otherwise than in philosophy.

§ 25.

If we call investigations of things' first presuppositions and knowledge's principles philosophical, there can doubtless be no doubt that philosophical investigations are necessary in the different sciences; but from this it does not yet follow that philosophy for itself is a peculiar science. One must, e.g., in every branch of science apply criticism, and also has in the different sciences a multitude of critical treatises, even treatises on criticism; but for that reason surely no one will maintain that criticism for itself is a peculiar science.

If there were now a circle of critical-philosophical treatises in which the diverse sciences had each in particular given an account of its principles and presuppositions; then one could doubtless, by working them all up according to one plan, unite them in a whole; but this whole would, by the finite connection of the parts, nonetheless just as little become a peculiar philosophical science as the collected treatises would become a peculiar science by being bound in one volume.

The objective Idealism would now at once lift philosophy's scientific independence by referring a) to the Dialectic in which the finite contributions are dissolved, melted down, and deposited as the mobile moments of an inner infinity; b) to the immanent method whereby the absolute content gets its moments laid out, separated, and gathered in the absolute form; finally c) to the deeper view of life, the solution of the riddle, which is attained only in Speculation.

But the presuppositions on which philosophy not long ago wished to ground its divine superiority unite here as flaws that threaten its scientific independence. Thus it holds with respect to a), that the dialectical dissolution may nonetheless not loosen principles and presuppositions from the soil of reality in which they have their root; with respect to b), that "the immanent method" must go the same way as "the absolute knowledge"; and finally with respect to c), that science does not concern itself with "the deeper view of life," since it on the whole does not sit in judgment over views of life, and that "Speculation" cannot solve "the riddle of existence."

How far philosophy is a science independent in itself is not dependent on arbitrary opinions or passing blunders, but on the nature of the matter. If philosophy is a science, then the peculiar form in which it develops its content must be of such decisive significance that scientificity would suffer an irreparable loss if there were no

work on bringing this form too to systematic completion. We must then here expressly bring out the influence philosophy exercises on the scientific life of consciousness, insofar as it is a matter of the appropriation of the material, the liberation of the concepts, and the inwardizing of the matter.

1) *Appropriation: an assimilating activity.* The outwardly given does not immediately go up in consciousness. In every subject-science there will always be a mass of constituents so heterogeneous with the life of consciousness that not even the strongest memory can gather and bear their quantity; such constituents are then preserved as materials in magazines, repertoires, and the like learned storerooms.

In relation to the real, scientific insight is conditioned by memory; in relation to the essential, by recollection; the real is remembered, the essential recollected. The appropriation under which the scientific material passes over into the flesh and blood of consciousness (*in succum et sanguinem* [into sap and blood]) is an intellectual process of digestion that works up the material, and takes on a different character according as it is the real in its particulars or the essential with its universal-determinations that becomes the main thing.

The more the side of reality predominates, the more closely thought joins itself to memory, and the more also the appropriation becomes restricted to a peculiar rubric of consciousness, determined for the peculiar subject. The more, on the contrary, the side of essence is brought out, the more intimately thought goes together with recollection; the finite constituents of the material then become real moments in the peculiar concept, and the appropriation of the peculiar concept sends out a light that shines through the whole consciousness.

When now these two modes of appropriation are referred each to its aim and its principle, a difference shows itself between subject-science and philosophy.

The carried-through assimilation relates to Speculation as real-philosophy to Idealism. In idealistic Speculation the given is volatilized into a priori essence; in real-philosophical assimilation the a priori essence is peculiarly determined by the given.

2) *Liberation: a humanizing activity.* Restricted to the different rubrics of consciousness, the won concepts are still so grown together with finite constituents, still so burdened with particular presuppositions, that they find themselves as in a state of unfreedom; they are here held as the material's prisoners within a narrow circle

of vision; they are, so to speak, the subject's serfs, who may not leave the definite workplace. This intellectual slavery of the concepts forms only a one-sided counterweight to the equally one-sided unboundedness that proceeds from superficial abstractions, and serves the ingenious play of the association of ideas. Every concept that is had completely is had with freedom; it then depends on seeing wherein the freedom consists.

That the concept of the mechanical belongs just as originally under mechanics as the concept of the chemical under chemistry is incontrovertible; when now Hegel in his *Logic* nonetheless deals with the concept's own mechanism and its logical chemism, then it is neither mechanics's nor chemistry's, but rather philosophy's task to investigate with what right such a transfer has happened. Word-explanations, acute interpretations of homonyms, synonyms, metaphors, and so on, do not mean much in this case. Here the question is not of word-explanations; here the question is expressly of the changes in the categories' validity that are brought about by a passage through different spheres. Such investigations demand expertise and Dialectic. In that a concept has passed over from one sphere to another, it is no longer the same, and yet still the same: "function" in mathematics and "function" in physiology is not the same, and yet the same; "approximation" in morals and "approximation" in mathematics is not the same, and yet the same; "reflection" in optics and "reflection" in logic is not the same, and yet the same. Here one can continue to infinity.

Philosophy's humanizing activity does not immediately coincide with a striving toward the popular. The dialectical liberation of the concept, which at every point must be exactly conditioned by what is given by assimilation, belongs exclusively to science. This humanizing activity, a necessary condition for consciousness's being able in all things to be itself, and to master its many rubrics, is possible only in that philosophy steps forth independently, and lifts its own method.

3) *Inwardizing: a totalizing activity.* What difference there is between knowing something when consciousness especially enters into the learned rubric, and knowing something in such a way that on a given occasion it instantly presents itself, and stands clear to thought, is generally known. The same content of knowledge can in the one form fall so heavy that each time a peculiar exertion of force is required when the knower is to lift it; in another form, when it, without losing its intellectual specific gravity, has by assimilation been separated from the all-too-circumstantial admixtures,

it can on the contrary be so light that consciousness, freed from the material's burden, bears its wealth as though it bore only itself: *omnia mea porto mecum* [all that is mine I carry with me].

The totalizing activity is conditioned by a corresponding method. To the finite separation of thinking and being corresponds a *critical* method; reflection moves about its object, points out the dogmatic proposition's untenability, and makes a subjective transition from the one to the other. Thought moves, but the thing stands still; here one thinks about the object, but the object does not come to unfold its moments: here the immanent Dialectic of the development of the matter is lacking. To Idealism's absolute knowledge corresponds a *dialectical* method. Here it is too little to think about the matter, for the matter is thought through in such a way that it looks as though it thought itself. The objective then does not properly express the real; it is an abstraction in which the subjective comes objectively to itself. The objective is not an existing object, it is a concept; it goes by "der Begriff" [the concept] and "der Begriff"; the pure concept entertains itself: here criticism is lacking. To a synthetic unity of thinking and being, a unity of essence that cannot suspend the real difference, corresponds a method in which objective development of the matter and subjective reflection determine each other mutually; such a method is *critical-dialectical*: criticism clarifies the finite conditions, Dialectic opens up the inner infinity.

The methodical movement is in philosophy a double one. On the one side the special in the content, the finite with its particulars, is worked up and assimilated in such a way that the moments go back into the universal, that the fundamental cognition is enriched, and great totalities are set in motion. In this respect the Hegelian method has an imposing superiority. But since the real, with its finite members, its angular forms, and existential punctuality, necessarily belongs to the development of the matter, it is on the other side an equally essential task to analyze the individual, separate the members, penetrate the entanglements, and bring the individual to dialectical vibration. In this respect the Kierkegaardian analyses furnish apt examples. On the heights of totality the telescope is applied, for immersion in the analysis of punctuality the microscope is used.

In the outer world two such opposed modes of viewing cannot possibly be applied simultaneously; to look at once through a telescope and a microscope is doubtless an impossibility. In knowledge's inner world the matter stands somewhat otherwise. If

already the subject-science, with its whole detail and its rubrics, can work itself up to higher reciprocal relations of insight into the individual and survey of the whole, then philosophy, with its assimilating power and liberating activity, must be able to carry it still further. The critical conditions the dialectical. The more dialectical insight and survey — insight into the special and survey of the total, insight into the total and survey of the special — alternate with each other, and in the dialectical refraction awaken each other, sharpen each other, and penetrate each other, the greater is the intellectual tension, the more the concept's force grows, the higher the waves of the movement beat, the stronger the stream of cognition goes.

Method relates to system as vital activity to the living organism. In the subject-science, where it always depends on getting the material placed, the schema can play a role. In philosophy, which is not burdened by material, schematism is intolerable. Every principled thought is at bottom always systematic; if the systematic form will not at once present itself, the thought ought to be developed with freedom; it will then afterward be seen from the development in what comprehensive connection it belongs.

The finite knowledge has many particular points of departure and many particular systems; the "absolute" knowledge will develop all from one principle in one continuous system. The synthetic knowledge, which can neither be satisfied by relative points of departure, nor reach the absolute principle, gets no absolute system, but within its bound an essential relation of relative systems.

The more the individual elements of knowledge are worked together, organized, and clarified, the more intimate the connection between the whole and its individual parts becomes, the more intimately also the content of knowledge goes together with the knowing consciousness. The assimilating and humanizing activity that is here demanded shows clearly that the complete inwardizing is dependent on an independent philosophy.

If we grasp the particular sciences as definite organs in the great organism of scientificity, then we must evidently grasp philosophy as the soul of the organism. Here too it will then repeat itself, that a rising separation and more definite development of the individual organs goes together with a corresponding concentration. The principled unity of the functions will here too become more independent, according as it becomes more intimate, and, though in the finite sense a product of the organs'

cooperation, nonetheless with inner infinity assert a subsistence in itself over against the particular organs.

III. Doctrine of Ideas.

§ 26.

What is an idea? Every science presents a content of the Idea, a side of the Idea, a peculiar idea, but mediately; only philosophy sets itself to present the Idea as Idea. One says in daily speech that something is so unknown to one that one cannot even form an idea of it; one speaks of giving the idea for a building, the idea for an undertaking, and so on. A certain general-representation, a plan, a living leading thought are the categories that, according to this usage, hover before consciousness. Among the Sensualists the word “idea” occurs in a loose, superficial sense.

Although Kant grants the Idea only subjective validity, he has nonetheless most forcefully asserted its right and pointed to Plato’s merit with respect to the doctrine of Ideas (*Kritik der r. Vern.* S. 289.)

“Plato made use of the expression Idea in such a way that one sees well that he understood under it something that is not merely not borrowed from the senses, but that even far surpasses the concepts of the understanding with which Aristotle occupies himself, since in experience one meets nothing congruent with it... Plato remarked very rightly that our power of cognition feels a far higher need than to spell out phenomena according to synthetic unity in order to be able to read them as experiences, and that our reason in a natural way soars up to cognitions that go far beyond anything that any object experience can give could ever congrue with, but which nonetheless have their reality, and are by no means mere figments of the brain.” “He who would draw the concept of virtue from experience, he who would make (what indeed many have really done) that into a model and source of cognition which in each case can serve only as example, he would make of virtue an ambiguous

absurdity, changeable according to time and circumstances, usable for no rule. On the contrary, everyone notices that when someone is presented to him as a model of virtue, he nonetheless always has the true original, with which he compares this alleged model, only in his own head, and merely values it accordingly.”

“The Platonic Republic has, as an allegedly striking example of dreamt perfection that can be found only in the idle thinker’s brain, become a proverb, and Brucker finds it laughable that the philosopher maintains that a prince would never rule well if he did not become a partaker in the Ideas. But one would do better to trace this thought more closely (where the excellent man leaves us without help), and by new exertions set it in the light, than, under the very wretched and shameful pretext that it is not executable, set it aside as useless.”

With Plato, then, Kant agrees that the Idea expresses the unconditioned, that which is exalted above experience. “The concept is either an empirical or a pure concept, and the pure concept, insofar as it then has its origin in the understanding (and is not a pure sense-image), is called *Notio*. The Idea, or the concept of reason, is a concept of notions that surpasses the possibility of experience. He who has once accustomed himself to this distinction must find it intolerable to hear the representation of the red color named as Idea.”

A significant contribution to the doctrine of Ideas Kant has furthermore made in his “*Kritik der Urtheilskraft*.” If the theoretical reason has taught us to know a world in which all goes according to natural laws, if the practical reason has opened us an insight into a world of ethics, in which all is determined by free- dom, then there would be an all-too-insurmountable chasm between the realm of nature and the realm of freedom, if this chasm were not filled in by the judgment’s setting up a ground of unity for both realms. (Anf. Skr. *Einleitung*.)

“The judgment, which is the faculty of thinking the particular as included under the universal, has for its object the concept of nature’s *purposiveness*. The purpose (aim) is, namely, the supersensible unity that contains the ground of the objective’s reality; every realization of a purpose is bound up with pleasure: the judgment contains the laws for the feeling of pleasure and displeasure.”

“Nature’s purposiveness can be represented either subjectively or objectively. In the first case I sense pleasure and displeasure immediately at the representation of an object, before I have formed a concept of it; my joy then refers only to a harmonious

relation of purpose between the object's form and my power of intuition. The judgment in this subjective sense is called the *aesthetic*. In the second case I form beforehand a concept of the object, and now assess how far the object corresponds to this concept. To find a flower beautiful for my power of intuition, I need have no concept of it; but to find the flower purposive requires a concept. The judgment in this objective sense is called the *teleological*."

"The objects of nature please us not only by a subjective purposiveness; they are at the same time in a purposive relation to one another mutually. The objective, material purposiveness is of a twofold kind: the *outer* and the *inner*."

"The *outer purposiveness* is, since it designates only a thing's utility for another, something merely relative. Snow secures the seed in cold lands against frost, and eases men's communication by sleds. The sand deposited by the seashore is beneficial for fir forests. The means is in this outer purposiveness not in and for itself, not according to its concept, but only contingently purposive. The deposition of the sand on the seashore is not grasped by saying it is a means for the fir forests; it is understandable in and for itself, quite apart from the concept of purpose: the outer, relative purposiveness lets itself also be grasped out of *natural mechanism alone*."

"It is otherwise with the *inner purposiveness*, which consists in this, that something is in itself both means and end, that its end therefore does not fall outside. The organic natural product is such a one, in which each of its parts is aim and at the same time means or instrument. In propagation the natural product is produced as species; in growth the natural product is produced as individual; in formation each part of the individual produces itself. This natural organism does not let itself be explained by merely mechanical causes; it must be explained by purpose-causes, it must be explained *teleologically*."

"The earlier systems have treated the concept of natural purposiveness dogmatically; they have either ascribed (Hylozoism, Theism) or denied (Fatalism) to nature as thing in and for itself this purposiveness. But we, who hold that teleology is only a regulative principle, decide nothing concerning the question whether inner purposiveness belongs to nature in and for itself, or not. Our understanding thinks discursively, always starts from the parts, and grasps the whole as result of its parts; it can therefore not grasp the organic natural products — in which, conversely, the whole is the parts' ground of coming-to-be and prius — in any other way than under the point of view

of the concept of purpose. If, on the contrary, there were an intuitive understanding (*intellectus archetypus*) that in the universal knew the particular, in the whole the parts as already co-determined, then such an understanding would grasp the whole of nature out of one principle, without needing the concept of purpose.” As teleological principle the Idea is essentially characterized; the next step is to get this subjective Idea to hold also as objective, to transform this in Kant merely regulative principle into a constitutive one, a principle developing its content out of itself, and “the Absolute” will be reached.

How Schelling has prepared this transition Hegel acknowledges (*Geschichte der Philosophie*, 3 Th., Berlin 1844, S. 613) “In the first place, in Schelling the Idea itself is to be brought out, that he has grasped the True as the Concrete, as unity of the subjective and the objective. It is thus the main thing in the Schellingian philosophy that in it it is a matter of this deep speculative content, which, as content, is the content of which it has been a matter in the whole history of philosophy. The thinking that is freely for itself, not abstract, but in itself concrete, grasps itself in itself as the intellectual-real world; and this is nature’s truth, nature in itself. The second great thing in Schelling is to have demonstrated the forms of spirit in nature: electricity, magnetism, and so on, are to him only the Idea’s outer modes. The defect is that this Idea as such, its separation into the ideal and the natural world, and thus the totality of its determinations, are not pointed out and developed as they follow with necessity from the concept. In that Schelling has not grasped this side, thought has in him been lost; the work of art thus becomes in him the highest and only mode in which the Idea is for spirit. But the Idea’s highest mode is much rather its own element; thought, the conceived Idea, is thus higher than the work of art. The Idea is the truth and all that is true is Idea; the Idea’s systematizing into world must be proved as necessary unveiling and revelation.”

If the philosophy of the Absolute had now really mastered its task of proving how the Idea systematized itself into world, then not only would philosophy in all essentials be concluded, but with the Idea’s “unveiling” and “revelation” the riddle of existence would at the same time be solved once and for all. How little this is the case has already been shown in the preceding sections.

In his “*Kritik der Urtheilskraft*” Kant sets up an antinomy under the following form: “*Thesis*: All productions of material things are possible according to merely me-

chanical laws.”

“*Antithesis*: Some productions of the same are not possible according to merely mechanical laws.”

This antinomy is still so far from having found its solution in favor of the cognition of an objective Idea, that an interpretation of reality according to merely mechanical laws rather threatens to rob the Idea as Idea of all objective reality.

“Sound sense,” which feels itself drawn to both sides, can decide nothing; we get — herein Kant is incontrovertibly right — in this respect antinomic impressions of the objective world.

In astronomy, e.g., the motions of the heavenly bodies are indeed explained by mechanical causes; nonetheless we receive, by looking into the structure of the world, an impression as though reason itself were the master-builder who used the individual in order to realize its plans in the whole, an impression as though we saw the Idea.

The physiologist must undeniably strive, as far as possible, to explain the expressions of life from the standpoint of “natural mechanism”; nonetheless “sound sense” cannot withdraw itself from the impression of the purpose for whose sake the organs are formed. It looks as though the fundamental parts with their unchangeable character were the real first, and life itself a blind product of these parts’ law-governed reciprocity; and it looks again as though life itself were the proper prius, and the parts came into being only to serve the whole; the whole looks like a realized Idea.

We look at the individual members, and we lose the Idea; we look at the inner whole, and the Idea stands living before us. We dissect a plant, and we lose the Idea; we let the plant, as it lives and grows, make its typical impression, and we meet the Idea. We analyze the proportions of measure in one of Raphael’s pictures mathematically, and we lose the Idea; we let the mathematical vanish into the aesthetic total impression, and we win the Idea.

This antinomic is now so much the harder as we, according to what has here already been shown, can neither with Kant take refuge in the merely subjective, nor with Hegel deny the real in order to think the objective. The problems belonging here will in this connection require a peculiar treatment of *the Idea in its Essentiality*, of *Reality against the Idea*, and of *the Validity of the Idea*.

A.

The Idea in its Essentiality.

§ 27.

“The most heterogeneous things Plato seems to gather under the name Idea. He refers thereto ideal (aesthetic and moral) concepts, e.g., the Idea of the Good, the Idea of Injustice, pure thoughts, such as the thought of unity and plurality, concepts of living species (e.g., the Idea of a bull), of individuals, of their states and expressions of life, the Idea of voice, of color. Indeed, he speaks of the Idea of human manufactures, of the Idea of a bed, of a table. No object of cognition seems then to be excluded from participation in the Ideas, and thus it really stands with it. All that has reality, existence, can be thought in such a way that it is referred under an Idea. Therefore Parmenides corrects the young Socrates’s opinion when he hesitates to know Ideas in things that are regarded as low and insignificant, e.g., hair or filth, since such things are low and insignificant only from a restricted human standpoint.” (Poul Møller).

That Plato’s doctrine of Ideas stands in intimate connection with his doctrine of knowledge and the essence of things is incontrovertible, and often enough brought out. A reference to the demand of knowledge and science in general is, on the contrary, insufficient, since there can undeniably be science and scientific penetration into the essence of things without Plato’s Ideas being for that laid at the foundation. It must expressly be brought out that it is only the Platonic science that is inseparable from the Platonic Idea, and that Plato himself gave this science (ἡ περὶ τὸ ὄν καὶ τὸ ὄντως καὶ τὸ κατὰ ταυτὸν ἀεὶ πεφυκὸς ἐπιστήμη [the science concerning being and the truly-existent and that which is by nature always self-same]) the name: Dialectic. One cannot therefore treat Dialectic for itself, and the doctrine of Ideas for itself; they relate as form and content: the Platonic Dialectic and the Platonic Idea are, essentially seen, two sides of the same.

1) *Ideas and concepts.*

Although Dialectic sees the essential, the universally valid, the True only in the concept (οὗ αὐτὸς ὁ λόγος ἄπτεται τῇ τοῦ διαλέγεσθαι δυνάμει [which reason itself grasps by the power of dialectic]), one cannot nonetheless say that Dialectic

occupies itself only with concepts, and not, on the contrary, with representations, sense-impressions, or sensible things. If Dialectic held only to the concepts insofar as these arose by removal of finite representations, i.e. by abstraction, the Idea in the objective, scientific sense would itself be an abstraction.

Here is a thoroughgoing difference between the concepts of abstraction and the dialectical realities Plato calls Ideas. The abstracting thought lets the finite, the sensibly individual, remain standing as the in-itself-independent valid common mark, the empty, dependent universal-form. In this sense it is then not the bull's concept, but the bull-individual, that is the real. For the dialecticizing thought the finite in its finitude has no reality. Instead of abstracting from the representation, and letting the finite stand as the properly existing, Dialectic engages with the representation, and shows how the finite, the sensible in its becoming and changeability, is an in-itself-untrue, a fleeting, a self-consuming, a semblance whose nothingness thought sees through. The more the finite things show themselves in their untruth, the more the concept shows itself as the True, as the unchangeable in the changes, as the self-valid that bears and determines all finite existence, i.e. as Idea. Only when Dialectic has consumed the bull-individual's immediate, finite reality is the concept itself seen as the producing principle, as the eternal reality, as "the bull's Idea."

Of the sensibly becoming there is no knowledge (τῶν ῥεόντων οὐκ ἔστιν ἐπιστήμη [of things in flux there is no science]), consequently the Idea must show itself as the being; but the being in its abstract, differenceless unity is in infinite contradiction with itself, thus the Idea's being must have in itself an essential difference, take on movement, and show itself as a plurality of beings. The former is established against the Heraclitean (Theaetetus), the latter against the Eleatic (Sophist) one-sidedness.

Attaching himself to the Heraclitean doctrine of becoming, the Sophist Protagoras had set up the proposition that man is the measure of all things (πάντων χρημάτων μέτρον ἄνθρωπος, τῶν μὲν ὄντων ὡς ἔστι, τῶν δὲ οὐκ ὄντων ὡς οὐκ ἔστιν [man is the measure of all things, of the things that are that they are, and of the things that are not that they are not]). To becoming corresponds sensation; when one feels it warm, it is warm for him; if another finds it cold, it is cold for him. Sense-sensation is the only mark of truth. This view is described by the Platonic Socrates in the dialogue Theaetetus. If man merely as sensing individual were the measure of things, then an animal could just as fully as the wisest man, indeed as a god, know the True

(ὅτι πάντων χρημάτων μέτρον ἐστὶν ὅς ἢ κυνοκέφαλος ἢ τι ἄλλο ἀτοπώτερον τῶν ἔχοντων αἴσθησιν [that the measure of all things is a pig or a dog-faced baboon or some other still more absurd creature that has sensation]). If all men have equal insight into the True, then it is a contradiction to give oneself out as a teacher of wisdom (σοφίζειν); Protagoras must indeed grant everyone right who gives him wrong. He who assumes that not every man is the measure of things, and that sense-sensation is not science, is just as much right as he who assumes the opposite; thus it is, according to Protagoras's own doctrine, also true that his doctrine is not true. Protagoras makes every knowledge of the future, and therewith of the useful, impossible (ἔστι δέ που καὶ περὶ τὸν μέλλοντα χρόνον τὸ ὠφέλιμον [and the useful, surely, also concerns future time]). Even if one assumed that every judgment about the present were equally correct, one could nonetheless not assume that every judgment about the future were equally correct, that what everyone with respect to agriculture or the healing of an illness happened to assume to be useful would also really show itself to be the useful. How absurd it is to regard sense-sensation as science is finally seen from this, that the infinite becoming must make every judgment impossible. The sensing one and the sensed are both in motion. Sensation itself is precisely the product of these two motions. The becoming is, and is not, just as the sensation is and is not. Becoming is of a twofold kind: change of place and change of quality (ἀμφοτέρως τὰ πάντα κινεῖται, φερόμενόν τε καὶ ἀλλοιούμενον [all things move in both ways, both moving in place and altering in quality]). In both respects the sensed can be said to be and not to be. In saying: it is such, one can at once add: it is not such, and not not-such (οὔτε ἄρα ὁρᾶν προσρητέον τι μᾶλλον ἢ μὴ ὁρᾶν, οὔτε τιν' ἄλλην αἴσθησιν μᾶλλον ἢ μὴ, πάντων γε πάντως κινουμένων [one must therefore call it no more seeing than not-seeing, nor any other sensation more than not, since all things are in every way in motion]). What is stated of the sensation is thus, in the same moment as it is stated, untrue, and the alleged science consequently not science (οὐδὲν ἄρα ἐπιστήμην μᾶλλον ἢ μὴ ἐπιστήμην ἀπεκρινάμεθα ἐρωτώμενοι ὃ τί ἐστὶν ἐπιστήμη [so, being asked what knowledge is, we answered nothing more knowledge than not-knowledge]). By combating the sophistical Sensualism Plato approaches the Eleatics; but in the way in which the Eleatic proposition — *only the being is, the not-being is not* — was accented, the Platonic Dialectic likewise discovers contradictions, and corrects the Eleatics in the dialogue that is named the Sophist. What is one to judge of not-being? If the

not-being is not at all, then it can neither be thought nor stated; if the not-being is a semblance, then this semblance must nonetheless be. This is precisely the difficulty in thinking not-being, that both he who denies it and he who defends it is forced to contradict himself. If one grants that there exists a false opinion (ὄντος δέ γε ψεύδους ἔστιν ἀπάτη [since, indeed, there is falsehood, there is deception]), then one at the same time presupposes the representation of not-being; for only that opinion can be called false which either declares the not-being to be being, or the being to be not-being. (κινδυνεύει τοιαύτην τινὰ πεπλέχθαι συμπλοκὴν τὸ μὴ ὄν τῷ ὄντι [it seems that not-being has been interwoven with being in some such interweaving]). If one holds the One for itself absolutely, it is impossible to say anything about it. He who says that only the One is has already two names: the One and being; but if there is here a difference between the One and its being, then here in this difference there is at the same time a principle of plurality (τό τε δύο ὀνόματα ὁμολογεῖν εἶναι μηδὲν θέμενον πλὴν ἓν καταγέλαστόν που [to admit two names when one has posited nothing but the One is surely ridiculous]). That becoming is a semblance cannot be understood as though the being were to exclude all motion. If the essence of things were wholly immovable, one could ascribe to it neither life nor reason; it could then not be known either. The known and the knowing must stand in reciprocal relation to each other; the essence of things that is to be known must consequently be affected by the knower, and thus come into motion (μανθάνω, τόδε γε, ὥς τὸ γινώσκειν εἴπερ ἔσται, ποιεῖν τι, τὸ γινωσκόμενον ἀναγκαῖον αὖ ξυμβαίνει πάσχειν [I understand this at least, that if knowing is to be a doing something, then the known must correspondingly be affected]). That the being must be able to move is seen at the same time by a consideration of the nature of speech, since speech consists precisely in this, that one in propositions combines substantives (ὀνόματα [nouns]) with verbs (ῥήματα [verbs]), i.e. the abiding with the changing (Δηλοῖ γὰρ ἤδη που τότε περὶ τῶν ὄντων ἢ γιγνομένων ἢ γεγονότων ἢ μελλόντων, καὶ οὐκ ὀνομάζει μόνον, ἀλλὰ τι περαίνει, συμπλέκων τὰ ῥήματα τοῖς ὀνόμασι [for it then makes clear something about things that are or are coming to be or have come to be or will be, and does not merely name, but accomplishes something, weaving the verbs together with the nouns]). The two concepts being and not-being are both difficult to grasp, but for wholly opposite reasons. Not-being, with which the Sophist prefers to occupy himself, cannot be seen for sheer darkness, and being, philosophy's object, is sought (διὰ λογισμῶν [through

reasonings]) in those regions of light where the eye is dazzled. What not-being is can, however, be known from the mutual relation of concepts. Some concepts, namely, can be combined either wholly or in part with each other, while others exclude each other. As for the general concepts being, rest, and motion, we see that rest and motion do indeed mutually exclude each other, but that they nonetheless both, each for itself, let themselves be united with being. The being of rest is the not-being of motion, and conversely; what here, then, in relation to itself, is a being, is, in relation to its other, a not-being (τί ποτ' αὖ νῦν οὕτως εἰρήκαμεν τό τε ταὐτὸν καὶ θάτερον [what, then, have we now thus called the same and the other]): being and not-being unite and determine each other mutually."

b) The Ideas themselves and the sensible things.

In that being, according to the concept, presupposes a not-being, the true original essence must develop itself in a system of essences and essence-determinations. What was here with respect to this point prepared in "Theaetetus" and "the Sophist" is now further carried through in "Parmenides," "Philebus," "the Republic," and "Timaeus."

In "Parmenides" the investigation is introduced with the remark, set forth by Socrates, that the being, if it is a plurality, must at once be a like and an unlike. But in that case the like would have to be an unlike, and the unlike a like; which is absurd (οὔτε γὰρ τὰ ἀνόμοια ὅμοια οὔτε τὰ ὅμοια ἀνόμοια οἷόν τε εἶναι [for neither can the unlike be like nor the like unlike]). There must then be a likeness for itself, an unlikeness for itself, a greatness for itself, a smallness for itself, and so on. The Ideas must have a being in itself unconditioned (αὐτὰ καθ' αὐτά [themselves by themselves]), so that the essence and properties of the finite things can be determined by a participation therein (οὐ νομίζεις εἶναι αὐτὸ καθ' αὐτὸ εἶδος τι ὁμοιότητος καὶ τῷ τοιούτῳ αὖ ἄλλο τι ἐναντίον, ὃ ἔστιν ἀνόμοιον; τούτοις δὲ δυοῖν ὄντοις καὶ ἐμὲ καὶ σὲ καὶ τὰ ἄλλα, ἃ δὴ πολλὰ καλοῦμεν, μεταλαμβάνειν [do you not hold that there is a certain form of likeness itself by itself, and again another contrary to such a one, which is unlikeness; and that, these being two, both I and you and the other things, which we call many, partake of them]).

By participating in likeness the things must then become like, by participating in unlikeness unlike. That the things, by participating in both concepts, become mutually like and unlike, is not astonishing; but if anyone could prove that the like in itself were the unlike, and the unlike the like, that would be astonishing. That all is one, insofar

as it participates in the One, and at the same time a manifold, insofar as it participates in plurality, is not to be wondered at; but if anyone could prove that the One precisely in itself is a manifold, and the manifold one, then one would have to wonder greatly.

But that which participates in the Idea must participate either in the whole Idea, or in a part of it. That the whole Idea should be whole and undivided in each of the many objects participating therein is unthinkable. One can spread a sheet over several people, but each single one then comes into contact only with a part, not with the whole; each object thus gets only a part of the Idea, not the Idea in its wholeness. But by being distributed among the objects the one Idea would fall away from itself, and be parceled into plurality, which is likewise absurd. To wish, e.g., to divide greatness itself, and to suppose that each of the many great things should be great by a part of greatness which is smaller than the great, is foolish. But when the objects can get neither a part of the Idea, nor the whole Idea (μήτε κατὰ μέρη μήτε κατὰ ὅλα τῶν εἰδῶν μεταλαμβάνειν δυνάμενα [being able to partake of the forms neither part by part nor as wholes]), then they can not participate in the Idea at all.

Perhaps the Idea is only a subjective concept, a thought in the soul? If the Ideas are only subjective concepts (νοήματα [thoughts]), and thought must necessarily have its object, then one must either assume that each object consists of thoughts and that all things think, or else that there are indeed thoughts, but no thinking (ἀνάγκη, εἰ τὰλλα φῆς τῶν εἰδῶν μετέχειν, ἢ δοκεῖν σοι ἐκ νοημάτων ἕκαστον εἶναι καὶ πάντα νοεῖν, ἢ νοήματα ὄντα ἀνόητα εἶναι [it is necessary, if you say the other things partake of the forms, either that it seem to you each is composed of thoughts and all things think, or that, being thoughts, they are unthinking]). Since now this assumption is absurd, the Ideas must be nature's prototypes, and the objects that resemble them their images (τὰ μὲν εἶδη ταῦτα ὥσπερ παραδείγματα ἐστάναι ἐν τῇ φύσει, τὰ δὲ ἄλλα τούτοις εἰκέναι καὶ εἶναι ὁμοιώματα [these forms stand as it were as paradigms in nature, and the other things resemble them and are likenesses]).

The concept of participation is then hereby transformed into the concept of resembling something, of taking on likeness with something (ἢ μέθεξις αὕτη τοῖς ἄλλοις γίνεσθαι τῶν εἰδῶν οὐκ ἄλλη τις ἢ εἰκασθῆναι αὐτοῖς [this participation of the other things in the forms is nothing other than being made like them]). Yet neither does this turn satisfy; the like must indeed be like its like and participate in the same Idea. But that by which the like becomes a like is precisely the Idea itself. Should something

now be like the Idea, or the Idea like something, then beside the Idea there would have to appear another, likeness-effecting Idea; and, when this has likeness with something, again another Idea, and so on to infinity. From whatever side the Idea and its other are touched, the contradiction breaks forth; it is then of importance to investigate the Idea-essence, namely with respect to the One (τὸ ἓν [the One]) and the manifold (τὰ πολλά [the many]).

α) If it is assumed that the One is (εἰ ἓν ἔστιν [if the One is]) in sharp separation from all possible plurality, then this One cannot in the first place be a whole; for, if it were a whole, it would have to have parts, and thereby come to consist in the plurality of the parts (Ἀμφοτέρως ἄρα τὸ ἓν ἐκ μερῶν ἂν εἴη, ὅλον τε ὄν καὶ μέρη ἔχον. Ἀμφοτέρως ἂν ἄρα οὕτω τὸ ἓν πολλά εἴη, ἀλλ' οὐχ ἓν [in both ways, then, the One would be of parts, both being a whole and having parts. In both ways, then, the One would thus be many, and not one]). If the One has no parts, then it has neither beginning (ἀρχή [beginning]) nor end (τελευτή [end]) nor middle (μέσον [middle]), for such moments would then be its parts. But beginning and end are indeed the limits (πέρας [limit]) of every thing; if the One has neither beginning nor end, it must be unlimited (ἄπειρον [unlimited]), and without shape (ἄνευ σχήματος [without shape]). It can then be a partaker neither in the round nor in the straight, can be surrounded neither by another nor by itself, can be neither in motion nor in rest, can in its infinite self-contradiction tolerate no predicate whatever, can, in one word, not be at all.

β) If it is nonetheless assumed that the One is, but in such a way that the difference between the One itself and its being is brought out (ἄλλο τι σημαῖνον τὸ ἔστι τοῦ ἓν [“is” signifying something other than “the One”]); then the being One is the whole, and the One itself and its being the two parts of the whole. But in neither of these two parts does the One let itself be torn loose from the being, or the being from the One. Each part then comes again to consist of two parts, and each of these again of two; so that the being One becomes an unlimited plurality (ἄπειρον ἂν τὸ πλῆθος οὕτω τὸ ἓν ὄν εἴη [the One that is would thus be unlimited in multitude]). With the difference between the One and its being there now arises a series of different, mutually conflicting predicates. The being One now becomes at once both limited and unlimited; with respect to likeness it becomes both like and unlike itself and its other; with respect to magnitude it becomes both greater and smaller than itself and its other, and, with respect to time, both older and younger than itself and its other,

and yet again neither older nor younger either than itself or the other. One can now state everything possible of the One; which then can also signify that this One, by being all in all, shows itself as the essence of the manifold, as the aggregate of all realities, as the primal ground of existence.

γ) If, finally, it is assumed that the One is not (εἰ ἐν μὴ ἔστιν; [if the One is not]), yet in such a way that this not-being of the One can be thought, stated, and determined by opposition to the Not-One, then the Not-One, the One's other, will in all its indeterminacy at the same time become more closely determined. Although the One indeed cannot be, insofar as it is a not-being, it can nonetheless participate in many properties (μετέχειν δὲ πολλῶν οὐδὲν κωλύει, ἀλλὰ καὶ ἀνάγκη [nothing prevents it from partaking of many things, but it is even necessary]). By reason of the unlikeness with its other, the One must namely be like itself; but, in that the other is posited as unlike the One, this at the same time comes to participate in unlikeness, and thereby becomes at the same time unlike itself. Indeed, the One must even, since it is not, participate in being. What is stated of the One must indeed be true; now here it is stated: the One is a not-being. If the One therefore gave off something of this its being, it would to a lesser degree be a not-being, and precisely thereby become a being (εἰ γὰρ μὴ ἔσται μὴ ὄν, ἀλλὰ τι τοῦ εἶναι ἀνήσει πρὸς τὸ μὴ εἶναι, εὐθὺς ἔσται ὄν [for if it will not be not-being, but relaxes something of being toward not-being, it will at once be being]). Participating both in being and in not-being, the One at the same time participates in change, motion, and so on. If one now reflects on the being other, this can not possibly be an other over against the One, since this One is not (τοῦ μὲν γὰρ ἑνὸς οὐκ ἔσται ἄλλα, μὴ ὄντος γε [for there will be no others of the One, since it is not]); it must then be an other over against another being other (ἕτερον δὲ γὰρ ποῦ φάμεν τὸ ἕτερον εἶναι ἑτέρου, καὶ τὸ ἄλλο δὴ ἄλλο εἶναι ἄλλου [we say, surely, that the other is other than an other, and the other is other than an other]): the otherness expresses that the being, in infinite self-separation, relates to itself by incessantly falling away from itself. The other is then not a being, but a something changing itself to infinity, a mass unlimited in the quantity of its parts (ὁ ὄγκος αὐτῶν ἀπειρὸς ἐστὶ πλήθει [their mass is unlimited in multitude]). In this mass each smallest part for itself may indeed seem to be a one; but it is as in a dream-vision (ὥσπερ ὄναρ ἐν ὕπνῳ [as a dream in sleep]): suddenly, instead of the apparent one, the manifold shows itself again, and, instead of a least, an infinitely great in relation to the continued parceling.

With the development of otherness in “Parmenides” the element of sensuousness is already intimated. This moment is especially brought out in “Philebus,” which distinguishes between three kinds of being, namely the limit (πέρας [limit]), the unlimited (ἄπειρον [unlimited]), and that composed of both (τὸ ἐξ ἀμφοῖν τούτων ἐν τι ξυμμισγόμενον [something one, mixed together out of both of these]), to which is still added a fourth principle, namely the cause of the mixture (αἰτία [cause]); likewise in “Timaeus,” where a distinction is made between that which truly is, which does not become, and that which always becomes, which has no true being (τὸ ὄν αἰεὶ, γένεσιν δὲ οὐκ ἔχον, καὶ τὸ γιγνόμενον μὲν αἰεὶ, ὄν δὲ οὐδέποτε [that which always is, but has no becoming, and that which always becomes, but never is]). The union of both points to a third, as that which takes up all becoming and all forms in its bosom, an invisible nurse of things (πάσης γενέσεως τιθήνη [the nurse of all becoming]), which in an inconceivable way has become a partaker in the rational (ἀνόρατον εἶδος τι καὶ ἄμορφον, πανδεχές, μεταλαμβάνον δὲ ἀπορώτατά πη τοῦ νοητοῦ [a certain invisible and formless kind, all-receptive, partaking in some most perplexing way of the intelligible]), the mass (ἐκμαγεῖον [matrix]) out of which all the sense-phenomena are formed. From matter, the not-being, which can neither be sensed nor thought, but is grasped only by a kind of spurious inference (μετ’ ἀναισθησίας ἀπτόν λογισμῶ τινι νόθῳ [graspable without sensation by a kind of bastard reasoning]), the things have their sensible character; from the Ideas they have their fundamental form and reality; as the Ideas’ imperfect images in the sensible, the things are intermediate links between being and not-being.

c) The mutual relation of the Ideas.

It is so far from the case that Plato, though he otherwise on the whole stands near the Eleatics, remains standing at being’s unity in the Idea, that he, as has also with right been remarked (Ritter), seldom names the Idea in the singular (εἶδος or ἰδέα, in whose place he then rather uses other expressions, such as: τὸ ὄντως ὄν [the truly-existent], τὸ κατὰ ταῦτα ἔχον [that which holds according to the same], ἡ οὐσία [being], and so on), but almost always uses the plural, and speaks of the Ideas.

How the Eleatic One here with inner necessity separates itself into ideal plurality is evident from the foregoing. In that each concept, so that the contradiction can be avoided, is to be held fast for itself — likeness for itself, unlikeness for itself, motion for itself, rest for itself, and so on — there arises an infinite manifold of substantial

units (ένάδες, μονάδες [henads, monads]) which, although each of them with respect to the changing phenomena is a universal (τὸ ἐπὶ πᾶσι κοινόν [that which is common to all]), and has a being independent of the sensible things (χωρίς ἐστὶ [is separate]), nonetheless mutually exclude each other, and cannot possibly go up as moments in a single, absolute unity.

Since, however, it is not the special but the universal concepts that are ascribed highest validity, it is at the same time seen that a system of superordinate and subordinate concepts must be able to be formed. It is the dialectician's task that he, with respect to each object in question, is to seek first the most universal concept, then divide it into its most nearly subordinate ones, and these in turn into their subordinate ones, until he has exhausted the whole number of concepts that lie in the middle between the One and the Unlimited. "Since the whole of nature is akin, nothing prevents one from starting from one thing and finding all, if only he is brave, and does not tire of seeking." (Meno). The graded series of the Ideas must then find its conclusion in a highest Idea, an Idea that, without supporting itself on any higher presupposition, grounds itself and all (τὸ ἰκανόν [the sufficient]). "In the sphere of cognition the Idea of the Good (ἡ τοῦ ἀγαθοῦ ἰδέα [the Idea of the Good]) is the highest, which is beheld only with difficulty, but which one cannot behold without at the same time having to know that it is for all the cause of everything right and beautiful, that in the visible world it has created the light and its lord the sun, and in the spiritual world, where it alone rules, has brought forth truth and reason, and that therefore everyone ought to have his gaze directed to it." (Republic).

The substantial discretion of the Ideas would, however, make a systematic derivation difficult to such a degree that one even understands from this what Aristotle brings out, that the Platonic doctrine of Ideas at last ended by passing over into a kind of Pythagorean number-doctrine. In the dialogues there is indeed nothing that expressly shows this. A distinction is indeed here made between pure and applied mathematics (ἀριθμοὶ μαθηματικοί ὁγ αἰσθητοί [mathematical and sensible numbers]), and likewise (Republic) a distinction between mathematical magnitudes and original number-concepts, but only in the same sense as when a distinction is otherwise made between the thing and the Idea, so that the Ideas of numbers also belong in the circle of the Ideas; on the contrary, there is here no thought that the Ideas on the whole should be represented by numbers. Nonetheless the transition is (Zeller)

explicable.

Since Plato, as especially “Timaeus” shows, intuitively the Ideas in the corporeal world under the form of mathematical determinations of relation, and comes further away from the dialectical, according as he goes more closely into the concrete, there can be contained herein a hint of how he, when the stricter Dialectic would not suffice, has been able to seize on the expedient of paralleling the Ideas with the numbers. The Ideas are then grasped as primal numbers or ideal numbers (ἀριθμοὶ νοητοί, εἰδητικοί, πρῶτοι [intelligible, ideal, first numbers]), and these primal numbers separated from the mathematical ones (συμβλητοί [combinable]) in that they are principles, and (ἀσύμβλητοι [non-combinable]) cannot, like the ordinary numbers, be added together, or flow together into larger, arbitrary units.

§ 28.

The weakness of the Platonic doctrine of Ideas becomes manifest from the critique Aristotle applies in order to ground his standpoint, and carry the development further. That knowledge presupposes the in-itself universally valid (τὸ ἐπίστασθαι πῶς ἔσται, εἰ μὴ τι ἔσται ἐν ἐπὶ πάντων [how will there be knowing, if there is not some one over all]), that the essence, not the sensibly becoming, is the True, about this Aristotle is agreed with Plato; but by the peculiar emphasis he lays on the single essence, the existing, the real concretion, he diverges from Plato. As little as the thing, separated from its concept, has essential form, so little has the concept, separated from the thing, real existence. With this stands another divergence in close connection. Whereas Plato sets Dialectic as knowledge’s highest form, as expression of the proper science, Aristotle, on the contrary, prefers the apodictic adducing of proof, and assigns the dialectical a rather subordinate place. In the so-called aporias (ἀπορία [aporia]) the dogmatic decision is introduced by dialectical considerations (ἔστι δὲ τοῖς εὐπορῆσαι βουλομένοις προὔργου τὸ διαπορῆσαι καλῶς [for those wishing to resolve difficulties it is advantageous to work through the perplexities well]). Dialectic is hereby reduced to a reasoning for and against, a supplement to the epagogic adducing of proof, an indication of the probable (διαλεκτικὸς δὲ συλλογισμὸς ὁ ἐξ ἐνδόξων συλλογιζόμενος [a dialectical syllogism is one that reasons from reputable opinions]). With such a reduction of the dialectical a deeper understanding of the Idea’s essence is doubtless impossible; but even under the presupposition of some misunderstanding, a critique

which with polemical acuteness lays bare the weak sides of the doctrine of Ideas, and thereby prepares a penetrating cognition of the reality of the single essences, must be regarded as an indispensable contribution to the further advancement of the matter (Μετὰ τὰ φυσικά, πρώτη φιλοσοφία [Metaphysics, first philosophy]).

a) Aristotle's critique of the doctrine of Ideas.

Although the Ideas, according to Plato's and the Platonists' assumption, are to be independent, for themselves subsisting unities, they nonetheless have no peculiar content. The ideal two and the empirical two have the same content. One speaks of man-itself, of horse-itself, of health-itself (αὐτὸ γὰρ ἄνθρωπον φασιν εἶναι καὶ ἵππον καὶ ὑγίειαν [for they say there is man-itself and horse and health]); but by setting "itself" before these head-words, nothing is won. The real horse has essentially the same conceptual determinations as the horse-itself; to define the real man is the same as to define man-itself (ποιοῦσιν αὐτοἰάνθρωπον καὶ αὐτοῖππον, προστιθέντες τοῖς αἰσθητοῖς τὸ ῥῆμα τὸ αὐτό [they make man-itself and horse-itself, adding to the sensibles the word "itself"]). Aristotle compares the Ideas thus made independent with the folk-gods; just as these are nothing other than mortal men whom one has in fantasy made into eternal beings, so the Ideas too are nothing other than perishable things that one has intuited in the form of the eternal and supersensible (οὔτε γὰρ ἐκεῖνοι, i.e. those who assume the folk-gods, οὐθὲν ἄλλο ἐποίουν, ἢ ἀνθρώπους αἰδίους, οὔθ' οὔτοι τα εἶδη ἄλλ' ἢ αἰσθητὰ αἰδία [neither did those make anything other than eternal men, nor did these make the forms other than eternal sensibles]). The "homonymy" (ὁμωνυμία [homonymy]) occurring between the Ideas and the single existences in question occasions the doubling.

When, then, the adherents of the doctrine of Ideas assume an Idea for each thing (εἶδη ἐστὶν ὅποσα φύσει [there are forms of as many things as exist by nature]), and thus double the number of substances, they appear to Aristotle like men "who supposed they could not reckon so well with few as with many numbers, and who therefore first multiplied the numbers before they began with their operations of calculation." Such a doubling of substance is idle. Why set the same thing twice? To what end, besides the sensible two and three, a two and three in the Idea? (ἄτοπος σώρευσις [absurd heaping-up]). The substance cannot possibly be separated from that of which it is the substance (δόξειεν ἂν ἀδύνατον εἶναι χωρὶς τὴν οὐσίαν καὶ οὗ ἢ οὐσία [it would seem impossible for the substance to be separate from that of

which it is the substance]). If the essence-determination is once and for all made into something for itself, then one gets not merely an Idea for each substance, but even for each accident; indeed the Ideas, whose meaning it is to be in the things, as in their other, become precisely thereby something accidental, for only the accidental can be in an other.

That the Ideas should constitute the essence of the corporeal things, and yet be incorporeal, for themselves existing substances, is a contradiction; partly, says Aristotle, Plato also speaks of an “Ideas’ matter,” which straightforwardly conflicts with the assumption that they should be outside space; partly in all natural objects matter and becoming belong to the objects’ essence and concept, and consequently this concept cannot be thought as a for-itself existing. One does not see how, and why, matter participates in the Ideas. To make this understandable, one would have to assume, besides the Ideas, a still higher principle as cause of this “participation” of things. The Ideas contain no principle for becoming and motion; but without a principle for becoming and motion it is not possible to explain to oneself what the things’ “participation” in the Ideas is to signify.

The assumption that the Ideas act as prototypes, and that the things in this respect have part in them, is an empty representation (τὸ δὲ λέγειν παραδείγματα αὐτὰ εἶναι καὶ μετέχειν αὐτῶν τᾶλλα, κενολογεῖν ἐστὶ καὶ μεταφορὰς λέγειν ποιητικὰς [to say that they are paradigms and that the other things partake of them is to speak emptily and to utter poetic metaphors]). Were there, e.g., a prototype for Socrates, then in this there would again have to be an infinite multitude of prototypes, a prototype for the animal, a prototype for the two-footed, and so on. The distance between the universal man-itself and the empirical single man is besides so great that here, as connecting link, an intermediate man is required (ὁ τρίτος ἄνθρωπος [the third man]).

Insofar as the Idea is to be the unconditionally single, it is not possible to define it; for the absolutely single does not let itself be defined.

If the Ideas are grasped as primal numbers, how can they then be causes of the sensible things (πῶς αἴτιοι ἔσονται; [how will they be causes])? The sensible things cannot possibly be given out for numbers, for the eternal ideal numbers cannot indeed produce perishable things. If it is the number-relations on which the emphasis is laid, then these, like the things themselves, presuppose a substratum whose relations they determine (δῆλον ὅτι ἔστιν ἓν γέ τι ὃν εἰσὶ λόγοι [it is clear that they are ratios of

some one thing that exists]). Out of several numbers a new number is formed, but how can out of several Ideas a new Idea be formed? Or is it perhaps not the numbers, but the units of which they consist, that constitute the essence of things; but then the question again arises how these units relate to each other. The triad (τριάς [triad]) is, e.g., man's number: what relation is there now between the units in this triad and the units in any other triad whatever?

It will already from these few traits be clear that the Aristotelian critique, with all the warrant it may otherwise have, does not properly set itself to lead us into Plato's original train of thought, but rather — this is precisely the uncritical thing in this critique — takes all that is had, loose and fixed, from Plato himself or his adherents, as given, and then sets forth the objections, partly in gathered, partly in scattered remarks. Only insofar as one looks to what Aristotle himself builds up, in tearing down the doctrine of Ideas, does the critique get its proper significance.

b) Aristotelian principles.

Substance, the original being (τὸ πρῶτως ὄν καὶ οὐ τι ὄν ἀλλ' ὄν ἀπλῶς ἢ οὐσία ἂν εἴη [the primarily existent — not a something-that-is, but simply being — would be substance]), is always a definite this (τόδε τι [a this]), a peculiar subject (πᾶσα οὐσία δοκεῖ τόδε τι σημαίνειν [every substance seems to signify a this]); substance in the proper sense (ἡ πρώτη οὐσία [the primary substance]) is individual; in a subordinate sense the genus-concept is substance (ἡ δευτέρα οὐσία [the secondary substance]); the other universal concepts are substance's attributes or accidents (μὴ οὐσῶν οὖν τῶν πρῶτων οὐσιῶν ἀδύνατον τῶν ἄλλων τι εἶναι [if the primary substances did not exist, it would be impossible for any of the others to exist]).

In the sensible single essence, matter (ὕλη [matter]) and form (μορφή, εἶδος [shape, form]) are distinguished. The way in which the reciprocal relation of matter and form is determined shows the difference between Plato and Aristotle. In Plato matter, as the infinite otherness, is the not-being, and form, as Idea, a for-itself-being; in Aristotle matter and form are two sides of the same. In its abstraction from form, matter is the indeterminate, the predicateless, that is not to be known (ἡ ὕλη ἄγνωστος καθ' αὐτήν [matter is unknowable in itself]). As the abiding that lies at the ground of all becoming, matter takes on the most diverse forms, but is itself in reality nothing. When there is talk of a determinate, a such (ἐκείνιν [a "made-of-that"]), every matter has a character; but universal matter is without any peculiar character. Matter is on the one

side opposed to form, on the other side one with form. As one with form, matter can pass over into form, lie at the ground of form, and give it content; as opposed to form, matter (στέρησις [privation]) hinders form from coming to complete actuality.

As the wood relates to the bench and the ore to the statue, so the first matter relates to all that is determinate. The Eleatics denied becoming, the Megarians denied possibility (ὅταν ἐνεργῇ, μόνον δύνασθαι. ὅταν δὲ μὴ ἐνεργῇ, μὴ δύνασθαι [that a thing can only when it is acting, and cannot when it is not acting]): with his concept of matter Aristotle has explained becoming, and asserted possibility. Out of the absolutely not-being nothing can become: out of nothing comes nothing; but out of the according-to-actuality not-being, i.e. out of the according-to-possibility being, i.e. out of matter, something can become. As form designates the actual (τὸ ἐνεργείᾳ, ὄν [that which is in actuality]), matter designates the possible (τὸ δυνάμει ὄν [that which is in potentiality]). By the actual Aristotle then understands the being developed to totality, the essence insofar as it has worked out all its determinations to existence; by the possible, the essence as the pure disposition, the in-itself-undeveloped, which can, but therefore not necessarily must, become a peculiarly determinate being (λέγομεν δὲ δυνάμει οἷον ἐν τῷ ξύλῳ Ἑρμῆν [we say “in potentiality” as, e.g., the Hermes in the wood]).

Just as Aristotle has determined the possible, he has also in corresponding manner determined the contingent (συμβεβηκός, τὸ ἀπὸ τύχης [the accidental, that which is from chance]). The objective ground of the contingent consists in this, that all that is not the completed actuality has in itself the possibility both of being and of not-being, or, in other words, that matter, as the indeterminate, makes possible opposite determinations (ἀόριστα μὲν οὖν τὰ αἷτια ἀνάγκη εἶναι, ἀφ’ ὧν γένοιτο τὸ ἀπὸ τύχης [the causes from which the chance-result comes must be indeterminate]). What nature produces in such a way that it does not let itself be determined beforehand is a contingent, which precisely for that reason also approaches the not-being (φαίνεται γὰρ τὸ συμβεβηκός ἐγγύς τι τοῦ μὴ ὄντος [for the accidental appears to be something close to not-being]).

Matter’s relation to form shows itself as a relation of potentiality to actuality (the Scholastics’ *esse potentia* [being in potency] and *esse actu* [being in act]). Form, which makes the indeterminate, differenceless material into a differentiated, a this (τόδε τι [a this]), an actual, is then the peculiar virtue, the completed activity, the soul in every

thing. “That which Aristotle calls form may therefore not be confounded with what we call shape [Façon]. A severed hand, e.g., still has the outer figure of a hand; on the Aristotelian conception, on the contrary, it is a hand only with respect to the material, not with respect to the form: a real hand, a hand with respect to the form, is only that which can carry out a hand’s peculiar work. Pure form is that which in truth is without matter (τὸ τί ἦν εἶναι [the essence]), or the concept of the essence, the pure concept. But such a pure form does not exist within the bound of determinate being. All that is a this is, on the contrary, a composite of material and form, a σύνολον [composite whole]. The individual is the harder to know, the more it bears the material in itself. But from what has here been said it follows at the same time that the opposition between matter and form is a fluid one; what in one respect is material is in another respect form: building-timber is, in relation to the finished house, material, in relation to the unhewn wood, form; the soul is, in relation to the body, form, in relation to reason — the form of form (εἶδος εἶδους [form of form]) — it is material. From this standpoint the whole of all existence must present itself as a ladder whose lowest rung is a first matter (πρώτη ὕλη [first matter]), which is not at all form, and whose highest rung is the last form, which is not at all matter, but pure form (the absolute, divine spirit). What is between these two extremes is in one respect matter, in another form, i.e. it is a continual passing-over from the first to the last.” (Schwegler).

Matter is that out of which the formed proceeds; but there must be something by which it proceeds: τὸ κινῆσαν ἐκ δυνάμεως εἰς ἐνέργειαν [that which moves from potentiality to actuality]. The transition from possibility to actuality is motion. The moving and the moved must be alike in genus, unlike in species. It is a matter that already has the form for which another matter has the disposition; the former then imparts to the latter its form. The moved and the movable (τὸ κινητόν ὁγ τὸ κινητικόν [the moved and the moving]) cannot be before the motion; nor can either of them come to be before the motion, for such a coming-to-be would precisely be a motion: every motion presupposes a preceding one, this again a preceding one, and so on to infinity. The same is seen from the concept of time. Time cannot be thought without a Now, but this is a middle between past and future. Time is without beginning; it is a determination by motion (πάθος τι κινήσεως [an affection of motion]); an absolute arising and passing-away, a coming-to-be out of nothing and a becoming-nothing, cannot be known (οὐ γίγνεται οὔτε ἡ ὕλη οὔτε τὸ εἶδος, λέγω δὲ τὰ ἔσχατα [neither

matter nor form comes to be, I mean the ultimate ones]).

As the efficient cause (τὸ αἷτιον [the cause]) is the beginning of motion, the aim (τὸ τέλος, οὗ ἕνεκα [the end, that for the sake of which]) is motion's terminus. Not only that which is undertaken in conscious intention, but also that which is produced by natural causes, has its purpose; the essence's actuality is motion's purpose, a purpose that lies in the motion itself. In relation to the purpose, necessity (ἀνάγκη [necessity]), which like chance has its ground in matter, is more closely determined. Matter is not, in Aristotle, as in Plato, the merely negative; matter is the means without which form could not possibly realize the finite things, the means without which the purpose could not possibly be attained. Here, then, with respect to the finite, there is a development both according to presupposition and consequence (τὸ ἐξ ὑποθέσεως ἀναγκαῖον [that which is conditionally necessary]) and according to the purpose (εἰσὶν ἄρα δύο αἰτίαι αὗται, τό θ' οὗ ἕνεκα καὶ τὸ ἐξ ἀνάγκης [there are then these two causes, that for the sake of which and that from necessity]). Matter, form, the efficient cause, and the purpose are the four so-called Aristotelian principles.

c) The Aristotelian entelechy.

"The Aristotelian system (Schwegler) is a system of becoming; in this system the principle of Heraclitus returns in richer and more mediated conception, as the Eleatics' principle in the Platonic. Aristotle has made a considerable step toward the overcoming of the Platonic dualism. If matter is form's possibility, becoming reason, then the opposition between Idea and the world of sight is at least in principle potentially overcome, insofar as it is one being that presents itself in both, in form and material, only at different stages of development."

"The relation of the potential to the actual Aristotle makes visible by the relation of the unworked to the worked, the master-builder to the one occupied with the building, the sleeping to the awakened. The seed is potentially a tree, the full-grown tree is it actually; a potential philosopher is also he who at this moment does not philosophize; a potential victor is the better general even before the battle; potentially space is divisible to infinity; potentially, according to possibility, is on the whole all that has a principle of motion, development, change, being-other in itself."

"Actuality or entelechy, on the contrary, designates the perfect action, the attained goal, the completed actuality (the full-grown tree, e.g., is the seed-corn's entelechy), the activity in which the action and its completion coincide in one, e.g. seeing, thinking:

he sees, and he has seen, he thinks, and he has thought — is one and the same, whereas in the activities that are bound up with a becoming, e.g. learning, walking, healing, both — action and completion — fall apart” (κίνησις ἐντελής as opposed to κίνησις ἀτελής [complete motion as opposed to incomplete motion]).

The perfect actuality, the absolute entelechy, is then the concept in which the Aristotelian doctrine of essence comes to conclusion. This concept’s reality is in the *Metaphysics* illuminated from different sides.

α) The first mover. The finite motions have beginning and end, but motion itself is eternal like time. If there is an eternal motion, there must also be a moving first. The first mover cannot be according to possibility; but, if its essence is to be the completed actuality, we must convince ourselves that the actual is always the presupposition of the possible. In the relative sense the actual does indeed proceed from the possible: the plant as actuality proceeds from the seed-corn, as from its possibility; but the possibility is nonetheless not the first: the seed-corn, the possible plant, is itself a product of a real plant (against the Pythagoreans and Speusippus: τὸ σπέρμα ἐξ ἐτέρων ἐστὶ προτέρων τελείων, καὶ τὸ πρῶτον οὐ σπέρμα ἐστίν, ἀλλὰ τὸ τέλειον [the seed is from other prior complete things, and the first is not seed but the complete]). On actuality and activity Aristotle lays — it is precisely one of his deepest thoughts — an unconditioned emphasis; the actual is the original (διὸ καὶ τοῦνομα ἐνέργεια λέγεται κατὰ τὸ ἔργον, καὶ συντείνει πρὸς τὴν ἐντελέχειαν [wherefore the name “actuality” (energeia) is said with reference to “work” (ergon), and tends toward entelechy]).

That actuality in essence and substantiality is the first is, with respect to time and motion, expressed thus, that the one actual overreachingly presupposes the other, until one comes to the first mover (τοῦ χρόνου ἀεὶ προλαμβάνει ἐνέργεια ἑτέρα πρὸ ἑτέρας ἕως τῆς τοῦ ἀεὶ κινουντος πρώτως [in time one actuality always precedes another, up to that of the primary ever-mover]). The more something is in actuality, the more energy it has; the more energy something has, the less material it is, and the less it is according to possibility. Therefore one need not, like some physicists, fear that the heaven with the sun and the fixed stars should ever grow tired of moving.

If the first mover were according to possibility, it would also have to have been possible that no motion had come about (ἐνδέχεται γὰρ τὸ δύναμιν ἔχον μὴ ἐνεργεῖν [for that which has potentiality may not be acting]). There was then not either eternal chaos or eternal night (ὥστε οὐκ ἦν ἄπειρον χρόνον χάος ἢ νύξ [so there

was not chaos or night for infinite time]) when the motion began; but the same has always been, either in constant alternation, or in another way, as truly as actuality is prior to possibility (εἴπερ πρότερον ἐνέργεια δυνάμεως [if indeed actuality is prior to potentiality]). If now the same is always to act in constant alternation, then there must be something that always acts in the same way (δεῖ τι ἀεὶ μένειν ὡσαύτως ἐνεργοῦν [something must always remain acting in the same way]); if there is to be arising and passing-away, then an other must incessantly act otherwise and again otherwise (ἄλλο δεῖ εἶναι ἀεὶ ἐνεργοῦν ἄλλως καὶ ἄλλως [another must always be acting otherwise and otherwise]). The first mover is thus on the one side, in relation to itself, always like itself, on the other side, in relation to another, always acting otherwise: τί οὖν ἄλλας δεῖ ζητεῖν ἀρχάς; [why then must one seek other principles?]

The first heaven is eternal (δῆλον, ὥστε αἰδῖος ἂν εἴη ὁ πρῶτος οὐρανός [clearly, then, the first heaven would be eternal]); the first heaven is only one. Were there several heavens, as there are several men, then their principles would indeed be identical in concept, but different in number. What is thus in number a plurality must have matter; but the first essence (τὸ τί ἦν εἶναι τὸ πρῶτον [the primary essence]) can have no matter. The first mover, which is itself unmoved, since it is always like itself, is then one both in concept and in number: ἐν ἄρα καὶ λόγῳ καὶ ἀριθμῷ τὸ πρῶτον κινεῖν ἀκίνητον ὄν [the first mover, being unmoved, is thus one both in account and in number].

β) The Good. Every possibility is determined by contradiction; for what is not possible cannot be, but the possible need not in actuality be. The possible can then be and not be, and this holds of all that is finite. What is according to possibility can be changed, and in the change become the opposite of what it was. The healthy can become sick, the sick healthy; the better can become worse, this again better. The eternal, on the contrary, cannot be according to possibility (οὐθὲν ἄρα τῶν ἀφθάρτων ἀπλῶς δυνάμει ἐστὶν ὄν ἀπλῶς [nothing imperishable is simply in potentiality simply]); the eternal is with necessity what it is: the eternal is the unchangeable (ἀπαθὲς καὶ ἀναλλοίωτον [impassive and unalterable]). In this unchangeableness the entelechy is the Good. The changeable can fall away from its concept, miss its goal, and therein show itself as the bad, the evil; but the eternal, the unchangeable, has the terminus in itself, cannot fall away from its concept, and is precisely thereby the perfect Good. There is then in the principles and in the eternal no evil, no fault, and

no corruptibility; corruptibility belongs to the evil (οὐκ ἄρα οὐδ' ἐν τοῖς ἐξ ἀρχῆς καὶ τοῖς αἰδίοις οὐθέν ἐστιν οὔτε κακὸν οὔτε ἀμάρτημα οὔτε διεφθαρμένον. καὶ γὰρ ἡ διαφθορὰ τῶν κακῶν ἐστιν [so neither in the first principles nor in the eternal things is there anything evil or faulty or corrupted; for corruption belongs to evil things]). As the eternal, perfect Good, the entelechy is that which moves in that it is desired; for all in nature is laid out toward a highest common goal (πρὸς μὲν γὰρ ἐν ἅπαντα συντέτακται [for all things are ordered together toward one]). As the in-itself perfect Good, the entelechy can move all without itself being moved; for that which is desired and thought moves without itself being moved. What shows itself beautiful is desired, what is beautiful is willed. We desire the beautiful because it shows itself, but it does not show itself because we desire it. The purpose moves because it is loved, and what is moved by the purpose then moves in turn the rest. The divine then consists in eternal, infinite energy, thus in eternal living, for life is energy (ὥστε ζωὴ καὶ αἰὼν συνεχὴς καὶ αἰδῖος ὑπάρχει τῷ θεῷ [so that life and continuous, eternal duration belong to God]). This life is blessed; such a joy is granted us only in short moments; but all joy (ἡδονή [pleasure]) is an effect of this principle.

γ) Reason. Thinking and the thought are in reason one (ὥστε ταὐτὸν νοῦς καὶ νοητόν. τὸ γὰρ δεκτικὸν τοῦ νοητοῦ καὶ τῆς οὐσίας νοῦς [so that reason and the object of thought are the same; for that which is receptive of the intelligible and of substance is reason]); but the thought is, as the thought-content, higher than the thinking; he who thinks the worst is in reality just as much thinking as he who thinks the best. The true, eternal thinking must then be determined by the true, eternal content. In the finite, material existence thinking cannot become identical with itself and its content, but where there is no matter, thinking is one with the thought (οὐχ ἑτέρου οὖν ὄντος τοῦ νοουμένου καὶ τοῦ νοῦ, ὅσα μὴ ὕλην ἔχει τὸ αὐτὸ ἔσται καὶ ἡ νόησις τοῦ νοουμένου μία [the thought and its object not being different, in things that have no matter it will be the same, and the thinking of the object one]). If the thought is composite, then thinking changes itself by passing over from one part to another; but what has no matter is indivisible like the human understanding, or as when a composite has reached its completion. The purpose is then not in this or that particular, but in a whole the Good, as an essence different from the piecework (οὐ γὰρ ἔχει τὸ εὖ ἐν τῷδὶ ἢ τῷδί, ἀλλ' ἐν ὅλῳ τινὶ τὸ ἀριστον, ὃν ἄλλο τι [for the good is not in this or that particular, but the best is in a certain whole, being something else]). So it

stands with the thought that thinks itself in all eternity. It is the divine understanding that eternally thinks itself, and this thinking is the thinking of thinking (αὐτὸν ἄρα νοεῖ εἴπερ ἐστὶ τὸ κράτιστον, καὶ ἔστιν ἡ νόησις νοήσεως νόησις [it thinks itself, then, if it is the most excellent, and the thinking is a thinking of thinking]).

By impressing energy, pointing out the principle of motion, developing the immanent teleology, and analyzing the essence of the Godhead, Aristotle has continued Plato's work, and carried the doctrine of Ideas further.

§ 29.

When the pulse stops, life ceases; when the development stops, the Idea vanishes. If we now regard the given traits in the Aristotelian entelechy as given, existing, stationary, the physiognomy is dead, and the worm of contradiction develops itself in the dead.

“One does not see (Schwegler) how something can be a moving cause and yet itself unmoved, cause of all becoming, i.e. of all passing-away and arising, and yet an energy that remains like itself, a principle of motion without possibility (potentiality): for the moving must after all stand in a relation of suffering and acting to the moved.”

That here in each of these points a contradiction shows itself cannot be denied; but it depends on whether such a contradiction is applied critically or dialectically. If one sets up, according to the logical principle of exclusion (*principium exclusi medii inter duo contradictoria* [the principle of the excluded middle between two contradictories]), the proposition: “The first mover must necessarily be either moved or unmoved,” one can easily prove that the Aristotelian philosophy culminates in an unsolved and insoluble contradiction.

Outside the Idea the contradiction is critical; in the development of the Idea it is dialectical. To grasp the relation of the unmoved and the moved in such a way that they immediately exclude each other is to grasp it without Idea: in the Idea the unmoved is itself moved. The Idea is the inner unity of the otherwise mutually excluding determinations, the identity of the otherwise mutually contradicting moments. This conception of the contradiction is in express conflict with the Aristotelian logic, but in intimate understanding with the Aristotelian entelechy. When Aristotle determines his πρῶτον κινουῦν [first mover] as an ἀκίνητον [unmoved], it is everywhere clear from the connection that by this ἀκίνητον he cannot possibly think an idle resting

that should exclude self-motion.

In the imperfect energy rest and motion fall apart: the resting does not move; the self-moving does not rest; the resting must be set in motion, the moved go to rest. In the perfect energy the motion coincides with the rest: here is rest in the motion, motion in the rest. By ἀκίνητον, then, according to the meaning, only the finite in the motion (e.g. local motion) is excluded, not the inner self-motion. But that the meaning has not found its conceptual expression in the presentation proves at the same time that the form of presentation is inadequate: the dialectical is lacking.

If a glance at the Idea, immediate intuition of the Idea, could satisfy philosophy's demand, the demand would already often and in many different ways have been satisfied. Not only Plato and Aristotle, but also Jacob Böhme would then have satisfied the demand. But if the task is to give the Idea its pure, clear, articulated conceptual form, we must, after having heard Plato and Aristotle, especially turn to Hegel as the one who, by his glance at the contradiction and his dialectical method, occupies the highest place.

a) The Dialectic of contradiction.

That black is white, that good is evil, that round is square, are contradictions that are just as little tolerated in the doctrine of Ideas as in existence; but that with respect to the Idea itself there are contradictions of another significance can among other things be seen from the Platonic dialogues. In "the Sophist" and in "Parmenides" it is indeed expressly set out to keep the contradiction out, and yet the contradiction breaks through, and that with so much the greater force, the more the acuteness exerts itself to withstand it.

"The not-being is not" — that is the proposition that is unconditionally to be held fast, and then it turns out that this not-being nonetheless has a being. The One is not a manifold, and then it turns out nonetheless that precisely this One makes itself a manifold. In this transition is the Idea. The Idea is not the contradiction; but, in that the thoroughgoing contradiction dissolves all finite difference into the differenceless, the Idea is itself the solution of the contradiction. The manifold is, in its finite and precisely thereby infinite manifoldness, not the One, as little as the One in and for itself either is or can be the manifold; that the One by its own contradiction nonetheless makes itself manifold means that it cannot hold itself in the idealess. The idealess One, the Eleatics' τὸ ἓν [the One], is in its solitude barren, unfruitful, and cannot endure

the manifold. When now the Idea approaches, this in-itself-hardened One is attacked by the contradiction, compelled to give up its enclosedness, and pass over into the manifold. Seen under the critical angle of the contradiction, this transition is the One's downfall: the One now is not, for now the manifold is; but this critical is lifted by the fact that the contradiction serves the Idea. The One then does not perish, no, it goes up into the Idea, sucks in the Idea's force, produces the manifold, and shows itself now as the manifold's One, as the Idea's own true, content-rich essence. The more thinking is sharpened, the more distinctly it is seen that the carried-through contradiction is precisely the contradiction's solution — in the Idea.

From this point of view the relation between Plato and Hegel can be more closely illuminated. In the Platonic Dialectic only the critical side of the contradiction is yet brought to consciousness. Concept-analyses distinguish themselves precisely by the fine, microscopic separation of the moments; the effort is to avoid the contradiction by help of the distinction: *qui bene distingvit, bene docet* [he who distinguishes well teaches well]. The subtle, apt distinction now doubtless brings with it a transition from the one conceptual determination to the other; but this transition is only something that happens, something wonderful that befalls the thinker; it is the contradiction that is after him, and drives him from the one position to the other, without the proper essence of the transition being seen through. Only in Hegel is it brought to consciousness what the transition signifies, and in what relation it stands to the contradiction. If we call the Platonic Dialectic a dialectic of distinction, we must call the Hegelian a dialectic of identity. In the dialectic of distinction the antinomic rules; the dialectic of identity is laid out to master the antinomic.

“Die wahre und positive Bedeutung der Antinomien besteht nur überhaupt darin, daß alles Wirkliche entgegengesetzte Bestimmungen in sich enthält. [The true and positive meaning of the antinomies consists only, in general, in this, that everything actual contains opposed determinations in itself.]”

“Was das Vorkommen der Dialektik in der geistigen Welt, und näher auf dem Gebiet des Rechtlichen und Sittlichen anbetrifft, so braucht hier nur daran erinnert zu werden, wie, allgemeiner Erfahrung zufolge, das Äußerste eines Zustandes oder eines Thuns in sein Entgegengesetztes umschlagen pflegt, welche Dialektik dann auch vielfältig in Sprichwörtern ihre Anerkennung findet. So heißt es z. B. summum jus summa injuria, womit ausgesprochen ist, daß das abstrakte Recht auf seine Spitze getrieben in Unrecht

umschlägt. Eben so ist es bekannt, wie im Politischen die Extreme der Anarchie und des Despotismus einander gegenseitig herbeiführen pflegen. [As for the occurrence of Dialectic in the spiritual world, and more nearly in the domain of the juridical and ethical, one need here only be reminded how, according to general experience, the extreme of a condition or of a doing is wont to turn round into its opposite, which Dialectic then also finds its recognition manifoldly in proverbs. So it is said, e.g., *summum jus summa injuria* (the highest right is the highest wrong), whereby it is expressed that abstract right, driven to its peak, turns round into wrong. Likewise it is known how in the political the extremes of anarchy and of despotism are wont mutually to bring about each other.]”

“Das Bewußtseyn der Dialektik im Gebiet des Sittlichen in seiner individuellen Gestalt finden wir in jenen allbekannten Sprüchwörtern: Hochmuth kommt vor dem Fall — Allzuscharf macht schartig u. s. w. — Auch die Empfindung, die leibliche sowohl als die geistige, hat ihre Dialektik. Es ist bekannt, wie die Extreme des Schmerzes und der Freude in einander übergehen; das von Freude erfüllte Herz erleichtert sich in Thränen, und die innigste Wehmuth pflegt unter Umständen sich durch Lächeln anzukündigen [The consciousness of Dialectic in the domain of the ethical, in its individual shape, we find in those well-known proverbs: pride goes before a fall — too sharp makes notches, and so on. — Feeling too, the bodily as well as the spiritual, has its Dialectic. It is known how the extremes of pain and of joy pass over into each other; the heart filled with joy relieves itself in tears, and the most intimate melancholy is wont under circumstances to announce itself through a smile.]” (Hegel).

In Plato and Aristotle the contradiction is a being that occasions the dialectical: either there is in this a contradiction, or there is none; either the contradiction is here or there. In Hegel the contradiction itself is the dialectical: it is, and yet is not; it is, for the finite is; it is not, for the Idea is. Here is in this finite a *summa injuria*; it is finitude’s ethical contradiction, and this contradiction is man’s torment. But *summa injuria* says precisely that here is a *summum jus*; for only in that righteousness is, can unrighteousness be. So it is then not a torment to suffer wrong for righteousness’ sake, but a blessedness, for righteousness is in the Idea, and the Idea is the blessed; it is thus a torment to suffer, and yet blessed to suffer: the contradiction is, for the finite is; the contradiction is not, for the Idea is.

b) The infinite negativity.

One hears not seldom the opinion expressed that negative knowledge does not satisfy, that philosophy ought to be positive. It now depends on how one understands the positive. If the positive is grasped so immediately that it has the negative outside itself, it is grasped in finite form, and is then in this its finitude fallen prey to the contradiction, and thereby to the negative. A philosophy that is dogmatically positive by keeping the negative out is without Idea, and the idealess positivism precisely by the Idea destined for dissolution into the negative. Therefore Dogmatism is always attacked by Scepticism. Dogmatism is idealess, because it lacks the negative; Scepticism is idealess, because it lacks the positive. The positive is in the Idea infinitely negative, and this negative precisely therefore the true positive. The objection that a philosophy is not positive enough is then, in the concept, to be understood so, that it is not negative enough. Scepticism is, in the same degree as it has the negative, nearer to reaching the true positive than Dogmatism.

“Der Skepticismus darf nicht blos als eine Zweifelslehre betrachtet werden, vielmehr ist derselbe seiner Sache, d. h. der Nichtigkeit alles Endlichen schlechthin, gewiß. Wer nur zweifelt, der steht noch in der Hoffnung, daß sein Zweifel gelöst werden könne, und daß das eine oder das andre Bestimmte, wozwischen er hin und herschwankt, sich als ein Festes und Wahrhaftes ergeben werde. Dahingegen ist der eigentliche Skepticismus die vollkommne Verzweiflung an allem Festen des Verstandes, und die sich daraus ergebende Gesinnung ist die der Unerschütterlichkeit und des Insichberuhens. Dieß ist der hohe antike Skepticismus, wie wir ihn namentlich beim Sextus Empirikus dargestellt finden, und wie derselbe als Komplement zu den dogmatischen Systemen der Stoiker und Epikuräer, in der spätern Römerzeit seine Ausbildung erhalten hat [Scepticism may not be regarded merely as a doctrine of doubt; rather it is certain of its cause, i.e. of the nullity of everything finite absolutely. He who only doubts still stands in the hope that his doubt can be resolved, and that the one or the other determinate, between which he sways back and forth, will turn out to be something fixed and true. On the contrary, the proper Scepticism is the complete despair of everything fixed of the understanding, and the disposition resulting therefrom is that of imperturbability and of resting-in-itself. This is the high ancient Scepticism, as we find it presented namely in Sextus Empiricus, and as it, as complement to the dogmatic systems of the Stoics and Epicureans, received its development in the later Roman period.]” (Hegel).

To carry through the negation is to negate it; in the negation of the negation the

positive returns. The ancient Scepticism has the infinite negation, but in a finite way. The art consists in shaking all assumptions and overthrowing all positive propositions; one goes from denial to denial without halting at any affirmation whatever. But since each denial for itself is only a denial of this or that finite, and thereby itself finite, the continuation of the partial negations is an endless finitude that lacks essentiality. The inner infinity of the negation, on the contrary, is essentiality's own immanent reflection, is as the negation of the negation the positive.

“Denn das Wesen ist das Selbstständige, das ist als durch seine Negation, welche es selbst ist, sich mit sich vermittelnd; es ist also die identische Einheit der absoluten Negativität und der Unmittelbarkeit. — Die Negativität ist die Negativität an sich; sie ist ihre Beziehung auf sich, so ist sie an sich Unmittelbarkeit; aber sie ist negative Beziehung auf sich, abstoßendes Negiren ihrer selbst, so ist die an sich seyende Unmittelbarkeit das Negative oder Bestimmte gegen sie. Aber diese Bestimmtheit ist selbst die absolute Negativität und dieß Bestimmen, das unmittelbar als Bestimmen das Aufheben seiner selbst, Rückkehr in sich ist [For the essence is the independent, which is as mediating itself with itself through its negation, which it itself is; it is thus the identical unity of absolute negativity and of immediacy. — Negativity is negativity in itself; it is its relation to itself, so it is in itself immediacy; but it is negative relation to itself, repelling negating of itself, so the in-itself-being immediacy is the negative or determinate against it. But this determinacy is itself the absolute negativity, and this determining, which immediately as determining is the sublating of itself, is return into itself.]” (Hegel).

And what then is the whole Idea? No thing! And the properties of things? No thing! And the thing in itself? No thing! By establishing each of these negations we are led into Scepticism, and thereby away from — i.e. on the way to — the Idea.

When it is here said of the Idea that it is no thing, the meaning is this, that the thing is, and is in all immediacy the positive. We get by this denial of the Idea a world of things; but the ideal has in itself no validity: only the sensible is; the non-sensible is not. Hereby is characterized the modern Scepticism, *“welcher bloß darin besteht, die Wahrheit und Gewißheit des Uebersinnlichen zu leugnen, und dagegen das Sinnliche und in der unmittelbaren Empfindung Vorhandene als dasjenige zu bezeichnen, woran wir uns zu halten haben.* [which consists merely in denying the truth and certainty of the supersensible, and on the contrary in designating the sensible and that present in

immediate feeling as that to which we have to hold.]”

When it is here said of the thing’s properties that they are not things, the meaning is this, that the property is precisely the thing’s negation, in that it is nonetheless a positive determination of the thing. To be property (predicate) is, as Aristotle too impresses, not the same as to be thing (subject). Man is white, snow is white: the white is, as property, not a thing. The white has in its pure whiteness no, absolutely no real being; the whiteness is only insofar as it is in the white substance. As it now goes with the whiteness, so it goes with the hardness, softness, sourness, sweetness, and, in short, with all possible properties. The many different properties seem at first glance to be sheer positive determinations, but dissolve, when thinking tests them, into pure negations. But, in that the properties’ reality in this way vanishes, the reality of things must also vanish; the thing in itself is a phantom: the thing is only what it is, in and by its properties.

To let the dialecticizing thought transform all properties into figments, all positive determinations into partial negations, is the distinguishing mark of the ancient Scepticism, which then also lets the reality of things be consumed in the same negations.

In that the thing is in this way negated, and becomes no thing, the Idea comes forth as Idea. The Idea is not the thing; but the thing dissolved into negations is restituted in and by the Idea. That the idealess immediacy bears the negativity in itself is a proof that it is subject to the Idea. To see the properties’ immediate plurality, not as an aggregate of positively different ones, but as negations of the thing’s own essential unity, is to see the properties in the Idea; to see the thing’s immediate unity not as substratum, but as negation of the properties’ plurality and positive difference, is to see the thing in the Idea. The immediately positive is in its essence negative; only in the negation of the negation is the Idea, the true positive essence.

c) The speculative form.

The positive in the Idea is, with another expression, at the same time the speculative. As the positive proceeds from the negation, the speculative proceeds from the reflection.

How Hegel grasps the relation between immediacy, reflection, and speculation becomes manifest especially from his “Wissenschaft der Logik.” In the form of immediacy being is one, and not-being an other; likewise essence is one and semblance an other. In this immediate separation it looks as though one could hold fast to the one of

the juxtaposed concepts, and give up the other. If now this attempt is made in earnest, the reflection begins. In the reflection it shows itself that the separated concepts determine each other mutually. Being has as positive being a reflex of its negation, not-being; conversely there is here in not-being, as the expression itself shows, a reflex of being. It is not merely we who make our reflections about being and not-being, no, the concepts mirror each other mutually: the reflection is objective. Would one, in order to hold to the pure essence, think all semblance away; then the pure essence in its infinite emptiness would itself be semblance: by denying the semblance, we would precisely get the semblance confirmed. Would one conversely, instead of knowing an essence in the semblance, deny the essence, and hold fast to the semblance alone, then we would thereby get semblance of semblance, a semblance shining through itself, i.e. the essence: by negating the essence we get the essence confirmed. The unity of the mutually negating and thereupon affirming concepts is seen in speculation. Thus becoming is a speculative unity of being and not-being, as the phenomenon is a speculative unity of essence and semblance; the speculative unity is dialectical.

*“Ein spekulativer Inhalt kann deshalb auch nicht in einem einseitigen Satz ausgesprochen werden. Sagen wir z. B. das Absolute sey die Einheit des Subjektiven und des Objektiven, so ist dieß zwar richtig, jedoch insofern einseitig, als hier nur die Einheit ausgesprochen und auf diese der Akcent gelegt wird, während doch in der That das Subjektive und das Objektive nicht nur identisch, sondern auch unterschieden sind. [A speculative content can therefore also not be expressed in a one-sided proposition. If we say, e.g., the Absolute is the unity of the subjective and the objective, then this is indeed correct, yet one-sided insofar as here only *the unity* is expressed and the accent laid on this, while in fact the subjective and the objective are not only identical but also distinguished.]”*

By setting speculation in intimate, inseparable connection with the dialectical reflection, Hegel has secured for the speculative a scientific character; whereas Schelling, who remained standing at “the intellectual intuition,” let the speculative go together with the mystical, and approach the apocalyptic, whereby all manner of fantasticalities could gain entrance. The Idea’s speculative form is in Hegel in inseparable unity with the dialectical method. Although immediacy, reflection, and speculation penetrate each other at all points of the development, the great tripartition — being, essence, and concept — is nonetheless laid out in such a way that being on the whole corresponds

to immediacy, essence to reflection, and the concept to speculation. The Idea, which is especially dealt with in the last part of the doctrine of the concept, is therefore not a special part of the Logic's content; the whole Logic is a doctrine of Ideas: the Idea is the concept's absolute content, as the concept is the Idea's absolute form.

“Die Idee ist die Wahrheit; denn die Wahrheit ist dieß, daß die Objektivität dem Begriffe entspricht, — nicht daß äußerliche Dinge meinen Vorstellungen entsprechen; dieß sind nur richtige Vorstellungen, die Ich Dieser habe. In der Idee handelt es sich nicht um Diesen, noch um Vorstellungen, noch um äußerliche Dinge. — Aber auch alles Wirkliche, in sofern es ein Wahres ist, ist die Idee, und hat seine Wahrheit allein durch und kraft der Idee. Das einzelne Seyn ist irgend eine Seite der Idee, für dieses bedarf es daher noch anderer Wirklichkeiten, die gleichfalls als besonders für sich bestehende erscheinen; in ihnen zusammen und in ihrer Beziehung ist allein der Begriff realisirt. [The Idea is the truth; for the truth is this, that objectivity corresponds to the concept — not that external things correspond to my representations; these are only *correct* representations that I, this one, have. In the Idea it is not a matter of this one, nor of representations, nor of external things. — But also *everything* actual, insofar as it is a true, is the Idea, and has its truth solely through and by force of the Idea. The single being is some side of the Idea; for this it therefore needs still other actualities, which likewise appear as particularly for-themselves subsisting; in them together and in their relation alone is the concept realized.]”

In the Platonic presentation the subjective rules, insofar as now this (εἰ ἓν ἔστι [if the One is]), now that (εἰ ἓν μὴ ἔστι [if the One is not]) is arbitrarily assumed; so subjective a form cannot possibly be the Idea's adequate form. The Hegelian method, on the contrary, is objective, and can by its systematic unfolding of the content's moments furnish an objective, clear Idea-form. With his dissecting dialectic of distinction Plato could not get the individual Ideas to arise as moments in the Idea's unity; with his immanent dialectic of identity Hegel can point out the Idea's absolute unity in all the particular Ideas, in that these are at once independent and yet moments. *“Die Idee selbst ist nicht zu nehmen als eine Idee von irgend Etwas, so wenig als der Begriff bloß als bestimmter Begriff. Das Absolute ist die allgemeine und Eine Idee, welche als urtheilend sich zum System der bestimmten Ideen besondert, die aber nur dieß sind, in die Eine Idee, in ihre Wahrheit zurückzugehen. Aus diesem Urtheil ist es, daß die Idee zunächst nur die Eine, allgemeine Substanz ist, aber ihre entwickelte*

wahrhafte Wirklichkeit ist, daß sie als Subjekt und so als Geist ist. [The Idea itself is not to be taken as an idea of something, as little as the concept merely as determinate concept. The Absolute is the universal and one Idea, which, as *judging*, particularizes itself into the system of the determinate Ideas, which, however, are only this: to go back into the one Idea, into their truth. It is from this judgment that the Idea is *at first* only the one, universal substance, but its developed true actuality is that it is as subject and thus as spirit.]”

B.

Reality against the Idea.

§ 30.

The Idea is then the truth, and the truth identity of essence and reality. He who in the real sees only an essential, insofar as he therein sees something universal, sees only a half-truth; he who in reality looks only at the individual, the special, the merely factual, likewise sees only a half-truth. In that a half-truth is set as the whole truth, untruth arises. So long as Idealism as doctrine of essence and Empiricism as doctrine of reality keep themselves in their finite opposition, no cognition of truth, no true cognition of the Idea, will be possible.

That reality in the Idea is to correspond to the concept is, however, easily said; but the question is whether human knowledge now also really masters the true unity of concept and reality. One thing is my subjective concept of reality, another is reality's own objective concept; one thing is reality's objective concept, another is reality itself. Reality's objective concept is doubtless in the Idea one with its real content; but precisely because the identity is only in the Idea, it is in reality not for us, when none of us is the Idea. The real, as it is for us, is known precisely by this, that there is always a finite difference between content and form. Man's essence is doubtless in every real man; but nonetheless no real man can in reality be the same as man's Idea. If we think an ideal man, a man who could with truth say of himself: "I am man's

Idea,” as existing; then suffering and degradation would have to express that such a man’s reality was nonetheless in this present inadequate to its essence.

If the knower, by knowing the Idea, can really make himself one with the Idea, then there is in philosophy an adequate cognition of the Idea, “an absolute knowledge”; but, if every attempt to identify oneself with the Idea itself must be rejected by science as an attempt in the mystically fantastic, then “the absolute knowledge” is in reality unattainable.

Among the philosophers who have in earnest set themselves to grasp the real, to lift reality’s content to the form of the concept, and thereby to identity with the Idea, Aristotle and Hegel are without comparison the most considerable; but it is so far from the case that either Aristotle or Hegel should have solved the task, that the task, literally speaking, has not yet been present for either of them.

“The whole Aristotelian philosophy’s riddle,” says Apelt, “lies now properly in the abstraction εἶδος. First Aristotle teaches logically quite rightly that there are three principles (ἀρχαί): ὑποκείμενον and the two ἀντικείμενα: εἶδος and στέρησις, i.e. the subject and the affirming and denying predicate. Here, then, τὸ εἶδος is the predicate in the affirm- ing judgment. But metaphysically the matter turns round for him, τὸ εἶδος transforms itself into τὸ τί, i.e. into subject, it obtains essentiality. The metaphysical question concerning that which is to be τὸ ὑποκείμενον or οὐσία, i.e. the essence of things, he decides thus, that the mass (ὕλη) merely exists according to possibility (δυνάμει ὄν), the formed mass, on the contrary, in actuality (ἐντελεχεία, ὄν), the shape (εἶδος or μορφή) is thus the ground of actuality. In this comparison of the grounds of being with the categories of modality there is lacking the indication of the ground of necessity, i.e. the law...”

“This philosophy had two roots: ἐμπειρία and νοῦς, experience and pure reason. The Scholastics, who developed the doctrine from its principles, kept νοῦς as the possibility of principles, and let ἐμπειρία fall. Bacon, on the contrary, conversely rejects νοῦς, and thus ἐμπειρία becomes to him the source of principles by means of ἐπαγωγή. ‘Induction,’ so he teaches, ‘assumes no original concepts of the understanding and does not rest satisfied with the immediate instruction of the senses.’ The first is a consequence of his rejecting νοῦς; the second is the Aristotelian separation between αἴσθησις and ἐμπειρία...” Induction’s object is thus τὸ εἶδος, forma.... Τὸ εἶδος is, according to Aristotle, something single, actual, substantial; according to Bacon,

something universal and necessary, a universal rule without essentiality: natural law.

Insofar as the Baconian proposition about natural law as universal rule without essentiality is to hold for more than a phrase, it is (Cf. S. 60) only a proof of how far Empiricism in its strife against Idealism can remove itself from the Idea. Empiricism rests on this, that the for-us real is and remains inseparable from the finite, from the things and their parceling. If now reality's emphasis falls on the finite facts, then the laws valid for these facts must also be grasped in their pure facticity, not as metaphysical essentialities, but as universal rules found by induction. That laws can in reality be discovered factually, and known in factual universality, before there can yet be talk of any essential grounding, is shown by natural science in manifold examples. Held fast for itself, the Baconian view is idealess, but as counterweight against the merely metaphysical it has its great significance.

Reality wills to be known; if now the real belongs to the Idea's essence, then the cognition of reality too must be an inviolable condition for the cognition of the Idea, and Empiricism, even if it knows it not, find itself in the Idea's service. The factual relation between Speculation and Empiricism is, however, not understanding, but strife.

§ 31.

For the immediate consciousness it looks as though space and time properly did not concern each other, as though one could well have space without time, and conversely, live in time without living in space. In reality it is, however, otherwise. What time signifies is seen in motion upon space; what space signifies is seen in motion upon time: motion expresses an intimate, essential relation of space and time. This essential relation is specified into real relations, in that motion is separated into definite motions; the difference between the motions existing in reality is a difference between a slower and a faster, which is designated by velocity, in that velocity in turn designates the relation of the traversed space and the time applied thereto, $v = \frac{s}{t}$.

That a thus-beginning analysis of reality does not without more let itself be transformed into a logical, metaphysical analysis of essence is not merely in general clear from the Aristotelian physics, but can even in more special traits be seen from the Hegelian blunders. Instead of following the mathematical direction Hegel will explain not merely velocity, but even the fall-law, from the pure concepts: space, time,

and corporeality.

“In dem Satze, daß die Geschwindigkeit den Zeiten proportional ist, ist die Geschwindigkeit zunächst überhaupt gesagt; so wird sie überflüssigerweise mathematisch als $\frac{s}{t}$ die schlechtgleichförmige gesetzt, so die Kraft der Trägheit herein gebracht und ihr dieß Moment zugeschrieben. Damit aber daß sie den Zeiten proportional sey, ist sie vielmehr als die gleichförmig beschleunigte $\frac{s}{t^2}$ bestimmt, und jene Bestimmung von $\frac{s}{t}$ hat hier keinen Platz, und ist ausgeschlossen. [In the proposition that velocity is proportional to the times, velocity is at first stated quite generally; so it is superfluously set mathematically as $\frac{s}{t}$, the merely-uniform, so the force of inertia is brought in and this moment ascribed to it. But in that it is proportional to the times, it is rather determined as the uniformly accelerated $\frac{s}{t^2}$, and that determination $\frac{s}{t}$ has here no place, and is excluded.]” (Naturphil. S. 87.)

Now follows the speculative explanation:

“Das Gesetz des Falles ist gegen die abstracte gleichförmige Geschwindigkeit des todten von Außen bestimmten Mechanismus ein freies Naturgesetz; d. h. es hat eine Seite in ihm, die sich aus dem Begriffe des Körpers bestimmt. Indem daraus folgt, daß es aus diesem muß abgeleitet werden können, so ist dieses sich vorzusetzen, und der Weg anzugeben, wie das Galileische Gesetz, “daß die durchlaufenen Räume sich wie die Qvadrate der verflossenen Zeiten verhalten,” mit der Begriffsbestimmung zusammenhängt.

“Dieser Zusammenhang ist aber als einfach darin liegend anzusehen, daß, weil hier der Begriff zum Bestimmen kommt, die Begriffsbestimmungen der Zeit und des Raums gegeneinander frei werden, d. i. ihre GröÙbestimmungen sich nach denselben verhalten. Nun ist aber die Zeit das Moment der Negation, des Fürsichseyns, das Princip des Eins; und ihre GröÙe — irgend eine empirische Zahl — ist im Verhältnisse zum Raum als die Einheit oder als Nenner zu nehmen. Der Raum dagegen ist das Außereinandersehn, und zwar keiner andern GröÙe, als eben der GröÙe der Zeit; denn die Geschwindigkeit dieser freien Bewegung ist dieß, daß Zeit und Raum nicht äußerlich, nicht zufällig gegen einander sind, sondern Beider Eine Bestimmung ist. Die der Einheit, als der Form der Zeit, entgegengesetzte Form des Außereinander des Raums, und zwar ohne daß irgend eine andere Bestimmtheit sich einmischt, ist das Qvadrat: die GröÙe außer sich kommend in eine zweite Dimension sich setzend, sich somit vermehrend, aber nach keiner andern als ihrer eignen Bestimmtheit — diesem Erweitern sich selbst zur Grenze machend, und in ihrem Anderswerden so sich nur auf sich beziehend.

Dieß ist der Beweis des Gesetzes des Falls aus dem Begriffe der Sache.” [The law of *fall* is, against the abstract uniform velocity of the dead, from-without-determined mechanism, a *free* natural law; i.e. it has a side in it that is determined from the *concept* of the body. Since it follows from this that it must be able to be derived from this, it is to be set before oneself, and the way indicated how the Galilean law, “that the traversed spaces relate as the *squares* of the elapsed times,” hangs together with the conceptual determination. This connection, however, is to be regarded as lying simply therein, that, because here the concept comes to determining, the conceptual determinations of *time* and of space become *free* against each other, i.e. their *magnitude-determinations* relate accordingly. Now time is the moment of *negation*, of being-for-self, the principle of the one; and its magnitude — some empirical number — is to be taken, in relation to space, as the *unit* or as denominator. *Space*, on the contrary, is the *being-outside-one-another*, and indeed of *no other magnitude* than precisely the magnitude of time; for the velocity of this *free* motion is this, that time and space are not *external*, not contingent against each other, but are the *one* determination of both. The form of the *being-outside-one-another* of space, opposed to that of the *unit* as the form of time — and indeed without any other determinacy mixing in — is the *square*: magnitude coming outside itself, setting itself in a second dimension, thus multiplying itself, but according to *no other than its own* determinacy — making this expanding itself into its limit, and in its becoming-other thus relating only to itself. This is the proof of the law of *fall* from the *concept of the matter*.]

In this “Beweis aus dem Begriffe der Sache” [proof from the concept of the matter] there shows itself, as everywhere, the Hegelian eye for essential relations; “daß Zeit und Raum nicht äußerlich, nicht zufällig gegen einander sind” [that time and space are not external, not contingent against each other] is the midpoint of thought about which the whole turns; but that Hegel just as little here as elsewhere has an eye for the proper analysis of reality is at the same time conspicuous, and betrays itself involuntarily by his dismissal of the formula $\frac{s}{t}$.

Already at the beginning (S. 85), where he does concede, concerning the laws of motion: “es sind unsterbliche Entdeckungen, die der Analyse des Verstandes die höchste Ehre machen” [they are immortal discoveries that do the highest honor to the analysis of the understanding], he comes, by mistaking the mathematical-physical conception, into a skewed position toward the matter. “Die Voraussetzung bei diesem

apriorischen Beweise ist, daß die Geschwindigkeit im Fall *gleichförmig* beschleunigt ist; der Beweis aber besteht in der Verwandlung der Momente der mathematischen Formel in physicalische Kräfte, in eine *beschleunigende* Kraft, welche in jedem Zeitmoment einen (denselben) Impuls mache, und in eine Kraft der *Trägheit*, welche die in jedem Zeitmomente erlangte (größere) Geschwindigkeit fortsetze — Bestimmungen, die durchaus ohne empirische Beglaubigung sind, so wie der Begriff nichts mit ihnen zu thun hat.” [The presupposition in this a priori proof is that velocity in fall is *uniformly* accelerated; but the proof consists in the transformation of the moments of the mathematical formula into physical forces, into an *accelerating* force which in each time-moment produces one (the same) impulse, and into a force of *inertia* which continues the (greater) velocity attained in each time-moment — determinations that are utterly without empirical accreditation, just as the concept has nothing to do with them.] With this dismissal Hegel has broken the staff over every analysis of reality; for every such analysis is conditioned by this, that the factors are set apart in such a way that their cooperation lets itself be calculated.

The phenomenon dealt with is, as is known, this, that a body which under the fall-motion traverses in the first second the space g , traverses in the second second $3g$, in the third sec. $5g$, and so on. The emphasis thus falls here, not, as Hegel supposes, on a squaring of the time by means of the space, but on the evenly increasing velocity with which the motion takes place. Although a second is only a very small part of time, it nonetheless contains an infinite number of moments; in each moment there is now traversed an infinitely small space, r , so that the series becomes $r + 2r + 3r + \dots nr$. By integration one obtains $r \int_0^n p \, dp = \frac{n^2 r}{2}$, $r \int_n^{2n} p \, dp = 3 \frac{n^2 r}{2}$, $r \int_{2n}^{3n} p \, dp = 5 \frac{n^2 r}{2} \dots$ or, when $\frac{n^2 r}{2}$ is set equal to g , the series: $g, 3g, 5g$, as above indicated.

For the explanation of the phenomenon Hegel adduces a speculative ground instead of an efficient cause. The fall-motion takes place with evenly increasing velocity, in order that natural reason may hereby get occasion to square the time by means of the space. Such speculative motives can with right be reckoned to “Bestimmungen, die durchaus ohne empirische Beglaubigung sind” [determinations that are utterly without empirical accreditation]. Just as little can the addition “Nur das Suchen des Centrums ist im Fall die absolute Seite” [Only the seeking of the center is in the fall the absolute side] be said to hit the real. When physics, on the contrary, seeks the efficient cause in the cooperation of matter’s inertia and the earth’s attraction,

the explanation has not only an empirical confirmation in the circumstance that the calculation built thereon really holds, but has even brought out a character in matter which in all experiments has shown itself constant, namely inertia. That inertia is not sluggishness, but on the contrary that property of matter in virtue of which it is in equal degree disposed both to motion and to rest, is seen precisely from the example before us. Set in motion by an external cause, the earth's attraction, the falling body takes up the imparted motion into itself, and keeps it; since now the continued attraction in each Now imparts to matter an equal motion, the motion must grow evenly in each moment, and the velocity thus be evenly increasing. If there is now indicated the space traversed in each finite definite part of time, in the 1st, 2nd, 3rd, etc. second, then it shows itself rightly enough that the traversed spaces relate as the squares of the applied times; but this relation is by no means restricted to the fall-motion, or "ein Suchen des Centrums" [a seeking of the center], because the fall-motion is one thing falling thereunder.

If we now return to the formula $v = \frac{s}{t}$, it is clear that velocity is not held fast abstractly as an empty concept, but that the real velocity presupposes a matter whose velocity can be determined. In the unchangeable velocity, inertia then expresses the maintenance of the speed imparted by the force in the moment of impact; whereas the changing velocity, on the contrary, presupposes a peculiar influence of a moving force.

In that the analysis of reality in this way separates the factors, these factors cannot indeed, like the logical reflexes of the analysis of essence, be transformed into immanent difference-determinations in an absolute identity; but nonetheless these factors in their cooperation do furnish peculiar conceptual determinations: reality is not conceptless, only that a separation of finitude belongs essentially in the concept of reality.

Insofar as s is a space already traversed in the time t , one cannot yet know whether this space is traversed with changing or unchanging velocity. To determine the velocity, one must give the time an increment, that is to say: procure time for continued motion. If the increment to the time is now called h , one has the series:

$$f(t + h) = s + \frac{ds}{dt} \frac{h}{1} + \frac{d^2s}{dt^2} \frac{h^2}{1.2} + \frac{d^3s}{dt^3} \frac{h^3}{1.2.3} + \dots$$

If the increment h is a finite magnitude, and the velocity constant, the whole series will restrict itself to $f(t + h) = s + \frac{ds}{dt} \frac{h}{1}$; if h is infinitely small, equal to dt , the same will hold, even when the velocity is variable. On the contrary, when h is finite, and the velocity variable, the members of the series can be increased to infinity.

Seen from the concept's point of view, the series indicates a totality of objective reflection- determinations. The individual members can, in that s means the space, v the velocity, and φ the force, be rendered by different expressions corresponding to one another:

$$\begin{aligned} f(t + h) &= s + \frac{ds}{dt} \frac{h}{1} + \frac{d^2s}{dt^2} \frac{h^2}{1.2} + \frac{d^3s}{dt^3} \frac{h^3}{1.2.3} + \\ &= .. + v \frac{h}{1} + \frac{dv}{dt} \frac{h^2}{1.2} + \frac{d^2v}{dt^2} \frac{h^3}{1.2.3} + \\ &= .. + .. + \varphi \frac{h^2}{1.2} + \frac{d\varphi}{dt} \frac{h^3}{1.2.3} + \end{aligned}$$

The categories *Motion*, *Velocity*, *Force*, and *Change in Force* are thus determined by objective reflection. Immediately s designates a length of space; but in the equation $s = f(t)$ this length of space is seen in reflection of a length of time; it owes indeed its existence to time, insofar as it is namely traversed in a given time. Immediately v designates a velocity, but in the expression $\frac{ds}{dt}$ the concept of velocity is determined by objective reflection of space and time in the given relation. Immediately the force is designated by φ , next by reflection of velocity and time in the definite relation $\frac{dv}{dt}$, finally, in that $v = \frac{ds}{dt}$ goes up in reflection of space and time, by a doubled reflection in the relation $\frac{d^2s}{dt^2}$. In this way the changes of force can, with respect to the relation of space and time, be determined as reflections of reflections to infinity.

The difference between analysis of essence and analysis of reality can be seen in “the moment.” In the analysis of essence the moment is the pure Now, a dialectical playing between being and not-being; in the analysis of reality the moment is, on the contrary, qualified: all the members following on s in the series set up are sheer qualified ones, each for itself designating a peculiar disposition, a peculiar energy, moments. In this qualification the separation is exact, and the concept of reality recognizable.

By excluding the speculative motives, asserting the incommensurable moment of finitude, and pursuing the exact discretion to infinity, the analysis of reality has broken with the pure essence, and can on that account not relate to the Idea as Idea.

§ 32.

In his *Physics* Aristotle (cf. Brandis 2ter Th., 2 Abth. 2) has with great circumspection proved that, since all motions are either circular or rectilinear or a mixture of both, the circular motion must be the perfect. But, in that the Greek astronomers started from the metaphysical presupposition that the circular motion, in which every point can be regarded as beginning, middle, and end, was the heavens' eternal motion, and that the earth was this motion's center, it is remarkable to see how observations and calculations gradually brought them into embarrassment with that speculative presupposition, and in manifold particulars forced them to a break with the metaphysical.

Already the observation that the sun in winter moves away from a fixed star by a larger arc in equal time than in summer was sufficient to prove that, if the motion were in itself even and uniform, the earth could not be center for the circle in which the sun's yearly motion took place. Ptolemy therefore proposed to explain the relation by letting the uniform motion take place, not about the earth as center, but about the so-called *punctum æquatorium* [equating point] or *p. æquans* [equant]. It has in recent times been proved that this *punctum æquans* corresponds fairly well to the planetary orbit's second focus, in that the motion, seen from there, approaches becoming uniform.

The decisive break between the astronomical and the speculative, which recent time has properly first carried through, by no means proceeds from this, that the Ptolemaic system has had to give way to the Copernican, or the Aristotelian circle to the Keplerian ellipse.

The Copernican world-structure is, if one will, an even more grandiose image of the Idea than the Ptolemaic; wherefore Copernicus too appeals to the Idea. "By no other ordering," he says enthusiastically (Humboldt: *Kosmos*, 2nd volume), "have I been able to find so admirable a symmetry in the universe, so harmonious a connection of the orbits, as when I set *the Sun*, this world-lantern (*lucernam mundi* [the lamp of the world]), steering the whole family of circling heavenly bodies (*circumagentem gubernans astrorum familiam* [governing and steering the circling family of the stars]) in the midst of the beautiful temple of nature as upon a royal throne. *Quis enim in hoc pulcherrimo templo lampadem hanc in alio vel meliori loco poneret, quam unde totum*

simul possit illuminare? [For who in this most beautiful temple would place this lamp in another or better place than whence it can at once illuminate the whole?]" (Nicol. Copern. *de Revol. orbium coelestium lib. I cap. 10*).

That Speculation can find the ellipse just as rational as the circle, indeed even more rational, Hegel proves.

"Was nun die Gestalt der Bahn betrifft, so ist der Kreis nur als die Bahn einer schlecht-gleichförmigen Bewegung zu faßen. Denkbar, wie man es nennt, ist es wohl, daß auch eine gleichförmig zu- und abnehmende Bewegung im Kreise geschehe. Aber diese Denkbarkeit oder Möglichkeit heißt nur eine abstracte Vorstellbarkeit, welche das Bestimmte, worauf es ankommt, wegläßt, und daher nicht nur oberflächlich, sondern falsch ist. Der Kreis ist die in sich zurückkehrende Linie, in der alle Radien gleich sind: d. h. er ist durch den Radius vollkommen bestimmt; es ist dieß nur Eine, und zwar die ganze Bestimmtheit. In der freien Bewegung aber, wo räumliche und zeitliche Bestimmung in Verschiedenheit, in ein qualitatives Verhältniß zu einander treten, tritt nothwendig dieß Verhältniß an dem Räumlichen selbst als eine Differenz desselben hervor, welche hiermit zwei Bestimmungen erfordert. Dadurch wird die Gestalt der in sich zurückgehenden Bahn wesentlich eine Ellipse; — das erste der Keplerischen Gesetze [As now for the shape of the orbit, the circle is to be grasped only as the orbit of a *merely-uniform* motion. *Thinkable*, as one calls it, it may well be that a uniformly in- and decreasing motion too takes place in a circle. But this thinkability or possibility means only an abstract representability, which omits the determinate on which it depends, and is therefore not merely superficial, but false. The circle is the line returning into itself, in which all radii are *equal*: i.e. it is completely determined by the radius; this is only *one*, and indeed the *whole* determinacy. But in the free motion, where spatial and temporal determination step into *difference*, into a qualitative relation to each other, this relation necessarily steps forth *on the spatial* itself as a *difference* of it, which hereby requires two determinations. Thereby the shape of the orbit returning into itself becomes essentially an *ellipse* — the first of the Keplerian laws]". (Naturphil. S. 104).

No, astronomy's and, on the whole, natural science's decisive break with Speculation is grounded in this, that natural law everywhere necessarily must look at the efficient causes, whereas Speculation with its metaphysical principles and essences will explain everything from purpose-causes. How little purpose-causes help physics, F. Bacon has already expressed in the well-known proposition: *Causarum finalium*

inquisitio sterilis est, et tanquam virgo Deo consecrata nihil parit (De Augm. Scient. Lib. III) [The inquiry into final causes is sterile, and, like a virgin consecrated to God, bears nothing].

Leibniz, though a naturalist, held to the final causes (*Bien loin d'exclure les causes finales — c'est de là qu'il faut tout déduire en Physique ... Extrait d'une lettre à Mr. Bayle sur un principe général utile à l'explication des loix de la nature* [Far from excluding final causes — it is from these that one must deduce everything in physics ... Extract from a letter to Mr. Bayle on a general principle useful for the explanation of the laws of nature]), because he held to Speculation, and Hegel has still, as far as the astronomical phenomena are concerned, such trust in the wisdom expressed in the concept that he even rebukes Newton and the astronomers, because they, with respect to the motions of the heavenly bodies, adduce the same physically efficient causes as those that hold in the finite on earth, whereas it must nonetheless in virtue of the concept be settled that there is another causality in the whole than in the parts, other causes in the heavenly than in the earthly.

“Da der Mond sechzig Durchmesser der Erde von der Erde entfernt ist, so wird also das Moment der Attraction in seiner Bewegung danach bestimmt. Es wird dann gefunden, daß das, was die Attractivkraft der Erde auf den Mond wirke (der Sinus versus, die Sagitta), zugleich den ganzen Umlauf des Mondes bestimme: er falle eben so. Das mag richtig seyn. [Since the moon is sixty diameters of the earth distant from the earth, the moment of attraction in its motion is accordingly determined. It is then found that that which the earth's attractive force acts upon the moon (the *sinus versus*, the *sagitta*) determines at the same time the whole revolution of the moon: it falls just so. That may be right.]” (Naturphil. S. 111).

The moon thus stands, although it otherwise in its motion floats freely, like other heavenly bodies, yet not high enough on the Idea's heaven to be emancipated from the finite, physically efficient causes.

“Das ist aber zunächst nur ein einzelner Fall, die Ausdehnung des empirischen Falls auf der Erde auf den Mond. Von den Planeten ist dieß nicht gemeint, oder gälte nur von ihnen im Verhältniß zu ihren Trabanten. Das ist also ein beschränkter Punkt. [But that is at first only a single case, the extension of the empirical fall on the earth to the moon. Of the planets this is not meant, or would hold only of them in relation to their satellites. That is thus a restricted point.]” (S. 111).

While, then, the bodies on earth are moved by physical causes, and the moon as satellite cannot either raise itself to the speculative, the planetary system is, in its wholeness, on the contrary subjected to a metaphysical causality in which the finite causes have vanished.

“Man sagt, den himmlischen Körpern kommt das Fallen zu. Sie fallen aber doch nicht in die Sonne; so giebt man ihnen noch eine andere Bewegung, welche den Fall aufhält. Das ist sehr einfach verendlicht. So schlagen Knaben mit dem Prügel einen Ball, der fallen will, auf die Seite. Es ist uns nicht geheuer, solche Knabenverhältnisse auf diese freie Bewegung angewendet zu sehen. [One says, falling belongs to the heavenly bodies. But they do not fall into the sun; so one gives them yet another motion, which halts the fall. That is very simply finitized. So boys strike with a club a ball that wants to fall, off to the side. It seems to us uncanny to see such boys’ relations applied to this free motion.]” (Naturphil. S. 111).

It is then not the relation between the planet’s own motion and the sun’s attraction, not a cooperation of centrifugal and centripetal force, that determines the orbit; no, it is reason itself, the concept in its system of judgments and inferences, in one word: the Idea, that moves the whole.

“Die Bewegung der Himmelskörper ist nicht ein solches Hin- und Hergezogenseyn, sondern die freie Bewegung; sie gehen, wie die Alten sagten, als selige Götter einher. Die himmlische Körperlichkeit ist nicht eine solche, welche das Princip der Ruhe oder Bewegung außer ihr hätte. Weil der Stein träge ist, die ganze Erde aber aus Steinen besteht, und die anderen himmlischen Körper eben dergleichen sind, — ist ein Schluß, der die Eigenschaften des Ganzen denen des Theils gleichsetzt. Stoß, Druck, Widerstand, Reibung, Ziehen und dergleichen gelten nur von einer andern Existenz der Materie, als die himmlische Körperlichkeit. Das Gemeinschaftliche Beider ist freilich die Materie, so wie ein guter Gedanke und ein schlechter beide Gedanken sind: aber der schlechte nicht darum gut, weil der gute ein Gedanke ist. [The motion of the heavenly bodies is not such a being-drawn-hither-and-thither, but the free motion; they go, as the ancients said, along like blessed gods. The heavenly corporeality is not such as would have the principle of rest or motion outside it. Because the stone is inert, but the whole earth consists of stones, and the other heavenly bodies are just the same — is an inference that equates the properties of the whole with those of the part. Impact, pressure, resistance, friction, pulling, and the like hold only of another existence of matter than

the heavenly corporeality. What is common to both is indeed matter, just as a good thought and a bad are both thoughts: but the bad is not good on that account, because the good is a thought.]” (Naturphil. S. 97).

So long as one held fast as a decisive dilemma: either nature’s order is an effect of blind chance, or else a work of God’s wisdom, it is comprehensible that men like Kepler, Newton, and Leibniz could not merely unite a penetrating study of nature’s laws with faith in a creating God, but even in decisive cases feel an urge to lay this their religious presupposition at the ground of a deeper interpretation of the phenomena. *Finalem causam non tantum prodesse ad virtutem et pietatem, in Ethicà et Theologia naturali, sed etiam in ipsa Physica ad inveniendum et detegendum abditas veritates. De ipsa natura*; Leibniz. [That the final cause is of use not only for virtue and piety, in ethics and natural theology, but also in physics itself for finding and detecting hidden truths. On Nature Itself; Leibniz.]

The more, on the contrary, the intermediate causes were traced out, the more the hypothetically necessary (τὸ ἐξ ὑποθέσεως ἀναγκάϊον [that which is necessary on a hypothesis]) took contingency in seizure, the more the thought pressed itself forward, that nature’s order itself must be a work of nature. It was then not a willing God, acting according to purposes, that gave the planet that first impulse, as little as it was a willing God who arbitrarily commanded matter to act attractively. No, that matter attracts follows from its nature. The original impulse, motion’s natural cause, is itself a natural effect of a preceding natural cause (the ring’s rotation and bursting, according to the hypothesis), this cause again an effect of a preceding one, and so on in infinity’s cycle.

“I have searched the whole heaven,” says the astronomer Lalande, “and nowhere found a trace of God.” — When the Emperor Napoleon asked Laplace why in his *Traité de mécanique céleste* there was nowhere talk of God, the famous astronomer is said to have given the answer: »Sire, je n’avais pas besoin de cette hypothèse!« [Sire, I had no need of this hypothesis!] From this answer one can conclude nothing concerning Laplace’s religiosity, but on the contrary much concerning his scientificity; he has not dwelt on edifying general considerations, but interpreted facts; he has not immersed himself in dialectical speculations, but in exact analyses, not intuited the pure essence, but tested the strict reality, not had need of *causæ finales* [final causes], but helped himself with *causæ efficientes* [efficient causes].

It is not only the application of the supernatural causality, or of the finite purposiveness (die äußere Zweckmässigkeit [the outer purposiveness]), that natural science must reject; even the immanent, speculative teleology has in the analysis of reality nothing to signify.

That the doctrine of “die besondern Formationen der Erde” [the particular formations of the earth] belongs under physical geography, and that in physical geography there can be no talk of any higher teleology, Hegel concedes. But a description of the physically mechanical is then also only of subordinate significance. “*Dieß erscheint zunächst als zufällig; aber die Thätigkeit des Begriffs ist, das als nothwendig bestimmt zu fassen, was dem sinnlichen Bewußtseyn als zufällig erscheint.*” (Naturphil. S. 441). [This appears at first as contingent; but the activity of the concept is to grasp as necessary that which appears to the sensible consciousness as contingent.]

The necessity of which there is here talk is not the blind fatal one that is dialectically one with the contingent; no, the necessity in which the concept is to sublimate the contingent has its deeper root in the teleological. In the separation of land and sea, in the division of the new and the old world, the Idea has revealed its wisdom:

“*Ueberhaupt (Naturphil. S. 442) stellt die neue Welt die unausgebildete Entzweiung dar, — einen nördlichen und einen südlichen Theil in der Weise des Magneten: die alte aber die vollkommene Entzweiung in drei Theile, deren Einer, Afrika, das gediegene Metall, das Lunarische, starr vor Hitze ist, wo der Mensch in sich selbst verdumpft, — der nicht ins Bewußtseyn tretende stumme Geist; der andere, Asien, ist die bacchantisch kometarische Ausschweifung, die wild nur aus sich ausgebährende Mitte, die formlose Erzeugung, ohne daß er über seine Mitte Meister werden könnte; der dritte aber, Europa, bildet das Bewußtseyn, den vernünftigen Theil der Erde, das Gleichgewicht von Strömen und Thälern und Gebirgen, — dessen Mitte Deutschland ist.* [In general, the new world presents the undeveloped division — a northern and a southern part in the manner of the magnet: the old, however, the complete division into three parts, of which one, Africa, the massive metal, the lunar, is rigid from heat, where man becomes dull within himself — the mute spirit not stepping into consciousness; the second, Asia, is the bacchantic, cometary excess, the wildly self-generating middle, the formless production, without his being able to become master over his middle; the third, however, Europe, forms the consciousness, the rational part of the earth, the equilibrium of streams and valleys and mountains — whose middle is Germany.]”

§ 33.

In the doctrine of Ideas matter is only a fleeting moment, a remainder of finitude vanishing in the negation of the negation. Although Hegel and Aristotle ascribe to matter's being another significance than Plato does, they are nonetheless all agreed in making matter into so formal a material that its being according to possibility only constitutes the determining form's passive side.

That Plato ascribes to the fundamental parts in the four elements definite mathematical figures — to the fire-parts the pyramid, to the earth-parts the cube, to the air-parts the octahedron, to the water-parts the icosahedron — cannot possibly be misunderstood as though he hereby thought of eternally existing, absolutely independent atoms; since he on the contrary grasps matter as the not-being. Aristotle does indeed let matter step forth with a certain independence against form, in that it makes form resistance, and brings it about that the one form falls out from itself into a plurality of forms; but the role that matter hereby plays is nonetheless, as in the Hegelian “Außersichseyn” [being-outside-itself], “Andersseyn” [being-other], only that of passivity and impotence: matter is in its being according to possibility a pure continuum, and its parts only points of transition for the form's, i.e. the Idea's, infinite self-mediation.

It is otherwise in the exact analyses of reality. Without deciding what matter is in metaphysical essentiality, Empiricism holds fast to the view that the real mass, though it may according to possibility be divisible to infinity, nonetheless in reality always consists and must consist of finite fundamental parts, since the presupposition of matter's finite discretion is an inviolable condition for every natural-scientific investigation grounded on calculations and on weight and measure (cf. § 10 S. 55 with § 11 S. 64).

To the difference that thus is between the idealistic and the empirical in the doctrine of matter corresponds a still more thoroughgoing difference in the doctrine of form. However much the Aristotelian form in its unity with matter otherwise diverges from Plato's for-itself-being Idea, it is nonetheless this, to furnish type, παράδειγμα [paradigm], common to both. The type, the definite form, whose realization is the last, is in concept and essence the proper first. In the formed matter the type is teleologically realized by the typically acting principle of unity: the material constituents with their

finite reciprocity are here the subordinate. From the empirical standpoint the matter is seen reversed; here it is precisely the material fundamental parts and their reciprocity on which the emphasis falls. Metaphysically seen, the type is principle, physically seen, it is result; metaphysically seen, the aggregation is subordinate to the paradigm, physically seen, the paradigm is subordinate to the aggregation.

“Die Thätigkeit in ihr Product übergegangen ist die Gestalt, und bestimmt als Krystall. In dieser Totalität sind die differenten magnetischen Pole zur Neutralität reducirt, die abstracte Linearität der ortbestimmenden Thätigkeit zur Fläche und Oberfläche des ganzen Körpers realisirt ... Es wirkt die Eine Form, indem sie α) die Kugel begrenzend, den Körper nach Außen krystallisiert, und β) die Punktualität gestaltend, seine innere Continuität durch und durch im Durchgang der Blätter, d. h. in der Kerngestalt, krystallisiert [The activity, having passed over into its product, is the shape, and determined as crystal. In this totality the different magnetic poles are reduced to neutrality, the abstract linearity of the place-determining activity realized into the surface and outer surface of the whole body ... The one form acts, in that it α) bounding the sphere, crystallizes the body outward, and β) shaping the punctuality, crystallizes its inner continuity through and through in the passage of the leaves, i.e. in the core-shape.]” (Hegel Naturphil. S. 265).

From this speculative interpretation of the crystal it is seen that the typical form is in Hegel just as independently self-active as in Aristotle. That this self-activity of the form must expressly be grasped teleologically goes without saying.

“Der Magnet ist noch nicht zweckmäßig; denn seine Entzweiten sind noch nicht gleichgültig, sondern nur rein Nothwendige für einander. Hier aber ist eine Einheit Gleichgültiger, oder Solcher, deren Daseyn in seiner Beziehung frei von einander ist. Die Linien des Krystalls sind diese Gleichgültigkeit: es kann eine von der andern getrennt werden, und sie bleiben; aber sie haben schlechthin Bedeutung nur in Beziehung auf einander, — der Zweck ist diese ihre Einheit und Bedeutung.

Indem der Krystall aber dieser ruhige Zweck ist, so ist die Bewegung ein Anderes, als sein Zweck; der Zweck ist noch nicht als Zeit [The magnet is not yet purposive; for its sundered ones are not yet indifferent, but only purely necessary for one another. But here is a unity of indifferent ones, or of such whose existence is in its relation free from each other. The lines of the crystal are this indifference: one can be separated from the other, and they remain; but they have significance simply only in relation to

one another — the purpose is this their unity and significance. But in that the crystal is this resting purpose, the motion is an other than its purpose; the purpose is not yet as time]”. (Naturphil. S. 267).

The real doctrine of crystals now knows indeed nothing at all of this “ruhige Zweck” [resting purpose], and attains nothing either by appealing to the typically active form. On the contrary, Empiricism, by seeing the crystallogonic in its intimate connection with the chemical, is everywhere — isomorphism, polymorphism — attentive to the atoms’ size, shape, number, and arrangement; Empiricism understands that it can make progress in the interpretation of the real crystal-formations only insofar as it gradually discovers the laws for the definite atoms’ definite aggregations.

Since the theory of aggregation plays so comprehensive a role in the science of experience, it goes without saying that the category *aggregation* must be taken now in a stricter, now in a wider, now in a mechanical, now in a chemical, now in an organic sense.

“All plants and animals begin ... in the form of a simple cell; the cell itself is formed out of a mother-substance (Cytoblastema), in such a way that several microscopic small grains gather in a group (the nucleus), around which lay themselves others, less firm parts of the mother-substance and an outer thin membrane. In crystallization there are separated out of a solution or, if one will, a mother-liquid, homogeneous firmer substances and around these others homogeneous, the smallest precipitated parts constantly observing a certain position to one another mutually, whereby there arises a regular figure, different according to the substance’s chemical nature. It is true that a certain difference between that cell-formation and this crystallization cannot be mistaken. The cells are round, soft, hollow, and filled with a liquid; the crystals angular, firm, only in certain cases containing a crystallization-liquid and in comparatively small quantity”. (D. F. Eschricht: *Læren om Livet* [The Doctrine of Life], S. 35).

The difference between the organic points and the crystal-points may otherwise be ever so thoroughgoing; so is nonetheless, according to what is hereby made clear, punctuality and aggregation a common condition for crystal- and plant-formation.

In that natural science looks at the types, it can in the different types very well see an expression of the teleological; but as natural science it must nonetheless in the very strictest sense assert the principle that the realization of any type whatever is explained by the mutually attracting and repelling fundamental parts’ separation,

union, and arrangement proceeding in such a way from the parts' original disposition that a type is realized only in that the strict law is by force of the hypothetically necessary fulfilled point for point.

If we now consider the difference, grounded in the nature of the matter, between an a priori doctrine of Ideas and an empirical doctrine of reality, we can understand that each of these directions must be related to a peculiar corresponding view of the world. That a one-sidedly continued Speculation must dull the eye for the real is just as psychologically explicable as that a persevering analysis of reality must make consciousness foreign to the Idea as Idea.

To this come still such idealistic blunders and fantastic encroachments as could not do otherwise than bring the again-refreshed Materialism into a rage.

Thus supposedly scientific efforts to hold fast to God's supernatural essence in the form of speculative Idea have, among Idealism's opponents, served only to nourish the delusion that science can in earnest disprove God's being. "*Der außer- und übermenschliche Gott ist nichts Anderes, als das außer- und übernatürliche Selbst, das seinen Schranken entrückte, über sein objectives Wesen gestellte subjective Wesen des Menschen.* [The extra- and superhuman God is nothing other than the extra- and supernatural self, the subjective essence of man withdrawn from its barriers, set above its objective essence.]" (Feuerbach).

Thus supposedly scientific attempts to prove the soul's immortality by force of the Idea's demand have served only to call forth proofs of the soul's mortality by force of reality's conditions. "*Es ist so unmöglich, daß ein unversehrtes Gehirn nicht denkt, wie es unmöglich ist, daß der Gedanke einen andern Stoff, als dem Gehirn als seinem Träger angehöre.* [It is as impossible that an undamaged brain does not think, as it is impossible that the thought belongs to another material than the brain as its bearer.]" (Moleschott).

Thus an unscientific application of the teleological has occasioned an equally unscientific denial of all teleology in nature. "*Wenn der Hirsch lange Beine zum Laufen hat, so hat er dieselben nicht deßwegen erhalten, um schnell laufen zu können, sondern er läuft schnell, weil er lange Beine hat.* [When the stag has long legs for running, it has not received them for that reason, in order to be able to run fast, but it runs fast because it has long legs.]" (Büchner).

Finally, in the strife for and against the reality of the ideal, the unconditional

Materialism has been occasioned to apply a means that will not thus rightly suit Idealism, namely a peculiar kind of natural pithiness. “*Die Gedanken stehen in demselben Verhältniß zu dem Gehirn, wie die Galle zur Leber oder der Urin zu den Nieren, um mich einigermaßen grob hier auszudrücken.* [The thoughts stand in the same relation to the brain as the bile to the liver or the urine to the kidneys, to express myself somewhat coarsely here.]” (C. Vogt).

C.

The Validity of the Idea.

§ 34.

To deny the Idea is to deny cognition's a priori presuppositions; to acknowledge these presuppositions is to acknowledge the Idea.

That Materialism in the 18th century must be regarded as a peculiar consequence of Locke's Sensualism has already been dealt with in the theory of knowledge (S. 14); in what relation the more recent Materialism, on the contrary, will set itself to the question of the a priori is still, as it seems, rather unclear.

“*Die Welt*” [The world], it is said in a work: “*Kraft und Stoff*” [Force and Matter] (von Dr. Büchner, Frankfurt a. M. 1855, S. 6) “*oder der Stoff mit seinen Eigenschaften, die wir Kräfte nennen, mußten von Ewigkeit sein, und werden in Ewigkeit sein müssen.* [or matter with its properties, which we call forces, must have been from eternity, and will have to be in eternity.]” Here the category “*Ewigkeit*” [eternity] is now used, with an emphasis so apodictically decisive as though this concept held a priori, and the a priori held unconditionally. Meanwhile the author does not seem to be quite sure in his matter; that is to say: he is, as a genuine Materialist, naturally eternally convinced of matter's eternity; but about the concept of the eternal he is nonetheless not quite clear with himself. “*Freilich*” [To be sure], he adds, “*ist der Begriff 'Ewig' ein solcher, der sich schwer mit unsern endlichen Verstandeskräften zu vertragen scheint* [is the concept ‘eternal’ one that seems to comport with difficulty with our finite powers

of understanding]”; ... here the materialist dogmatist seems to have a touch of the Humean scepticism.

If Materialism is really to assert the scientific superiority it ascribes to itself, then it must necessarily express itself clearly and definitely concerning the a priori. It must then be scientifically demonstrable how it properly comes about that one can a second time approach Locke without being troubled by Hume, a second time do homage to Sensualism without being assailed by Scepticism.

Materialism is in its immediate self-sufficiency dogmatic. So long as this Dogmatism straightforwardly builds on its own assertions, it naturally need not trouble itself the least about the results won in the theory of knowledge; but when it once again condescends to tread the scientific path, it will scarcely be able to avoid serious negotiations with Hume and Kant. That Materialism with its crudities must be to a thinker like Kant insufferable goes without saying. He himself says of his critique of reason in the preface to the second edition: “*Durch diese kann allein dem Materialismus, Fatalismus, Atheismus, dem freigeisterischen Unglauben, der Schwärmerei und Aberglauben, die allgemein schädlich werden können ... selbst die Wurzel abgeschnitten werden.* [Through this alone can the very root be cut off from *materialism, fatalism, atheism*, from freethinking *unbelief*, from *enthusiasm* and *superstition*, which can become generally harmful ...]”

But to wish to exclude Materialism by excluding Speculation and holding to Kant alone is nonetheless also unscientific. If one first grants the a priori validity, one is also compelled to grant the dialectical validity; but, when the Dialectic is carried through, the critique of reason is transformed. The rubrics set apart in the Kantian schema (cf. S. 24–26) contain, when the matter is seen essentially, by no means separate for-themselves-existing pieces of a finitely composite whole of consciousness, but dialectical moments of a rational totality-unity. The pure intuitions and the pure concepts of the understanding do not fall outside or beside the ideas of reason; for sense and understanding belong just as surely under reason as intuitions and concepts under the Idea.

To see how the critical bound at all points in the analysis of essence gives way to the dialectical, we can here, e.g., make a trial with the separation of analytic and synthetic judgments.

“Matter is an extended thing,” this judgment is analytic; for to deny a predicate that

straightforwardly follows from the subject's concept would indeed be a contradiction. But that in this analytic judgment there is at the same time a synthetic moment is just as incontrovertible; it follows, namely, from this, that two different intuitions — extension and matter — are in the judgment united into unity. By wholly removing the synthetic we got only the tautological: "Matter is matter, the extended is an extended thing," but in such tautologies the analytic too is then lacking.

"The straight line is the shortest path between two points," and " $5 + 7 = 12$," these propositions Kant grasps as synthetic judgments a priori. "*Denn ich nehme zuerst die Zahl 7, und indem ich für den Begriff der 5 die Finger meiner Hand als Anschauung zu Hilfe nehme, so thue ich die Einheiten, die ich vorher zusammennahm, um die Zahl 5 auszumachen, nun an jenem meinem Bilde nach und nach zur Zahl 7, und sehe so die Zahl 12 entspringen ... Der arithmetische Satz ist also jederzeit synthetisch .. Eben so wenig sind irgend ein Grundsatz der reinen Geometrie analytisch.* [For I first take the number 7, and in that, for the concept of the 5, I take the fingers of my hand as intuition to help, I add the units that I previously gathered together to make out the number 5, now to that image of mine little by little up to the number 7, and thus see the number 12 arise ... The arithmetical proposition is thus at all times synthetic .. Just as little is any principle of pure geometry analytic.]" (Krit. d. r. Vernft. Einleitung).

That in all mathematical judgments there is a synthetic element can so much the less be denied, as it has precisely shown itself that an analytic judgment is impossible when one excludes all co-determination by the synthetic; but that the mathematical judgments should therefore, as Kant supposes, straightforwardly exclude the analytic, is so striking a misunderstanding that Hegel in his displeasure hereat has even explained it by "a caprice."

What has moved Kant to so finite a separation of the analytic and the synthetic is now nonetheless no caprice; an unconditional holding-fast to the critical bound is, what Hegel could easily have seen, here as everywhere an inviolable consequence of Criticism itself.

As it goes with the reciprocal determination of the analytic and the synthetic, so it goes with all the other reciprocal determinations: from the side of reality the critical holds, but from the side of essence the dialectical.

The critical separation of the pure intuition without concept and the pure concept without intuition is idealess and in its essence untrue: "*Raum und Zeit sind besonders*

dann keine Gedankenbestimmungen, wenn man keine Gedanken dabei hat: aber ein Begriff, sobald man einen Begriff davon hat. [Space and time are especially then no thought-determinations, when one has no thoughts in the matter: but a concept, as soon as one has a concept of it.]” (Hegel). Concept and intuition are moments for one another mutually, their dialectical unity is their truth: they are in the Idea one. The critical separation of the categories of quantity, quality, relation, and modality has in the finite its significance, but is, when it is a matter of the essence itself, idealess and untrue. To criticism’s finite separation corresponds an equally finite connection; but the Idea is just as little a sum of finitely connected categories as a for-itself, i.e. over against the others existing, category of reason. The Idea is, as Idea, the categories’ inner, principled totality, the truth in the categorical determinations. The critical separation of a thing in itself without properties and a phenomenon of properties without a thing in itself is idealess and untrue; the truth is that the thing is determined by its properties and the properties by the thing: the infinite reciprocal determination grounds itself on the Idea. The critical separation of a subject for itself without object and an object for itself without subject is from the side of essence the highest idealessness. To deny the identity of the subjective and the objective is to deny the Idea and make all cognition of truth impossible.

As for the assessment and the use of the categories’ schema, an essential difference shows itself between one-sidedness and thoughtlessness. That Hume denies the Ideas’ a priori essentiality is one-sided, but consistent, since he on the whole denies the a priori; that Kant himself ascribes to the Ideas only subjective validity is likewise one-sided, but consistent, since he does not either ascribe to his a priori intuitions and categories objective significance. But to wish to use some of the a priori categories in a merely subjective, others on the contrary at the same time in an objective sense, is thoughtlessness.

If one considers the a priori categories of the understanding for themselves, scarcely anyone is so inconsistent as to ascribe to the ground objective validity when he grants the consequence only subjective validity, or to grant both ground and consequence objective significance when the concept of causality is to have only subjective significance; on the contrary, it is a not so seldom, though in its essence equally thoughtless, inconsistency that one without more applies the a priori intuitions and concepts of the understanding objectively, while one nonetheless regards the Ideas as something

so subjective that they exist only in our heads, without in the least concerning the essence of things.

Outside the analysis of essence it does indeed at first glance look as though the Idea were either merely a subjective point, or, as far as the objective is concerned, nonetheless so infinitely far away that the human understanding could scarcely descry it. But the matter stands reversed; the objective Idea is infinitely near. In every intellectual glance, in every a priori category, in every point where knowledge and being are united in truth, the Idea acts. He who in mathematics will deny the validity of the higher analyses must consistently end by denying the validity of the mathematical axioms; and yet there is such a distance between these axioms and those analyses that “sound understanding,” which involuntarily acknowledges the axioms, must work long before it raises itself to insight into the analyses’ inner infinity. In corresponding manner it stands with the analysis of the Idea and the a priori categories.

That it is one and the same fundamental thought that runs through the Platonic, the Aristotelian, and the Hegelian doctrine of Ideas, so that one cannot — what nonetheless sometimes happens in thoughtlessness — praise the Platonic Dialectic and cast contempt on the Hegelian, or, as far as the Idea is concerned, support oneself on Aristotle in order to combat Hegel, is, when we look at the principle, incontrovertible.

The Hegelian Logic is in its grandiose execution a scientific proof of the Idea’s validity. One can grant that there are in this work of genius, as far as individual points and portions are concerned, obscurities and blunders that need corrections, and that corrections will especially be conditioned by sharp preceding analyses of reality; but one cannot give up the fundamental thought without at the same time either wholly giving up philosophy, and thereby at bottom all the sciences’ ideal foundation, or making oneself guilty of a striking thoughtlessness.

Even if it is an error that the philosophizer should in Speculation be able to make himself one with the Idea, it is nonetheless no error that the Idea as Idea lets itself be presented only in dialectical scientific Speculation; even if it is a misunderstanding that philosophy should be able to give a knowledge absolute, corresponding to reality, it is nonetheless no misunderstanding that the absolute identity of essence and reality should be in the Idea itself; even if it is untrue that the universal concepts should not be able to hold except by force of dialectical speculative analysis, then it is nonetheless not for that reason untrue that the a priori categories indispensable in every science

are, when they are seen in their inner dialectical connection, sides of the objective essence, moments of the Idea itself.

§ 35.

To deny the Idea is to deny the real sciences' common ideal; to acknowledge this ideal is to acknowledge the Idea.

When it is said in the theory of science (S. 69) that only that science is cultivated which is cultivated for its own sake, this will in other words say that every true theory must seek its satisfaction in the Idea. The ground that the particular sciences do not relate to the Idea as Idea is by no means this, that they lack an eye for the Idea, or are indifferent to the Idea; on the contrary! it is precisely out of interest for the Idea that they choose their point of departure in the finite, and work themselves through the real; for only on these conditions can the cognition of the Idea get its real content.

Since now a real science cannot possibly take its standpoint in the pure Idea, and just as little can strive after any other completion than that which consists in true appropriation of the Idea's content, the Idea in its ideality must be the goal for its striving, the absolute goal, the ideal. Insofar as the Idea is the ideal, one does not begin with the principled, the teleological, the typical, but one strives onward toward the principled, the teleological, the typical. That the matter stands thus can here be shown by reference to definite ideals:

a) The cosmological ideal.

Astronomy has not begun, and cannot begin, either with a Platonic ἰδέα [idea], or an Aristotelian εἶδος [form], or a Hegelian "Begriff" [concept]; astronomy must, it lies in the nature of things, begin with observing facts and making calculations. Since now every science in a certain respect always continues as it has begun, astronomy will not merely begin, but also continue, with observations and calculations, and it could to that extent at first glance seem as though it, immersed in its observations and calculations, never heeded the Idea, never came nearer to the Idea.

It is, however, not phenomena at random, but a definite circle of definitely connected phenomena, with which astronomy occupies itself: the object of its observations and calculations is not this or that contingent, but a totality subsisting in itself, resting

on itself. The greater progress the science makes, the more penetratingly, all-sidedly, and comprehensively the mutual relation, ground, and character of the whole and the parts is known, the more distinctly the rational is seen, the more the merely contingent vanishes.

That the solar system is a realized plan cannot then, from the standpoint of the science, be doubted. How far this plan is realized with or without consciousness, with or without determining will, is a quite different question; but that the plan, as it now for once is realized, shows a thoroughgoing rationality, a wisdom, as though it were realized with divine knowledge, is from the side of the science incontrovertible.

That astronomy, in that it solves the phoronomic tasks, is at the same time occasioned to treat cosmological problems, can be seen from the connection between the theory of gravitation and the Kantian-Laplacean hypothesis of a rotating primal nebula. If it now, what surely no one can deny, lies in the science's interest to get this hypothesis worked out and grounded in such a way that the solar system's formation, from the first, principled disposition down to its present articulation, thereby lets itself be explained as a development typically-teleologically advancing toward completion; then it is at the same time established that the astronomical science presupposes the Idea's reality, and approaches the turning-point when the investigations' finite piecework goes up in the unity of the cognition of the Idea, as into its ideal.

b) The geological ideal.

In that geology presupposes geognosy, it must necessarily take its beginning in the finite, in reality's piecework.

To collect and investigate minerals, to characterize rock-species, to point out the relation between strata, broken-through masses, neptunic, plutonic, and volcanic formations, and so on, belongs to such a degree under the category of piecework that one would, on preliminary consideration, scarcely suppose that such heterogeneous constituents could be worked together into a scientific whole, let alone that their working-up should lead to a cognition of the Idea.

Nonetheless geognosy has become a science; the more this science masters the facts, the more thoroughly it works up the elements of finitude, the more it points out beyond itself: geognosy is there for geology's sake; but geology has the goal in itself, for its content is of the Idea. Geology is indeed at its present stage still far from

being able to present the earth's formation as the Idea's own realization; but that this is nonetheless the ideal toward which it strives is unmistakable.

Seen from geognosy's standpoint the earth is only a pieced-together whole; from geology's standpoint the earth-body is, on the contrary, to be grasped as an elementary organism, an essential whole whose parts have gradually developed themselves out of an inner disposition. In the disposition's originality we then have again the principled, in the parts' separation and the whole's planned articulation the typically-teleological, i.e. we have, if the cognition of the infinite mediation were only complete, the Idea realizing itself in the earth-formation.

c) The physiological ideal.

"One has often compared the organism and its individual parts with a machine or a similar physical-chemical apparatus. The heart resembles a pump-work furnished with fitting valves, the arteries elastic tubes, the eye a *Camera obscura*, the vocal organs a tongue-instrument; the bones relate as levers; the secretory organs in part as filtering apparatus; the lungs act as an evaporation apparatus furnished with a large surface." (Valentin, Physiologie, worked up by Hannover, Copenhagen 1857, S. 5.)

When the physiologist thus permits himself to grasp even the human organism as a machine, this has not only for the immediate self-feeling something offensive, but seems even to degrade science, and set it in a Materialism irreconcilable with the Idea of life. This offensiveness dissolves, however, into a nothing when one considers that a physiological investigation, just as little as any other investigation belonging under reality, can take its point of departure in the pure Idea. That science, however, does not, like Materialism, restrict itself to uttering an individual opinion, or content itself with a finite machinery, but on the contrary sets itself to work its way into an inner infinity, can already be seen from the way in which the above-adduced description is continued. "But our mechanical apparatus' arrangement is simple, poor, and imperfect; they consist of large parts, their mode of action is stable and bounded, and they can be kept going only by supply, help, and repair from without; in the opposite case they stop instantly. Our body's apparatus are, on the contrary, composed of an infinitely great number of microscopic parts, which form just as many small workshops, whose activity engages in one another, and according to circumstances is modified in manifold ways, since they are labile arrangements. They do indeed need nourishment from

without; but by reason of their independence they are able not only to maintain the activity of the parts already at hand, but they can themselves repair the worn-out parts, replace loss, and even create new workshops. The living beings play at the same time the role of the machine, the machine-master, and the master-builder.”

By this last addition the category of the machine is carried out beyond its bound; the organism is again organism, and physiology will, the more deeply it penetrates into the parts’ typically-teleological mediation, the more approach the cognition of the Idea.

The real sciences’ relation to the Idea can be exactly determined by their relation to the teleological: they cannot straightforwardly apply teleology, and they cannot do without teleology.

That the real knowledge must hold to the efficient causes has already been brought out. In the analysis of reality one can just as little explain something’s coming-to-be by appealing to its purposiveness as by appealing to the pure Idea. The application of the teleological has in this respect often served as a cover for ignorance.

It is a great misunderstanding to wish to help oneself with a purpose-cause instead of an efficient cause, or to suppose that, when something happens without our being able to discover the efficient causes, it is then for that reason immediately effected by an invisible will. The dilemma “either something happens in nature according to efficient causes, or it happens according to a purpose-cause” does not hold. Just as the purpose is realized only in the system of means, so the purpose-cause is realized only in the system of the efficient causes.

In the Idea as Idea the means are moments for the purpose, the efficient causes for the purpose-cause; in the finite reality the means, the efficient causes, step especially forth as existing members, while the purpose, the purpose-cause, dawns as a thought hovering over the development. From this it is understood that the real knowledge, which holds immediately to the means, the efficient causes, cannot look at the purpose, cannot immediately use the teleological.

The real knowledge must nonetheless not any the less presuppose and tacitly understand the teleological, just as it also must presuppose and tacitly understand the Idea.

In the hypothetical inference “If A is, then B must be; now A is, thus B is too,” B’s being is doubtless grounded as a consequence of A’s being, but A’s being is,

in the proposition “now A is,” only grasped as fact, i.e. as contingent. The whole grounding thus rests on something contingent; if we again found A’s being grounded as a consequence of a preceding member and went millions of members back, this would nonetheless not change the matter. In the infinite series of the members conditioning one another there rules a thoughtless play of the blindly contingent and the blindly necessary: “The thing is, because it is, and the thing is, because another thing is”; in this thoughtlessness there is no grounding.

If one now casts the grounding away, one can easily imagine oneself able to explain nature’s order without needing any wisdom of nature. The explanation is then this: “there is in nature an order of things, not because there is an ordering principle, but because a disorder of things is impossible. Nature is an infinite whole that consists in its parts. As the parts are, so too must the whole be, and conversely: as the whole is, so must the parts be. Had we another nature-whole, then we would also have other parts; nature’s order rests precisely on this, that the parts cannot do otherwise than correspond to the whole, and the whole to the parts. What must therefore happen, happens, and what does not, according to the connection of things, necessarily have to happen, does not happen either; see, that is the whole wisdom! If all that we now call order dissolved, and the world-structure collapsed into chaos, then this would nonetheless happen only because it necessarily had to happen, and this apparent disorder would thus nonetheless be in its order.”

But thus to dissolve the grounding into a play of the contingent-necessary is unscientific. The scientific grounding demands an essential first. The real knowledge must indeed start from the fact; but the fact is for that reason not the essential first; from the facts in their contingency there is striving inward toward the connection in the total, inward toward the original disposition, inward toward the ground, for the ground is the essential first. This essential first is in the disposition’s realization at the same time the last, and yet there is a difference between the first and the last.

The contradiction, that the first and the last are in essence one and yet in reality different, can be resolved only in the teleological: the teleological identity of the last and the first is not a superficial circular motion, but the Idea itself, which, in that it realizes its content, grounds all reality.

If now the real sciences, each from its point of departure, are in motion to find out the given facts’ ground and grounding, then they are also all, each from its point of

departure, in motion onward toward the all-grounding Idea; even if none of the real sciences can relate to the Idea as Idea, the cognition of the Idea is nonetheless none the less the common ideal for them all.

§ 36.

To deny the Idea is to deny Nature; to acknowledge Nature is to acknowledge the Idea.

Although the natural objects involuntarily offer themselves to immediate intuition, it is nonetheless a difficult task to develop the concept *Nature*, to state quite accurately and definitely what Nature is. Insofar as the realm of nature is a realm of facts, and facts must be grasped by experience, one would suppose that it properly fell to the science of experience to answer the question: what is Nature?

Nonetheless one will in vain seek the answer in the proper doctrine of nature. We meet here the wonderful thing, that the science which alone among all rightly understands how to investigate nature's objects, nature's forces, nature's laws, in short: all that nature's is, nonetheless becomes embarrassed, and begins to stammer and lisp, when it is to say definitely what nature is. Either it remains at loose examples and figurative descriptions of "the Mother of the All," of "the Bearing," of "the eternally Existing in becoming and producing," or else, when the concept is to be indicated a little more fully, and determined a little more sharply, it ends with a reference to some philosopher, e.g. to Aristotle.

That natural science cannot say what Nature is must, however, not be misunderstood as though it either did not acknowledge the reality of the concept of nature, or were not able to connect a definite thought with the word "Nature"; on the contrary! what nature is, natural science can in its way present clearly enough, but, be it well noted, thereby, that it presents, not Nature itself, but nature's facts and laws, thus not straightforwardly, but mediately, and that for the good reason that Nature is an Idea, and no science of experience able to present the Idea as Idea.

Insofar as Nature is Idea, the development of the concept of nature is a task for philosophy; but since the Idea as Nature is only what it is in its productions, phenomena, and laws, the development of the concept's content must necessarily belong under Empiricism. It is precisely a peculiarity of the concept of nature that it with equal force separates and connects philosophy and Empiricism.

a) Nature as Idea.

In the thinking grasp of the natural things Empiricism meets, according to Hegel (Naturphil., Einl.), with philosophy. *“Beim theoretischen Verhalten ist das Erste, daß wir von den natürlichen Dingen zurücktreten, sie lassen, wie sie sind, und uns nach ihnen richten. Wir fangen hierbei von sinnlichen Kenntnissen der Natur an. Wenn die Physik indessen nur auf Wahrnehmungen beruhte, und die Wahrnehmungen nichts wären, als das Zeugniß der Sinne: so bestände das physikalische Thun nur im Sehen, Hören, Riechen u. s. w., und die Thiere wären auf diese Weise auch Physiker. Es ist aber ein Geist, ein Denkendes, welches sieht, hört u.s.w. [In theoretical comportment the first thing is that we step back from the natural things, leave them as they are, and direct ourselves according to them. We begin here from sensible knowledge of nature. If physics, however, rested only on perceptions, and perceptions were nothing but the testimony of the senses: then physical activity would consist only in seeing, hearing, smelling, and so on, and the animals would in this way also be physicists. But it is a spirit, a thinking thing, that sees, hears, and so on.]”*

The next thing is then that the things, in their relation to us, receive the determinations of the universal, or that we transform them into something universal. The more thought there is in the representation, the more of the things' naturalness, finitude, and immediacy vanishes: *“durch den sich eindringenden Gedanken verarmt der Reichthum der unendlich vielgestalteten Natur, ihre Frühlinge ersterben, ihre Farbenspiele erblassen. Was in der Natur von Leben rauscht, verstummt in der Stille des Gedankens; ihre warme Fülle, die in tausendfältig anziehenden Wundern sich gestaltet, verdorrt in trockne Formen und zu gestaltlosen Allgemeinheiten, die einem trüben nördlichen Nebel gleichen. [through the intruding thought the wealth of infinitely manifold nature is impoverished, its springs die away, its plays of color grow pale. What in nature rustles with life falls silent in the stillness of thought; its warm fullness, which shapes itself in a thousandfold alluring wonders, withers into dry forms and into shapeless universalities that resemble a dull northern mist.]”*

These theoretical determinations are not only the opposite of the practical, but they are even in contradiction with one another mutually, in that theory immediately seems to effect the opposite of what it intends. *“Nämlich wir wollen die Natur erkennen, die wirklich ist, nicht etwas, das nicht ist; statt sie nun zu lassen, und sie zu nehmen, wie sie in Wahrheit ist, statt sie wahrzunehmen, machen wir etwas ganz Anderes daraus. Daburch, daß wir die Dinge denken, machen wir sie zu etwas Allgemeinem; die Dinge sind aber*

einzelne, und der Löwe überhaupt existirt nicht. Wir machen sie zu einem Subjectiven, von uns Producirten, uns Angehörigen, und zwar uns als Menschen Eigenthümlichen; denn die Naturdinge denken nicht, und sind keine Vorstellungen oder Gedanken. [Namely, we want to know nature, which really is, not something that is not; instead of now leaving it and taking it as it is in truth, instead of perceiving it, we make something quite other out of it. By thinking the things, we make them into something universal; but the things are single, and the lion in general does not exist. We make them into a subjective, produced by us, belonging to us, and indeed peculiar to us as men; for the natural things do not think, and are no representations or thoughts.]”

To resolve this contradiction, Hegel appeals to natural philosophy, whose determination and aim it is that spirit can in nature find its counter-image, its own essence, the existing concept. *“So ist die Natur die Braut, mit der der Geist sich vermählt. Aber ist diese Gewißheit auch Wahrheit? Indem das Innere der Natur nichts Anderes, als das Allgemeine ist: so sind wir, wenn wir Gedanken haben, in diesem Innern der Natur bei uns selbst ... Ich in meinem Wesen ist der Begriff, das mit sich selbst Gleiche, durch Alles Hindurchgehende, welches, indem es die Herrschaft über die besonderen Unterschiede behält, das in sich zurückkehrende Allgemeine ist. Dieser Begriff ist zugleich die wahrhafte Idee, die göttliche Idee des Universums, die allein das Wirkliche.* [Thus nature is the bride with whom spirit weds itself. But is this certainty also truth? In that the inner of nature is nothing other than the universal: so we are, when we have thoughts, in this inner of nature with ourselves ... I in my essence am the concept, the self-same, going through all, which, in that it retains the mastery over the particular differences, is the universal returning into itself. This concept is at the same time the true Idea, the divine Idea of the universe, which alone is the actual.]”

The thinking consideration of nature must now show how nature in itself is the process of becoming spirit, of sublating its being-other, — and how the Idea itself is present at every stage of nature; *“von der Idee entfremdet, ist die Natur nur der Leichnam des Verstandes. Die Natur ist aber nur an sich die Idee, daher sie Schelling eine versteinerte, Andere sogar die gefrorne Intelligenz nannten ... Ist die Natur die Idee in der Form des Andersseyns, so ist, nach dem Begriffe der Idee, die Idee darin nicht, wie sie an und für sich ist, obgleich nichtsdestoweniger die Natur eine der Weisen der Idee ist, sich zu manifestiren, und darin vorkommen muß.* [estranged from the Idea, nature is only the corpse of the understanding. But nature is only in itself the Idea, wherefore

Schelling called it a petrified, others even the frozen, intelligence ... If nature is the Idea in the form of being-other, then, according to the concept of the Idea, the Idea is therein not as it is in and for itself, although nonetheless nature is one of the modes of the Idea to manifest itself, and must occur therein.]”

b) The Idea as Nature.

The Idea of nature, as it is grasped in the above Hegelian presentation, is still grasped only according to its essentiality; but to the Idea of nature belongs the real, and in reality the doctrine of experience holds; the question thus becomes this, how far the science of experience, in that it penetrates more and more into the essence of the natural things, now also really finds such first presuppositions as are in their essence compatible with that natural-philosophical doctrine of Ideas.

Two presuppositions seem, when we in this connection consider the matter from Empiricism's standpoint, to offer insurmountable barriers to a cognition of the Idea as Nature, namely matter's *atoms* and their *aggregation*; these presuppositions ought therefore to be tested.

That natural science can neither give up the atomic theory, nor in its analyses of reality go out beyond it, has already been dealt with in the foregoing; but by the fact that Empiricism with its analyses must stop at material fundamental parts as at the separations' outermost bounds, the Idea's validity is not in the least weakened. Only when Materialism once in time gets occasion to furnish a scientific, irrefutable proof of the so-called elementary atoms' eternity and unchangeableness will all talk of Nature as Idea and the Idea as Nature have an end.

If trumping assertions were proofs, then Materialism might indeed already be said to have brought many and great and forceful proofs for the atoms' eternal unchangeableness; but insofar as a scientific proof is to be built on facts and grounds, there has here, so far as is known, hitherto no proof yet come forth.

If Materialism now did not wholly disdain every regard to the relation between experience and the a priori, it would surely not deny that the proposition about the atoms' eternity and unchangeableness cannot possibly be fetched from any finite experience; but, when it becomes evident that the proposition is a priori, belonging under metaphysics, one will not either be quite able to overlook the difficulties meeting here. To presuppose eternal, in themselves unchangeable atoms is, in other